

Identification of Rational Types in Unknown Environments: Evidence from Kidnapping in Colombia

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Abstract

This paper studies the design of a dynamic indirect mechanism to deter and dismantle criminal organizations under incomplete information, with application to kidnapping for ransom in Colombia. The empirical evidence provides the mechanism's prior and parameters: a vector error correction model characterizes the macroeconomic long-run trends of kidnapping and interest rates; multinomial logit and competing hazard models identify costs, durations, outcomes, and heterogeneity across criminal types. The central theoretical contribution is the Theorem on the Identification of Rational Types in Unknown Environments, which transforms recurrent separation between observable signals into posterior identification of the true type. Under consistent Bayesian updating in the prolonged captivity event, the State's beliefs converge almost surely to the true type of the captor. The paper also establishes conditional implementability in Perfect Bayesian Equilibrium when the feasible policy set is non-empty. The dynamic calibration illustrates the operational viability of the mechanism and shows how state instruments can accelerate learning and discipline optimal policy.

Keywords: mechanism design; Bayesian learning; competing risks; kidnapping; dual control; identification.

JEL Codes: C73, D82, K42.

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1 Introduction and Related Literature

Observable criminal behavior tends to group distinct types precisely when public policy needs to discriminate among them. A ransom demand, a period of silence, a partial proof of survival, or a violent escalation may each be compatible with different organizational technologies. When a single public signal can be produced by disciplined insurgent organizations, short-horizon criminal gangs, or fragmented structures, a purely passive policy faces an informational deadlock: the decision that minimizes immediate loss may preserve the pooling that prevents identifying the adversary in the future. Mechanism design provides the architecture to formulate rules that align private incentives with the State's normative objectives under voluntary participation and sequential rationality.

The central question is how to design an active screening mechanism when observable actions lie in pooling regions. The planner is the State; its continuous instruments are financial interdiction $\alpha_t \in [0, 1]$ and operational pressure $\gamma_t \in [0, 1]$. The adversary possesses a private type $\theta_K \in \Theta_K$. The problem therefore consists not only of choosing between rescue and negotiation, but of calibrating (α_t, γ_t) so that the induced law of daily signals is informative about the latent criminal technology.

This paper sits at the intersection of dynamic mechanism design, the economics of crime, and active learning. The Revelation Principle and incentive compatibility and individual rationality constraints provide the foundation of the implementation problem (????). The dynamic mechanism design literature extends this approach to intertemporal environments (???), while the dynamic information acquisition literature shows why an optimal policy must value the effect of present actions on the quality of future signals (?). The economics of crime and kidnapping studies document incentives, durations, and organizational heterogeneity (???????).

The contribution integrates three elements. First, the Mano de Dios–Guadalupe (MDG) process introduces stochastic implementation with full support, keeping the Bayesian operator well-defined off the equilibrium path. Second, a competing risks structure connects duration, mode of termination, and learning. Third, the Theorem on the Identification of Rational Types in Unknown Environments converts a local condition of recurrent separation into global posterior identification. The Implementability Theorem complements this result by establishing the conditions under which the State's rule can be sustained in Perfect Bayesian Equilibrium without violating IC^K , IR^K , or IR^F .

2 Institutional Framework and Stylized Facts

2.1 Kidnapping as a Concentrated and Heterogeneous Phenomenon

UNODC records (?) for 145 countries in 2023–2024 count 1,514,546 kidnapping cases and reveal a distributional asymmetry with direct implications for analysis. The median annual average per country stands at 46 cases, while the mean reaches 1,153—a mean-to-median ratio of approximately 25:1, ruling out the existence of a representative country and indicating that the distribution of national annual averages has a heavy tail. The 75th percentile is 109 cases, meaning three-quarters of countries record volumes below this threshold while the upper quarter concentrates the dominant share of the world total. This extreme concentration structure, analogous to income or firm-size distributions, invalidates interpretations based on average behavior and demands treating institutional, legal, and criminological heterogeneity across countries as a structural attribute of the phenomenon—motivating the heterogeneous-types approach adopted here.

The world map in Figure 1 shows that several developed countries rank among the highest in absolute kidnapping numbers, in some cases exceeding countries typically associated with lower living standards, high income inequality, elevated poverty, and prolonged armed conflict, such as India, Pakistan, Afghanistan, Somalia, Haiti, Ecuador, Venezuela, and Colombia. This configuration does not necessarily reflect greater physical risk for the general population; rather, it is driven by substantive differences in the legal definition of kidnapping, institutional capacity for data recording, and propensity to report crimes, as documented in ? and ?. In developed jurisdictions, official statistics encompass behaviors such as parental abduction, unlawful deprivation of liberty in family contexts, and certain forms of criminal exploitation, substantially expanding the statistical universe relative to countries where only ransom kidnappings are recorded.

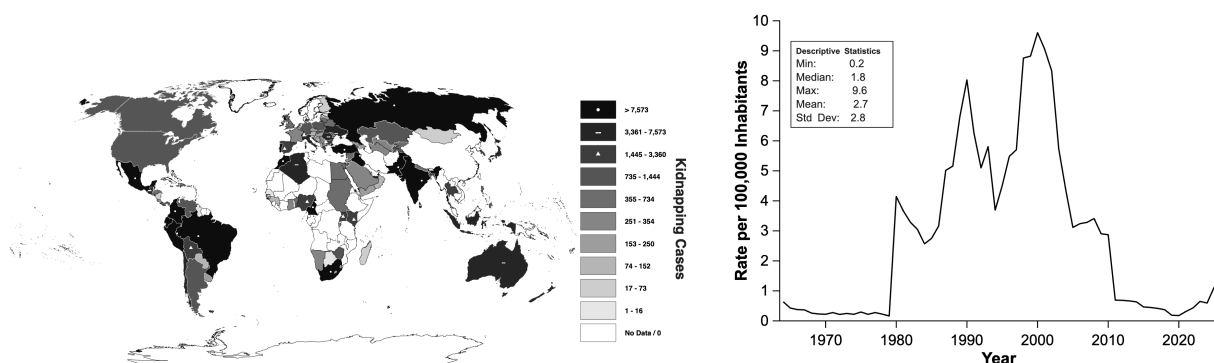


Figure 1. World distribution and trends of kidnapping in Colombia. (a) World distribution of kidnapping cases, 2003–2024. (b) Kidnapping rate in Colombia, 1964–2025.

Notes: Panel (a) shows the world distribution of kidnapping cases based on UNODC data; panel (b) shows the evolution of the kidnapping rate in Colombia during 1964–2025. *Source:* Authors' calculations based on ? and ?.

In Colombia, between 1964 and 2025, the rate of ransom and non-extortionary kidnapping per 100,000 inhabitants follows a non-linear trajectory reflecting the interaction of armed conflict dynamics, illicit economies, economic activity, and state capacity (Figure 1). From the mid-1960s through the late 1970s, the rate remained low and relatively stable at approximately 0.2–0.5 kidnappings per 100,000 inhabitants, associated with marginal forms of violence and the incipient stage of insurgent groups. A structural break emerged in the late 1970s, with the rate exceeding 4 kidnappings per 100,000 inhabitants in the early 1980s. Through the 1980s and 1990s, kidnapping exhibited a growth pattern with volatility, associated with the consolidation of guerrilla groups, narco-trafficking expansion, and the use of kidnapping as a financing and political coercion mechanism (??). This process culminated in a historical peak near 9.6 kidnappings per 100,000 inhabitants around 2000, coinciding with weakened state control and armed conflict escalation. From 2002 onward, a persistent reversal occurred, with a sustained decline reaching levels below 1 per 100,000 during the 2010s and historical lows after the 2016 Peace Agreement. A slight recent increase, still far below past critical levels, suggests a reconfiguration associated with fragmentation of armed actors.

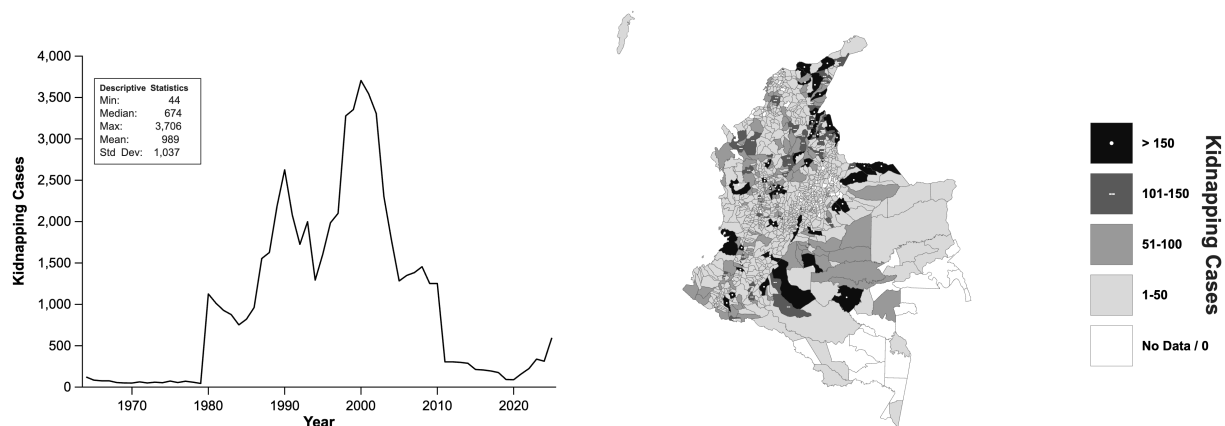


Figure 2. Temporal evolution and geographic distribution of kidnappings in Colombia. **(a)** Evolution of kidnapping cases in Colombia, 1964–2025. **(b)** Geographic distribution of kidnapping cases in Colombia, 1964–2025.

Notes: Panel (a) shows the evolution of total kidnapping cases in Colombia during 1964–2025, highlighting main escalation and decline phases. Panel (b) shows the geographic distribution across Colombian municipalities. *Source:* Authors' calculations based on ?.

In absolute terms (Figure 2), annual kidnapping records remained below 100 cases through the late 1970s. A structural break in the early 1980s pushed annual counts above 1,000. During the 1980s–1990s, volume grew sustainedly but with volatility, reaching a historical peak of approximately 3,700 cases around 1999–2002 amid conflict expansion and illicit economies. From 2002 a persistent reduction followed, with annual records below 500 during the 2010s. Spatially (Figure 2b), kidnappings occurred across most of the territory but were heavily concentrated: Bogotá D.C. (1,437 cases), Cali (784),

and Medellín (652) lead, followed by intermediate cities and strategic municipalities such as Valledupar (530), Argelia (452), Villavicencio (436), Aguachica (436), Ocaña (348), Bucaramanga (304), and Santa Marta (291). This spatial concentration, consistent with the crime-triangle framework of ?, reflects differentiated criminal opportunities, strategic corridors, and persistent state capacity limitations rather than a uniform distribution across the territory.

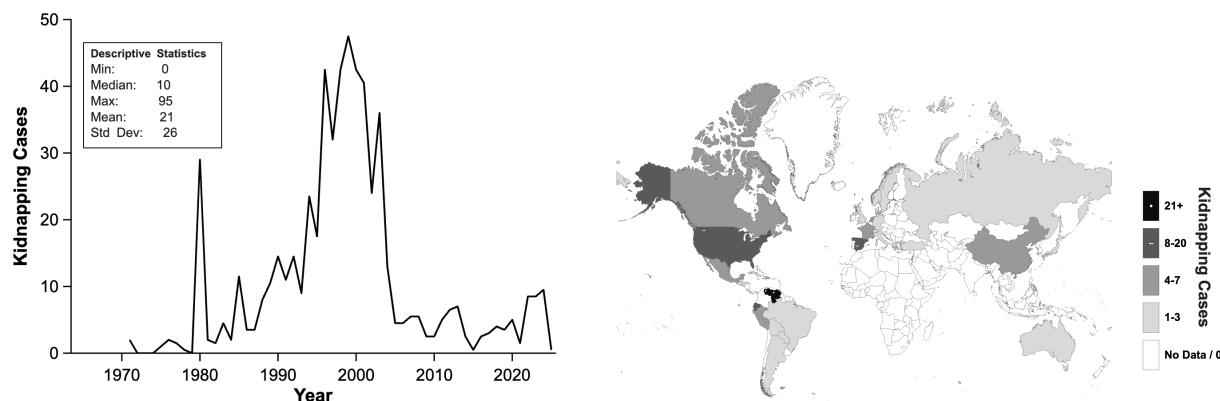


Figure 3. Temporal evolution and geographic distribution of kidnappings of foreign victims in Colombia. (a) Annual number of kidnappings involving foreign victims in Colombia, 1964–2025. (b) Geographic distribution of foreign victim kidnappings, 1964–2025.

Notes: Panel (a) shows the annual evolution of kidnapping cases involving foreign victims; panel (b) presents the geographic distribution by municipality. *Source:* Authors' calculations based on ?.

Kidnappings of foreigners followed a parallel trajectory to total cases, though at smaller scale (Figure 3). Annual cases were near zero through the late 1970s; peaks exceeding 40 cases per year (and a maximum of approximately 95) occurred during the 1990s and early 2000s. By nationality, Venezuelans represent the largest number of recorded cases (41), followed by Ecuadorians (20), Spaniards (15), and US citizens (12), reflecting a rationality of opportunity and strategic calculation configured by conflict intensity and the perceived ransom-paying capacity and international visibility of specific nationalities.

The establishment of the GAULA units (Grupos de Acción Unificados para la Libertad Personal), enacted by ?, marked a strategic transition from fragmented anti-kidnapping units toward a unified action doctrine, consolidated with the creation of the Police Anti-Kidnapping Directorate (?). Their deployment, illustrated in Figure 4, prioritized primary centers of criminal activity rather than arbitrary administrative distribution. A national network of 34 units covered strategic urban nodes; in several complex jurisdictions (Medellín, Valledupar, Cartagena), dual institutional presence was maintained. However, municipalities with high kidnapping intensity such as Argelia, Buenaventura, and Ocaña lack local GAULA units, highlighting a disconnection between reactive state capacity and crime persistence. The distribution of 3,949 arrests in ransom kidnapping episodes

(2010–2025) reveals that a substantial fraction concentrates in cities with GAULA presence: Bogotá (15.93%), Medellín (9.29%), Cali (7.39%), Villavicencio (2.38%), and Cúcuta (2.36%) lead, consistent with greater operational, investigative, and prosecutorial capacity at these metropolitan centers.

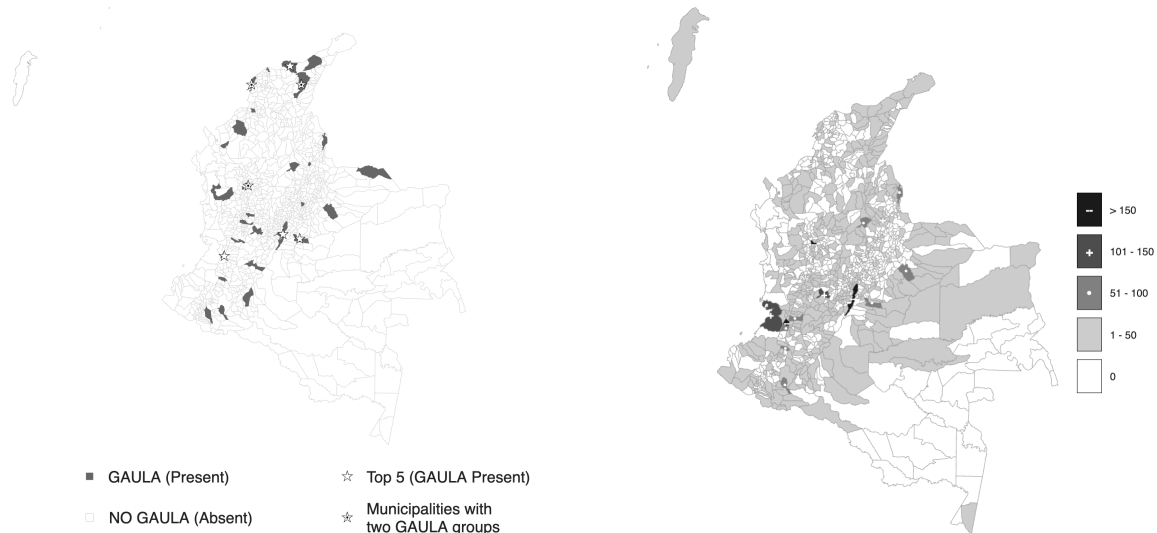


Figure 4. Spatial distribution of GAULA presence and kidnapping arrests in Colombia. **(a)** Geographic distribution of GAULA units in Colombia. **(b)** Arrests for ransom kidnapping by municipality, 2010–2025.

Notes: Panel (a) shows the geographic distribution of GAULA units; dark zones indicate GAULA presence, stars highlight the five leading municipalities and locations with dual units. Panel (b) shows the municipal distribution of arrests related to ransom kidnapping, 2010–2025; darker tones indicate more arrests. *Source:* GAULA location data from the National Police of Colombia; incidence data from ?.

Kidnapping in Colombia has been closely linked to the evolution of the internal armed conflict and the rise and consolidation of illegal armed actors. Figure 5 provides an integrated synthesis of this dynamic across successive presidential administrations, connecting changes in kidnapping patterns to distinct conflict phases and the main norms and institutional milestones shaping the state response. Through the 1960s–1970s, kidnapping appeared as a low-intensity, predominantly rural phenomenon consistent with the early stages of insurgent formation and a still-limited state capacity. A sustained escalation phase became evident beginning with the Turbay Ayala administration and more clearly under Betancur and Barco, coinciding with guerrilla expansion, narco-trafficking actors, and growing self-defense groups. This period also saw targeted normative responses including ? and subsequent decrees. The masification phase under Gaviria, Samper, and especially Pastrana saw kidnapping reach its highest levels—exceeding 13,000 cases in 1998–2002—coinciding with the 1991 Constitution, cartel pressure, paramilitary consolidation, and the Cauguán demilitarized zone, alongside ? and ? establishing the National Anti-Kidnapping Statute and GAULA. From 2002 onward, the Uribe administrations’ security policy, strengthened military and police capacity, and legal reforms including ?,

?, and ?—a penal tightening framework substantially raising the costs of kidnapping—drove a containment phase. The Santos administrations saw a readjustment phase at low kidnapping levels in the context of the peace process and the signing of the Final Peace Agreement with the FARC, supported by instruments such as ?, which introduced conditional sentence reductions for kidnapping in cases where perpetrators submitted to the JEP. Under Petro, a moderate resurgence is observed, associated with the fragmentation of armed actors, the re-emergence of dissident groups, and evolving conflict dynamics within the regulatory environment shaped by ?.

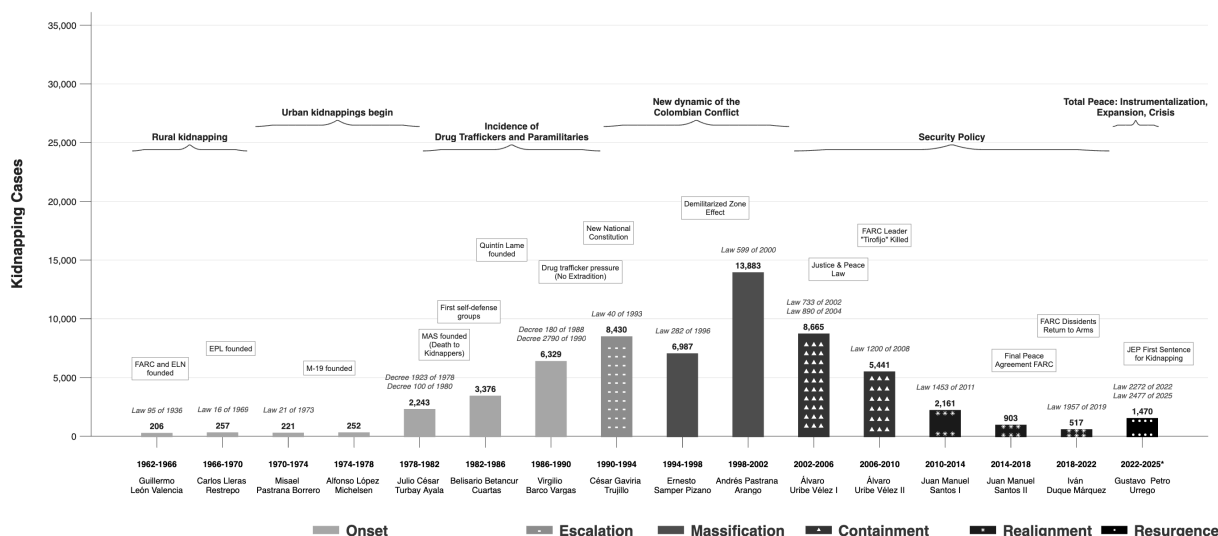


Figure 5. Evolution of kidnapping in Colombia: armed conflict, security policy, and institutional change (1962–2025). Each bar corresponds to a presidential administration; shading distinguishes different conflict phases; annotations highlight key events related to armed groups, legal reforms, and security policies. Total reported kidnappings over the indicated period: 61,341 cases.

Notes: The final interval reflects partial data for the ongoing Petro administration (term extending to 2026). *Source:* Authors' elaboration based on ? and ?.

2.2 Macroeconomic Analysis: Stochastic Trend

Under Colombian criminal law, kidnapping is classified into two types: ransom kidnapping and non-extortory kidnapping. Figure 6 shows the historical trajectory of these crimes from 1964 to 2025. The aggregate kidnapping volume has been driven largely by the dynamics of the extortory modality, a correlation especially pronounced during the 1996–2002 peak reflecting the instrumental use of kidnapping as a financial resource during armed conflict expansion. After 2010, a compositional shift emerges: while ransom kidnapping represented the majority of cases before 2000, its relative prevalence declined as overall figures fell, with the proportion of non-extortory kidnappings increasing toward a more diversified distribution.

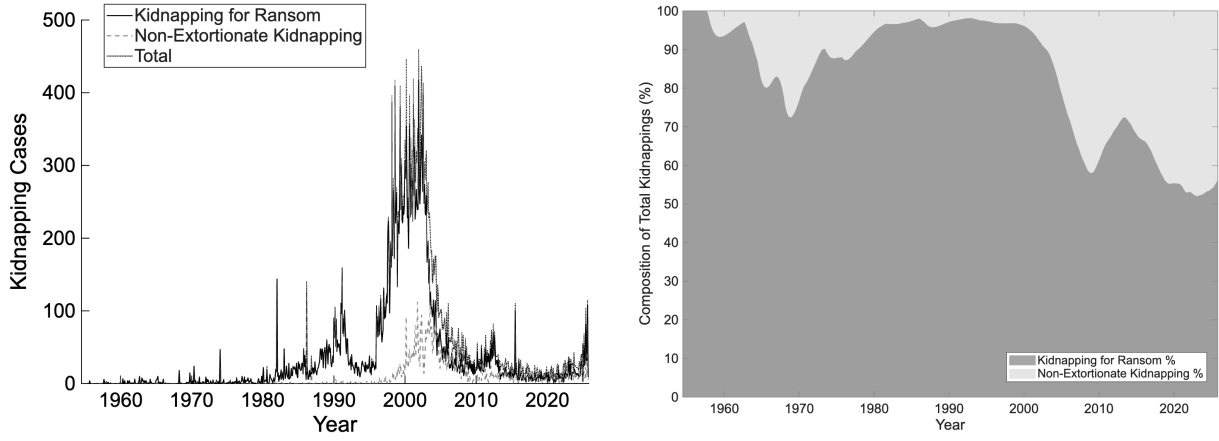


Figure 6. Evolution and composition of kidnapping modalities in Colombia. **(a)** Composition of kidnappings in Colombia by type, 1964–2025. **(b)** Percentage composition of kidnappings in Colombia, 1964–2025.

Notes: Panel (a) presents the annual number of kidnapping cases differentiating ransom, non-extortionate, and total; panel (b) shows the smoothed percentage composition using double exponential smoothing. *Source:* Authors' calculations using MATLAB R2025b and data from ? and the National Police of Colombia.

The use of a VEC model is methodologically appropriate for analyzing the joint dynamics of these variables, given their non-stationarity in levels under specifications including a constant. Augmented Dickey-Fuller (ADF) and KPSS tests confirm that the series exhibit non-stationary behavior consistent with an integrated process; analysis of first differences confirms stationarity, establishing that the variables are integrated of order one, $I(1)$. The Johansen cointegration test does not reject the null of at most one statistically significant cointegrating vector, providing robust evidence of a long-run equilibrium relationship. The VEC model thus allows simultaneous modeling of short-run adjustments and long-run error correction mechanisms.

For the VEC estimation, the analysis considers ransom kidnapping (K^E), non-extortionate kidnapping (K^{NE}), and the commercial bank lending rate (r). Data are monthly, covering July 1954 through December 2025 ($T = 858$ observations). The optimal specification corresponds to a VEC model with 36 lags, exhibiting the best performance in terms of AIC, absence of residual autocorrelation and heteroskedasticity, lower RMSE, and no evidence of ARCH/GARCH effects. The cointegrating vector β is reported in Equation 1:

$$\underset{(p<0.001)}{0.037} K_t^E - \underset{(p<0.001)}{0.072} K_t^{NE} - \underset{(p=0.022)}{0.049} r_t = \tilde{\varepsilon}_t, \quad \tilde{\varepsilon}_t \sim I(0) \quad (1)$$

Normalizing on K_t^E , the structural long-run equation is:

$$K_t^E = 1.93 K_t^{NE} + 1.31 r_t + \varepsilon_t \quad (2)$$

The normalized cointegrating equation identifies a long-run equilibrium in which non-extortionate kidnapping is a key determinant of ransom kidnapping. The positive and

statistically significant coefficient on K^{NE} indicates that, in the long run, a one-unit increase in this modality is associated with an approximately 1.93-unit increase in ransom kidnapping, *ceteris paribus*. Non-extortionary kidnapping reflects the accumulation of coercive and organizational capacity—persistent availability of organizational resources enabling territorial control, intimidation, and deprivation of liberty—reducing the marginal costs of shifting to the more profitable ransom modality. The two modalities should be understood not as substitutes but as complementary activities within a common criminal portfolio. The positive and statistically significant relationship between the lending rate and ransom kidnapping (coefficient 1.31) points to a mechanism consistent with liquidity constraints facing illegal organizations: periods of monetary tightening, characterized by higher interest rates and more limited access to financing, are associated with greater difficulties in financing operations through legal channels, making ransom kidnapping a means of generating immediate liquidity.

$$\Delta K_t^E = \underset{(p<0.01)}{-2.94} \tilde{\varepsilon}_{t-1} + \sum_{j=1}^{35} \Gamma_{1,j} \Delta \mathbf{X}_{t-j} + \varepsilon_{1,t} \quad (3)$$

$$\Delta K_t^{NE} = \underset{(p=0.17)}{0.36} \tilde{\varepsilon}_{t-1} + \sum_{j=1}^{35} \Gamma_{2,j} \Delta \mathbf{X}_{t-j} + \varepsilon_{2,t} \quad (4)$$

$$\Delta r_t = \underset{(p=0.10)}{-0.06} \tilde{\varepsilon}_{t-1} + \sum_{j=1}^{35} \Gamma_{3,j} \Delta \mathbf{X}_{t-j} + \varepsilon_{3,t} \quad (5)$$

where $\Gamma_{i,j(1 \times 3)}$ denotes row-parameter vectors capturing short-run dynamics, $\Delta \mathbf{X}_{t-j(3 \times 1)} = [\Delta K_{t-j}^E, \Delta K_{t-j}^{NE}, \Delta r_{t-j}]'$, and $\varepsilon_{i,t}$ are reduced-form white-noise residuals.

The adjustment coefficient for ransom kidnapping is $\alpha_{KE} = -2.94$ (significant at 1%), indicating that ransom kidnapping is the primary variable through which disequilibria are absorbed. The effective adjustment speed is $\alpha_{KE} \cdot \beta_{KE} \approx -0.11$, implying that approximately 11.0% of any deviation from the long-run equilibrium path is corrected each period through adjustments in ransom kidnapping—consistent with a gradual convergence process.

From the structural perspective, ransom kidnapping emerges not merely as an expression of violence but as an adaptive economic response within a broader criminal portfolio, linking fluctuations in conflict dynamics and macroeconomic conditions to sustained patterns of coercive rent extraction. Its persistence reflects dynamic incentives connecting criminal violence with organizational capacity and the macroeconomic environment (???)

To characterize the long-run stochastic structure, the permanent-transitory decomposition of ? is applied. The common stochastic trends (f_t) governing the long-run behavior are expressed as linear combinations of the observed variables capturing permanent sources of variation:

$$f_{1,t} = 0.12 K_t^E + 1.00 K_t^{NE} \quad (6)$$

$$f_{2,t} = -0.02 K_t^E + 1.00 r_t \quad (7)$$

The first common factor assigns unit weight to non-extortionary kidnapping and a small weight to ransom kidnapping, implying that long-run movements captured by this factor are dominated by the evolution of the non-extortionary modality—reflecting sustained criminal capacity as the persistent availability of organizational resources for territorial control and coercion. The second common factor primarily reflects permanent variation associated with macrofinancial conditions, responding nearly one-for-one to changes in the lending rate (Figure 7).

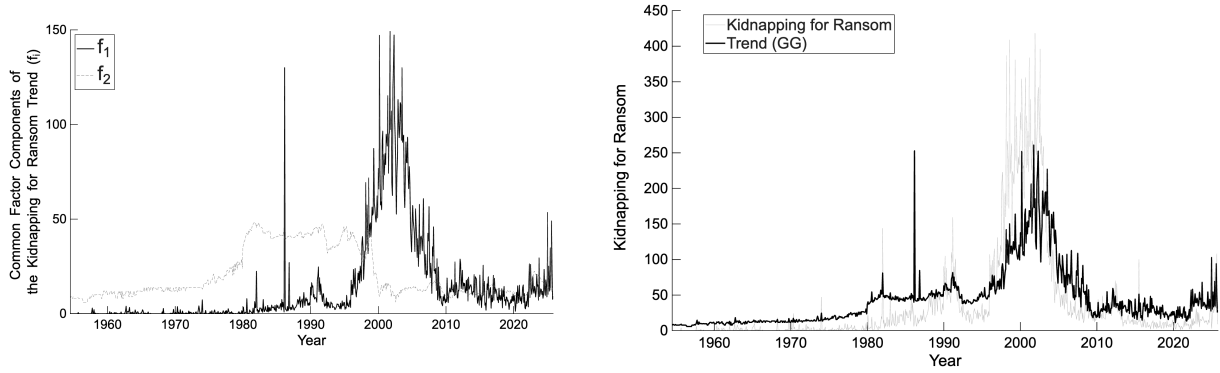


Figure 7. Kidnappings in Colombia: temporal and spatial patterns. (a) Common trend components of ransom kidnapping in Colombia, 1964–2025. (b) Ransom kidnapping in Colombia and the Gonzalo–Granger long-run trend, 1964–2025.

Notes: Panel (a) shows the common factor components (f_1 and f_2) extracted from the Gonzalo–Granger decomposition. Panel (b) shows the observed ransom kidnapping series together with the long-run trend from the Gonzalo–Granger methodology. *Source:* Authors’ calculations using MATLAB R2025b and ?.

The Gonzalo–Granger decomposition separates the observed series into permanent and transitory components.¹ The loading matrix representation is:

$$\begin{pmatrix} K_t^E \\ K_t^{NE} \\ r_t \end{pmatrix} = \begin{pmatrix} 1.60 & 1.09 \\ 0.81 & -0.13 \\ 0.04 & 1.02 \end{pmatrix} \begin{pmatrix} f_{1t} \\ f_{2t} \end{pmatrix} + \begin{pmatrix} \mu_{1,t} \\ \mu_{2,t} \\ \mu_{3,t} \end{pmatrix}, \quad \mu_t \sim I(0). \quad (8)$$

Ransom kidnapping is exposed significantly to both common trends, with loading coefficients of 1.60 on f_{1t} and 1.09 on f_{2t} . Exceeding unity, these coefficients indicate that

¹The decomposition is based on the orthogonal complements of the cointegrating vector β and adjustment vector α in a system with $k = 3$ variables and cointegration rank $\tilde{r} = 1$. The loading matrix is $A_{GG} = \beta_{\perp}(\alpha'_{\perp}\beta_{\perp})^{-1}$. The estimated adjustment vector $\alpha = (-2.9369, 0.3579, -0.0645)'$ indicates that ransom kidnapping is the main adjustment variable. The transitory component $\mu_t \sim I(0)$ captures deviations from common trends.

the evolution of ransom kidnapping amplifies the persistent shocks arising from both the accumulation of coercive capacity and long-run macroeconomic conditions (Figure 7b). In the long run, approximately 40% of ransom kidnapping variability is explained by permanent shocks originating in non-extortionary kidnapping, while the contribution of shocks associated with the interest rate decreases progressively (Figure 8).

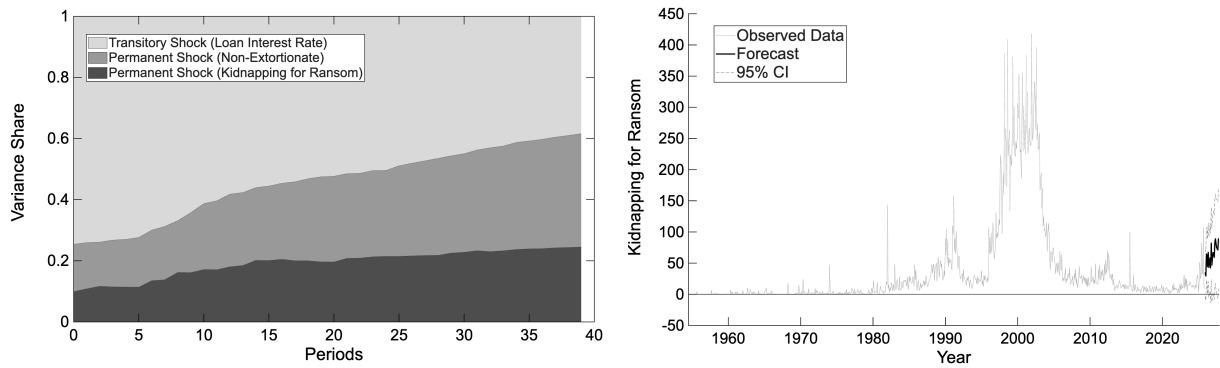


Figure 8. Dynamic analysis of ransom kidnapping: model forecasts and structural decomposition. (a) Forecast error variance decomposition under Blanchard-Quah restrictions. (b) Forecast of ransom kidnapping in Colombia using a VEC model.

Source: Authors' calculations using MATLAB R2025b and ?.

As a complement, the estimated Local Linear Trend (LLT) model offers a structurally and empirically coherent representation of the dynamics of ransom kidnapping by decomposing the observed series into a stochastic level trend, a persistent stochastic slope, and a transitory cyclical component, as formalized in Equation 9. Under this specification, ransom kidnapping is modeled as an $I(1)$ process, allowing the permanent movements of the series to be distinguished from temporary cyclical fluctuations. The estimation results indicate that all unobserved components are statistically relevant, as reflected in the significance of the variances associated with their innovations, with standard errors reported in parentheses.²

²The significance of the parameter is assessed using Wald tests based on the asymptotic normality of maximum-likelihood estimators, with inference conducted through Student- t statistics, as is standard in state-space models (??).

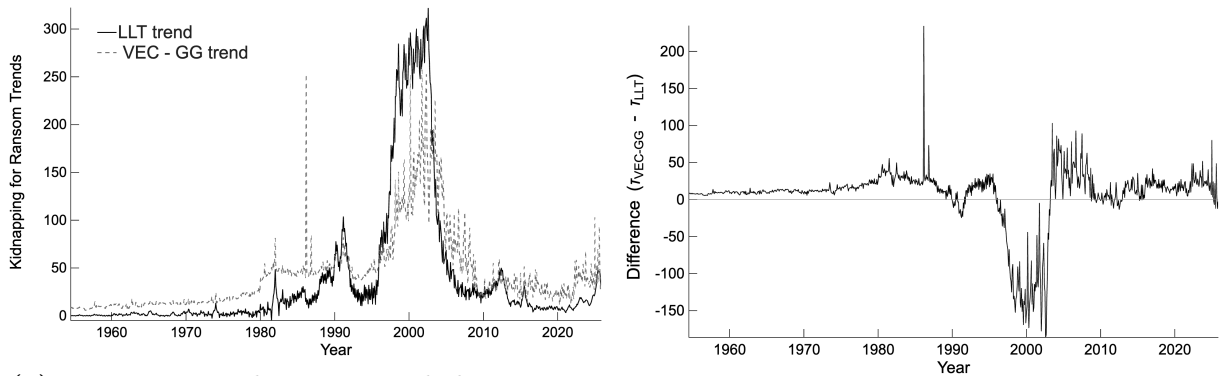
$$\begin{aligned}
\text{Observed series: } y_t &= \tau_t + c_t + \xi_t, & \xi_t &\sim \mathcal{N}(0, \sigma_\xi^2), & \hat{\sigma}_\xi &\approx 11.96 \text{ (0.024)} \\
& & & & & [p < 0.001] \\
\text{Stochastic trend: } \tau_t &= \tau_{t-1} + b_{t-1} + \eta_t, & \eta_t &\sim \mathcal{N}(0, \sigma_\eta^2), & \hat{\sigma}_\eta &\approx 7.13 \text{ (0.058)} \\
& & & & & [p < 0.001] \\
\text{Stochastic slope: } b_t &= \sum_{j=1}^8 \phi_j b_{t-j} + \zeta_t, & \zeta_t &\sim \mathcal{N}(0, \sigma_\zeta^2), & \hat{\sigma}_\zeta &\approx 3.51 \text{ (0.203)} \\
& & & & & [p < 0.001] \\
\text{Stochastic cycle: } c_t &= \sum_{k=1}^5 \rho_k c_{t-k} + \kappa_t, & \kappa_t &\sim \mathcal{N}(0, \sigma_\kappa^2), & \hat{\sigma}_\kappa &\approx 6.77 \text{ (0.083)} \\
& & & & & [p < 0.001]
\end{aligned} \tag{9}$$

In particular, the variance of the innovation in the stochastic level is estimated at $\hat{\sigma}_\eta \approx 7.12$ with $p < 0.001$, confirming that permanent shocks exert a substantial influence on the long-run level of ransom kidnapping. Likewise, the variance of the slope innovation, $\hat{\sigma}_\zeta \approx 3.51$ with $p < 0.001$, together with an autoregressive structure of order eight, provides strong evidence of persistent changes in the growth rate of the series, consistent with gradual adjustments in underlying criminal capacity rather than abrupt regime changes. The cyclical component is also statistically significant, with estimated variance $\hat{\sigma}_\kappa \approx 6.77$ and $p < 0.001$, capturing medium-run fluctuations around the long-run trend without distorting its path, while the irregular component exhibits considerable variance, $\hat{\sigma}_\xi \approx 11.96$ with $p < 0.001$, indicating the presence of short-run noise and idiosyncratic shocks. Taken together, these results show that ransom kidnapping is characterized by a dominant stochastic trend with persistent slope dynamics, complemented by mean-reverting cyclical movements, a structure fully consistent with the evidence obtained from the VEC model and the Gonzalo-Granger decomposition.

Joint examination of the estimated trends, namely the one obtained from the Gonzalo-Granger decomposition based on the VEC model and the one derived from the Local Linear Trend (LLT) model, reveals sustained coherence over the sample period despite being constructed from conceptually distinct econometric frameworks. From an empirical perspective, both trends consistently reproduce the main long-run movements in ransom kidnapping, including the sustained increase before the late 1990s, the peak associated with the intensification of the armed conflict, and the subsequent stabilization phase. The differences between these two trend estimates generate a stationary residual, $I(0)$. This result indicates that the discrepancy between the two approaches is transitory and mean-zero in nature, implying the absence of systematic long-run divergence. From a theoretical standpoint, this convergence is consistent with the corollary on the equivalence of stochastic trends (Corollary 2.1), according to which different valid decompositions of an integrated series identify the same underlying permanent component, regardless of the filtering method used.

The trend difference $d_t = \hat{\tau}_t^{VEC} - \hat{\mu}_t^{LLT}$ between the Gonzalo-Granger permanent component and the LLT trend is stationary: the ADF test rejects the unit root ($p = 0.001$) and the KPSS test fails to reject stationarity at 12 lags ($p = 0.071$). The following corollary formalizes this equivalence.

Corollary 2.1 (Equivalence of stochastic trends). The permanent component $\hat{\tau}_t^{VEC}$ extracted from the VEC-Gonzalo-Granger decomposition and the LLT trend $\hat{\mu}_t^{LLT}$ track the same underlying stochastic process: $d_t = \hat{\tau}_t^{VEC} - \hat{\mu}_t^{LLT} \sim I(0)$. Consequently, both representations provide an equivalent macroeconomic backbone for the structural prior calibration of the mechanism.



(a) Long-run trends in ransom kidnapping in Colombia: LLT and VEC-GG estimates, 1964–2025.

(b) Difference between the VEC-GG and LLT trends of ransom kidnapping, 1964–2025.

Figure 9. Comparative analysis of long-run trend estimates of ransom kidnapping.

Notes: Panel (a) compares the estimated long-run trends in ransom kidnapping obtained using a Local Linear Trend (LLT) model and the VEC-GG approach. The LLT specification assumes a stochastic trend with an ARMA(8,0) process for the slope component and an ARMA(5,0) process for the cyclical component. Panel (b) shows the difference between the VEC-GG trend and the LLT trend.

Source: Authors' calculations using MATLAB R2025b, based on data from ? and the Colombian National Police.

The cross-validation of the long-run trend in ransom kidnapping rests on the structural equivalence between the univariate stochastic component estimated through the LLT model and the common trend derived from the VEC framework. In particular, the projection of the common factors indicates that the permanent path of ransom kidnapping can be expressed as a linear combination of two latent forces shared with the rest of the system, $\tau_t^{GG} = 1.60 f_{1,t} + 1.09 f_{2,t}$. This representation implies that its long-run evolution is not idiosyncratic, but rather reflects exposure to structural shocks associated with the accumulation of coercive capacity and persistent macrofinancial conditions. This structure accounts for the magnitude of the level shifts captured by the LLT model and suggests that the univariate trend does not follow an isolated random walk, but aggregates the influence of the common determinants identified in the multivariate system. The stability of the difference between both trends, confirmed by the ADF and KPSS unit-root tests,

provides empirical evidence consistent with the *Stochastic Trend Equivalence Corollary* and reinforces the validity of the underlying cointegrating relationship. In this sense, the convergence between the trend estimated agnostically by the LLT model and the trend constructed from interactions with non-ransom kidnapping and the real interest rate supports the interpretation that these variables operate as true structural determinants of the long-run dynamics of ransom kidnapping, rather than as spurious or purely transitory relationships.

2.3 Microstructure and Competing Risks in Colombia

2.4 Data Management and Sample Construction

The data used in this section come from anonymized individual-level microdata on ransom kidnappings reported by the National Center for Historical Memory [?]. This institution documents kidnapping through the Information System on Events of Violence in the Colombian Armed Conflict (SIEVCAC), which integrates and validates multiple institutional sources, such as the Attorney General's Office and the National Police, as well as social sources, including NGOs, human-rights archives, and national and local press. Rather than generating primary data, the methodology compiles, harmonizes, and systematically reviews records on a case-by-case basis in order to eliminate duplicates, individualize victims, and structure each event by perpetrator, victim, date, location, and modality. The records are subsequently coded, subjected to rigorous quality-control procedures, and statistically validated before publication, producing a unified historical repository that prioritizes documentary rigor and the reconstruction of conflict-related patterns.

Starting from an initial universe of $N = 39,369$ observations, the data-cleaning process summarized in Table 1 produced a final analytical sample of $n = 1,125$ cases, equivalent to 2.9% of the total. The sample is restricted to events with confirmed ransom-related typology, valid outcome information, complete calendar records, and the key explanatory variables required for the joint estimation of the Multinomial Logit (MNL) model and the duration models. Although these restrictions imply a cumulative reduction of 97.1% of the original universe, reflecting both the structural limitations inherent in data collection in armed-conflict settings and the high prevalence of unreported or undocumented outcomes in peripheral regions, the statistical validation exercises reported in Table 2 indicate that this reduction does not compromise econometric consistency or the internal validity of the results.

In particular, to ensure the consistency of the MNL estimators, a three-stage robustness-validation strategy was implemented, with the results summarized in Table 2. First, following the specification test proposed by [?] and its robust generalization based on seemingly unrelated estimation (SUEST) developed by [?], there is no statistical evi-

Table 1. Sample attrition and data availability

Filtering stage	Observations	Share of total	Cumulative loss
Initial universe (CNMH)	39,369	100.0%	—
Confirmed extortive typology	11,270	28.6%	71.4%
Observable valid outcome (Y)	1,133	2.9%	97.1%
Complete covariates (pre-estimation)	1,125	2.9%	97.1%
Final estimation sample	1,125	2.9%	97.1%

Note: The sharpest reduction occurs because of the “No Data” (ND) classification in the outcome variable, reflecting the incidental truncation inherent in conflict data from remote areas. The final reduction from 1,133 to 1,125 observations, 0.7% of the valid sample, is due to random missing values in the geographic-location variable (*Zona Geografica*) during maximum-likelihood estimation, without introducing systematic bias.

Source: Authors’ calculations using Stata 19.

dence to reject the null hypothesis of Independence of Irrelevant Alternatives (IIA), ($\chi^2(1) = 0.24$, $p = 0.625$), confirming the structural validity of the modeled discrete-choice framework. Second, estimator stability is assessed through a sensitivity analysis that compares the baseline specification with alternative model formulations, showing that the key variables associated with armed actors and territorial conditions consistently retain their statistical significance and sign. This stability indicates that the economic interpretation of the model does not depend on a specific specification, in line with standard criteria for correct specification and robustness in the models discussed by ?. Finally, a difference-in-proportions test confirms statistically significant structural heterogeneity in lethality across armed actors ($z = -5.71$, $p < 0.001$), while the attrition analysis identifies non-random information-loss mechanisms (MNAR), especially in peripheral areas. However, rather than invalidating the empirical exercise, this reporting bias is addressed transparently and consistently with the literature on missing data in complex settings, as documented by ?, thereby strengthening the credibility of the main results.³

2.5 Strategic Determinants and Temporal Dynamics of Kidnapping

Based on a final sample of 1,125 observations, the MNL model is estimated to analyze the determinants of kidnapping outcomes. This specification is particularly suitable for the

³Although the attrition analysis indicates that the probability of observing the outcome is not random, this does not compromise the consistency of the main estimators. Reporting bias operates on the *observability* of the outcome rather than on the underlying decision process that determines the kidnapping outcome. In particular, the factors associated with missing outcomes affect whether an event is recorded, but they do not systematically alter the strategic incentives faced by the actors involved. Therefore, the stability of the core coefficients across alternative specifications suggests that the estimated relationships reflect behavioral structural mechanisms rather than artifacts induced by the data-cleaning process, consistent with standard arguments in the literature on missing data in complex settings.

Table 2. Statistical validation of the econometric models

Statistical test	Observed result	Economic interpretation	Verdict
Independence of Irrelevant Alternatives (IIA) test	$\chi^2(1) = 0.24$ $p = 0.625$	Independence of alternatives: removing the “Death” outcome does not alter the relative probabilities of “Rescue” or “Payment”, supporting the validity of the multinomial logit specification.	Passed ($p > 0.05$)
Difference-in-proportions test (lethality)	$z = -5.71$ $p < 0.001$	Structural heterogeneity: common criminal groups show a significantly higher lethality rate (7.1%) than the FARC (0.2%), revealing differences in the costs associated with criminal failure.	Significant ($p < 0.01$)
Attrition / missingness logit test	significant odds ratios ($p < 0.05$)	Reporting bias: the probability that kidnapping-outcome information is missing is not random and increases in peripheral regions and in contexts with lower state capacity.	Limitation (reported)

Note: The IIA test evaluates the assumption of independence of irrelevant alternatives by comparing the coefficients of the full model with those of a restricted model using the SUEST approach. The lethality test corresponds to a difference in proportions across armed groups. The attrition test uses a logit model to estimate the probability that the kidnapping outcome is not observed.

Source: Authors’ calculations using Stata 19.

multicategorical and unordered nature of the dependent variable, which captures mutually exclusive outcomes such as Payment, Death, Rescue, and Escape or Release. Unlike binary or linear regression models, the MNL framework is explicitly designed for risk environments in which perpetrators and the State interact strategically. Its relevance in this context lies in its ability to quantify how a vector of explanatory variables changes the probability that a negotiation departs from the baseline outcome of Payment toward failed or violent resolutions. By decomposing the net effects of organizational, territorial, and institutional determinants, the model provides a granular view of the incentive structure governing kidnapping behavior, allowing an empirical assessment of whether ransom kidnapping operates according to a rational market logic or is shaped by criminal frictions.

The estimated econometric model is presented below. The estimation of the parameters specified in Equation 10 is carried out by relaxing the assumption that the error terms are independent and identically distributed, in light of the spatial nature of the armed conflict. Under this premise, kidnapping events occurring within the same jurisdiction are assumed to share unobserved institutional and operational determinants, such as local law-enforcement capacity, topographic conditions, or the historical presence of armed groups, which may induce correlation in the error terms. Consequently, the variance matrix is adjusted using robust standard errors clustered at the municipal level, covering a total of 373 territorial units, in order to correct for intraclass correlation and ensure statistically robust inference.⁴

To analyze the determinants of kidnapping outcomes, an MNL model is estimated, where $Y_{i,m}$ denotes the outcome of kidnapping i in municipality m . The dependent variable is numerically coded and takes one of four mutually exclusive outcomes indexed by $j \in \{0, 1, 2, 3\}$. Specifically, $j = 0$ denotes escape or release without payment, $j = 1$ corresponds to ransom payment, $j = 2$ indicates the death of the victim, and $j = 3$ represents GAULA rescue. This categorical structure ensures that each kidnapping event is uniquely classified into a single outcome. Taking payment, $j = 1$, as the base category, which represents successful rent extraction by the criminal, the model estimates the probabilities of observing each alternative outcome $j \in \{0, 2, 3\}$ relative to payment as a function of the relevant covariates and municipal characteristics.

⁴Clustering standard errors at the municipal level accounts for the fact that kidnapping acts within the same municipality may share unobserved characteristics, such as local institutional capacity, geographic constraints, or the presence of armed groups. These factors may induce correlation in the error terms across observations within the same group, violating the assumption of independent errors. For example, a temporary weakening of law-enforcement capacity in a municipality may increase the probability of ransom payments in multiple cases. Ignoring this intramunicipal correlation may lead to underestimated standard errors and overstated statistical significance. Clustering addresses this dependence and supports consistent inference.

$$\ln \left(\frac{\Pr(Y_{i,m} = j \mid \mathbf{X}_{i,m})}{\Pr(Y_{i,m} = 1 \mid \mathbf{X}_{i,m})} \right) = \alpha_j + \underbrace{\mathbf{G}'_i \boldsymbol{\beta}_{1,j}}_{\text{Perpetrator}} + \underbrace{\mathbf{V}'_i \boldsymbol{\beta}_{2,j}}_{\text{Victim}} + \underbrace{\mathbf{C}'_{i,m} \boldsymbol{\beta}_{3,j}}_{\text{Context}} + \underbrace{\mathbf{T}'_i \boldsymbol{\beta}_{4,j}}_{\text{Tactics}}, \quad j \in \{0, 2, 3\}. \quad (10)$$

Where α_j denotes the alternative-specific intercept for outcome j , and $j = 1$ is the base category. The full vector of covariates is defined as the concatenation $\mathbf{X}_{i,m} = [\mathbf{G}'_i, \mathbf{V}'_i, \mathbf{C}'_{i,m}, \mathbf{T}'_i]' \in \mathbb{R}^{26}$. The model is estimated by maximum likelihood under the standard MNL assumption that the unobserved components of utility are independently and identically distributed with a Type-I extreme-value distribution. Standard errors are clustered at the municipal level to account for possible intramunicipal correlation in unobserved determinants.

- $\mathbf{G}_i \in \mathbb{R}^3$ (*Perpetrator*): A column vector that identifies the armed organization. It includes specific variables for the ELN, paramilitary groups, and common crime. The estimates are relative to the FARC baseline.
- $\mathbf{V}_i \in \mathbb{R}^6$ (*Victim*): Captures victim characteristics. The vector comprises four dummy variables for socioeconomic profiles, including middle class, vulnerable population, public-sector employees, and unknown profiles relative to the high-income baseline. It also includes indicators for female victims and missing gender information relative to the male baseline.
- $\mathbf{C}_{i,m} \in \mathbb{R}^8$ (*Context*): Controls for the geostrategic environment. It includes four regional indicators for Andean, Caribbean, Pacific, and Eastern territories relative to metropolitan areas. In addition, it incorporates dummies for the 1998–2002, 2003–2010, and post-2011 periods relative to the pre-1998 era, together with the continuous measure of the log of total victims.
- $\mathbf{T}_i \in \mathbb{R}^9$ (*Tactics*): Accounts for operational heterogeneity. This vector contains four indicator variables for capture methods, including roadblocks, armed incursions, deception tactics, and unknown methods relative to direct attacks. It also includes an indicator for collective kidnapping structure, three indicator variables for short, long, and missing duration categories relative to express captivity, and a variable for GAULA intervention.

The coefficient vectors $\boldsymbol{\beta}_{k,j}$ associated with each structural component $k \in \{1, \dots, 4\}$ are estimated by maximum likelihood. To ensure valid inference, standard errors are clustered at the municipal level (m), accounting for unobserved spatial correlation in institutional capacity and local conflict dynamics.

The main estimation results are reported in terms of Relative Risk Ratios (RRR) in Table 3, while the full set of estimated coefficients is presented in Appendix B. In the MNL

framework, estimated coefficients represent changes in the log odds of observing outcome j relative to the base category, while the RRR corresponds to exponentiated coefficients and measures multiplicative changes in relative probabilities, offering a more intuitive interpretation. Standard errors are clustered at the municipal level, accounting for 373 distinct groups, to address possible intramunicipal correlation and heteroskedasticity. The model is jointly statistically significant, as indicated by a Wald statistic of $\chi^2(78) = 5,250.34$ with $p < 0.0001$, rejecting the null hypothesis that the explanatory variables are jointly irrelevant. The Pseudo R^2 of 0.1847 indicates a reasonable goodness of fit for a discrete-choice model in a conflict setting.

For a correct interpretation of the findings, it is crucial to understand that the coefficients are presented as RRR. These do not indicate absolute probabilities, but rather quantify how many times more likely a specific outcome is to occur relative to the normative base category of the model: Payment ($Y = 1$). Likewise, categorical variables are interpreted relative to a reference group. For example, when analyzing the “Perpetrator” group, the ELN or common-crime coefficients show how relative risk changes compared with the behavior of the reference group, the FARC. An RRR greater than 1 indicates that the event is more likely than payment compared with the FARC, whereas an RRR below 1 suggests a stronger tendency toward effective ransom extraction.

In this context, during the contemporary period (Peace/Current), the relative probability of a “failed kidnapping” (Escape or Release without payment) increases by a factor of 11.28 relative to the payment base, compared with the pre-1998 era. This effect is statistically significant ($p < 0.01$) and provides empirical support for the deterrence framework proposed by ?, suggesting that the consolidation of specialized GAULA units has substantially weakened the financial incentives underlying the crime. In practical terms, this implies that under current institutional conditions, kidnapping is much more likely to end in forced release or abandonment, constituting a clear financial loss for the perpetrator, than in successful ransom extraction. Consequently, the evidence reflects a structural shift from income-generating criminal activity toward a high-risk environment characterized by a markedly greater probability of failure.

It is worth emphasizing that market equilibrium is altered by adverse-selection mechanisms inherent in certain victim profiles. Target-population heterogeneity introduces non-trivial distortions; in particular, the selection of public-sector officials systematically degrades the probability of success, altering the criminal’s risk calculation. The model indicates that for this group, the relative probability of “Escape or Release” is 3.92 times higher than that of a successful payment ($p < 0.01$) compared with upper-class victims. This finding corroborates that political or state-related kidnapping operates under incentives distinct from purely economic ones: the State, by maintaining a strict policy of non-monetary negotiation and imposing a high political and media cost on the captor, makes these hostages financially unviable. This pushes outcomes toward release or es-

cape due to institutional pressure, rather than toward the monetary transaction typical of extortive kidnapping.

However, despite these improvements in state capacity and the distortion of the market, the model identifies a critical limit in rescue technology governed by information asymmetry. The relative probability of rescue decreases monotonically as captivity duration extends, with the RRR falling from 0.18 in the short run, less than one month, to 0.08 in the long run ($p < 0.001$). This decay indicates that state efficiency is strictly time-dependent; a victim is more than twice as likely to be successfully rescued during the first weeks of the event as in later months. This evidence corroborates the hypothesis that rescue operations depend on “hot intelligence” during the initial urban or transit phases. By contrast, once the victim is absorbed into the geographic “rear guard” controlled by the perpetrator, information asymmetry becomes insurmountable, making military exfiltration increasingly unlikely and shifting the bargaining advantage to the captors.

Within this framework, public policy must prioritize neutralizing factors that increase lethality and criminal resilience, focusing specifically on the perpetrator’s logistical bottlenecks. The analysis of lethality determinants highlights a critical inverse relationship between organizational complexity and violence: common crime exhibits an RRR for death of 63.63 ($p < 0.01$) relative to the FARC baseline. This finding supports the hypothesis that lethality is a function of logistical constraints. The lack of infrastructure for long-term retention pushes these actors into a compressed time horizon where the absence of immediate payment precipitates the elimination of the victim to mitigate detection risks. For mechanism design, the State must increase the cost of immediate retention, forcing the criminal to release the victim early before the execution incentive is triggered.

The spatial analysis reveals a consistent center-periphery pattern. Using the Metropolis as the reference category, peripheral regions such as the Andean and Caribbean regions present RRR values consistently below one. This indicates that failed kidnapping outcomes are considerably less likely in peripheral regions than in major urban centers. The Caribbean region provides a clear example of the state-capacity hypothesis. While deterrence and GAULA response infrastructure are effective in the metropolis, forcing operational failures, the periphery offers an environment of lower institutional friction favorable to rent extraction. In addition, the findings indicate that the time factor undermines the State’s response capacity. While express kidnapping entails high exposure to immediate operational friction, once captivity exceeds one month the criminal successfully consolidates control over the victim, reflected in a coefficient of 0.079. In this context, prolonged duration acts as a stabilization mechanism that minimizes operational risk and substantially reduces the probability of successful rescue.

Finally, regarding institutional response, GAULA intervention exhibits an RRR of 1.39 for the rescue outcome, representing a strategic factor that, while showing a p -value

of 0.34, underscores a critical area for policy improvement. Although this coefficient does not reach conventional levels of statistical significance, its magnitude suggests positive pressure on tactical outcomes by increasing the risk of capture during the initial stages of kidnapping. This lack of statistical significance likely reflects the high variability of operational success and the “cold intelligence” barrier inherent in later stages of captivity. Consequently, strengthening GAULA intervention mechanisms is essential to transform this latent tactical pressure into a robust deterrent effect, effectively shifting the risk-benefit ratio against the perpetrator from the very beginning of the criminal act.

Table 3. *Multinomial logistic regression results for kidnapping outcomes*

Predictors	Escape / Release (RRR)	Death (RRR)	Rescue (RRR)
<i>Perpetrator group (Ref.: FARC)</i>			
ELN	0.800 (0.531)	4.500 (0.315)	0.840 (0.459)
Paramilitaries	0.932 (0.876)	20.306* (0.017)	0.680 (0.295)
Common crime	0.493 (0.135)	63.629** (0.002)	0.494* (0.045)
<i>Victim profile (Ref.: upper class)</i>			
Middle class	1.868 (0.119)	0.515 (0.696)	1.277 (0.350)
Vulnerable population	0.624 (0.291)	7.056** (0.008)	0.800 (0.420)
Public sector	3.917** (0.004)	0.000*** (0.000)	0.654 (0.323)
<i>Geographic zone (Ref.: Metropolis)</i>			
Andean region	0.637 (0.313)	0.560 (0.591)	0.800 (0.587)
Caribbean	0.496 (0.143)	0.000*** (0.000)	0.426 (0.061)
Eastern Plains/Jungle	0.727 (0.561)	0.624 (0.688)	0.740 (0.501)
<i>Modus operandi (Ref.: direct attack)</i>			
Roadblock (illegal checkpoint)	2.287 (0.111)	0.000*** (0.000)	1.513 (0.348)
Deception	0.878 (0.814)	0.537 (0.632)	1.232 (0.645)

Continued on next page

Table 3 – continued from previous page

Predictors	Escape / Release (RRR)	Death (RRR)	Rescue (RRR)
<i>Kidnapping characteristics</i>			
Group structure (vs. individual)	0.760 (0.498)	11.744** (0.001)	0.972 (0.921)
GAULA intervention (vs. none)	0.942 (0.873)	0.536 (0.514)	1.390 (0.341)
Ln(total victims) (<i>continuous</i>)	0.764 (0.297)	0.084*** (0.000)	1.116 (0.534)
<i>Captivity duration (Ref.: express ≤ 2 days)</i>			
Short (1 month)	0.322 (0.065)	0.580 (0.562)	0.183*** (0.000)
Long (> 1 month)	0.235* (0.029)	0.477 (0.517)	0.079*** (0.000)
<i>Historical period (Ref.: pre-1998 rise)</i>			
Crisis/Caguan (1998–2002)	1.506 (0.360)	1.230 (0.834)	1.028 (0.935)
Democratic Security (2003–2010)	1.235 (0.658)	1.405 (0.732)	1.554 (0.249)
Peace process/Current (2011+)	11.283*** (0.000)	10.008* (0.017)	3.849** (0.006)
<i>Model statistics</i>			
Observations (N)		1,125	
Pseudo R^2		0.18	
Wald $\chi^2(78)$		5250.34	
Prob χ^2		0.0000	

Note. Coefficients are reported as Relative Risk Ratios (RRR). The base outcome category for the dependent variable is **Payment** ($Y = 1$). Robust p-values clustered by municipality are shown in parentheses. * $p < .05$. ** $p < .01$. *** $p < .001$. *Source:* Authors' elaboration using Stata 19.

The preceding analysis is further reinforced by the estimated marginal probabilities from the MNL.⁵ Specifically, when the probability of state rescue is examined as a function of captivity duration, a clear declining pattern emerges that is consistent with the “critical window” hypothesis. As shown in Figure 10a, the estimated rescue probability is approximately 0.61 for express kidnappings lasting less than two days, decreases to 0.31

⁵The figures report average predicted probabilities obtained from the MNL model. For each observation, the model calculates the predicted probability of a given outcome, and the reported value corresponds to the sample average of those probabilities within each category. These are unconditional mean probabilities evaluated over the observed distribution of the explanatory variables, not cumulative probabilities or marginal effects at the mean. Standard errors are calculated using the delta method and clustered at the municipal level.

for short-term events of up to one month, and declines further to 0.18 for kidnappings extending beyond one month. This trajectory indicates that the effectiveness of state intervention is considerably higher during the initial phase of the event, when information remains recent and operational capacity is more agile. As captivity persists, the probability of rescue decreases substantially, suggesting that time functions as a mechanism through which perpetrators consolidate control and gradually shift the strategic balance in their favor.

Another result of special interest is that the victim profile emerges as a structural determinant of the probability that a kidnapping concludes in escape or release without ransom payment. The average predicted probabilities derived from the MNL model show a clear upward gradient, increasing from vulnerable profiles (0.05) to public-sector victims (0.26), with high-income victims (0.08) and middle-class victims (0.12) located between them, as shown in Figure 10b. This pattern indicates that as institutional visibility and the potential political costs associated with kidnapping increase, the probability that a ransom is paid declines. From a strategic standpoint, the kidnapping of public officials appears to entail greater operational risks, an intensified state response, and reduced incentives to complete the negotiation successfully, thereby lowering the expected profitability for the perpetrator.

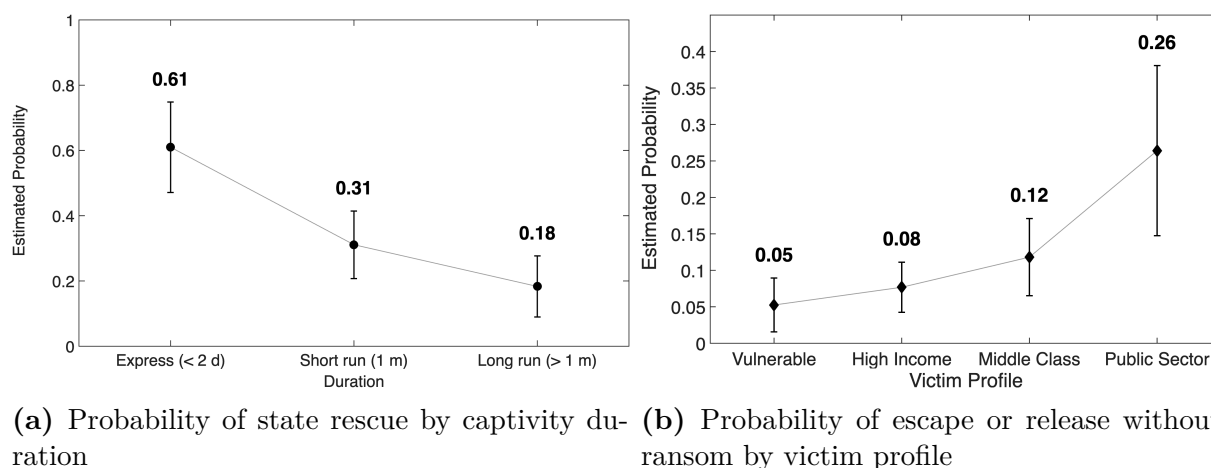


Figure 10. Predicted probabilities of alternative kidnapping outcomes.

Notes: Panel (a) shows the average predicted probability that a kidnapping results in state rescue, conditional on duration category. Panel (b) shows the average predicted probability of escape or release without ransom payment across victim profiles.

Source: Authors' calculations using Stata 19, based on data from ?.

For ransom-kidnapping events, metropolitan areas present the highest average predicted probability that the event concludes in rescue, with a value of 0.465, followed by the Andean region at 0.447, while the Caribbean region shows the lowest estimated probability at 0.343. The Pacific region records a probability of 0.414 and the Eastern Jungle region 0.424, both occupying intermediate positions, as shown in Figure 11a. All estimates are statistically significant at the 1% level, with confidence intervals that remain

well above zero, indicating that the territorial environment constitutes a relevant determinant of the outcome in the framework under analysis. Substantively, these findings suggest that geographic structure and the degree of institutional presence systematically influence the probability of rescue, reflecting differences in state capacity, accessibility, and operational conditions that shape the strategic dynamics of kidnapping.

A relevant scenario for the analysis concerns the average predicted probability of death in captivity, which shows an increasing gradient across armed actors. The FARC exhibit the lowest estimated probability of lethality (0.002), followed by the ELN (0.007); paramilitary groups reach 0.026, while common crime or other actors register the highest probability at 0.071, as shown in Figure 11b. This ordering indicates that execution risk is not homogeneous, but varies according to the organizational structure and strategic incentives of each group. In particular, the greater lethality associated with common criminal groups is consistent with lower levels of territorial control, greater exposure to operational failures, and a greater propensity for violent responses under pressure. By contrast, insurgent organizations with hierarchical structures show lower probabilities of death, which may reflect more defined strategic, reputational, and internal-disciplinary constraints. Taken together, the evidence supports the theoretical distinction between structured armed organizations and decentralized actors, showing that the cost of logistical failure in kidnapping varies systematically according to the organizational profile of the perpetrator.

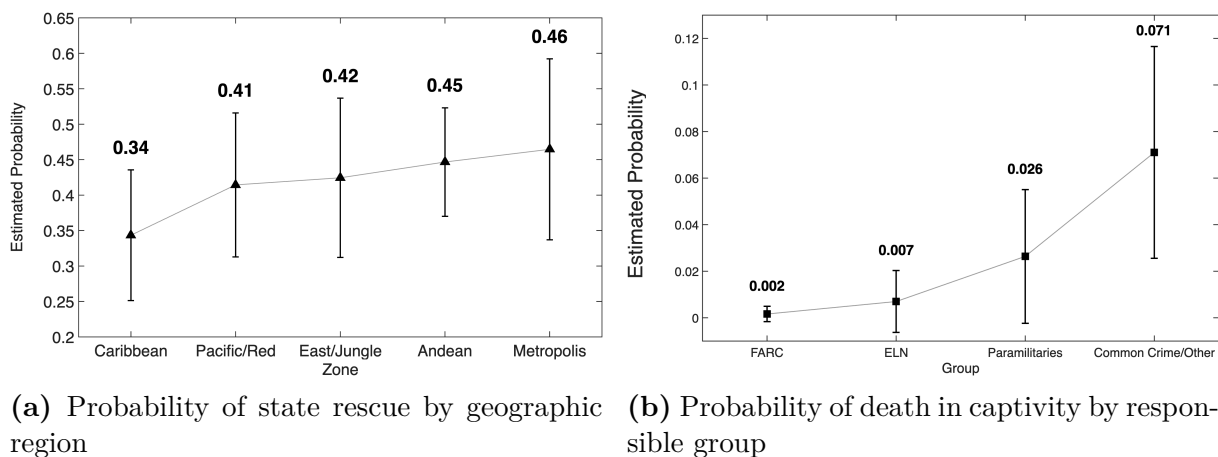


Figure 11. Predicted probabilities of rescue and death.

Notes: Panel (a) shows the average predicted probabilities of state rescue. Vertical bars represent 95% confidence intervals clustered at the municipal level. Panel (b) represents the average predicted probabilities of death in captivity.

Source: Authors' calculations using Stata 19, based on data from ?.

Figure 12a presents the average probabilities of state rescue, differentiated by captivity duration and the presence of GAULA intervention. In the absence of intervention, the probability of rescue reaches 0.64 in express kidnappings, declines to 0.28 in short-duration events, and falls further to 0.14 in prolonged kidnappings. When GAULA intervention

is present, these probabilities increase systematically to 0.72, 0.35, and 0.19, respectively. This pattern reflects two complementary conclusions. First, the critical-window hypothesis is confirmed, since the probability of rescue declines as captivity duration increases. Second, specialized intervention consistently increases the probability of rescue across all temporal segments, shifting the entire probability structure upward. The difference is especially visible during the initial stages, when recent information and operational capacity amplify the effects of state action. From a strategic perspective, Figure 12a suggests that GAULA intervention not only increases the absolute probability of state success, but also alters the equilibrium of the game by narrowing the time window that allows perpetrators to consolidate control over the event.

The analysis of kidnapping duration requires the explicit incorporation of the temporal dimension, recognizing that the strategic dynamics of negotiation are evolutionary rather than static processes. Within this framework, survival curves constitute an appropriate methodological tool for addressing temporality, since they describe the probability of remaining in a given state over time, as formally articulated by ?. In the context of ransom kidnapping, these functions estimate the conditional probability that a victim remains in captivity at each point in time without ransom payment having occurred. Tracing this trajectory makes it possible to identify structural heterogeneity in captivity persistence across different armed actors, thereby providing a dynamic perspective on outcomes that extends beyond the limitations of traditional static models.

Formally, the analysis is based on the proportional hazards model proposed by ?. This framework models the hazard rate $h(t | \mathbf{X})$, defined as the instantaneous probability that the event occurs at time t conditional on survival up to that moment. The rate is expressed as the product of two specific components: an unspecified nonparametric baseline hazard $h_0(t)$, which captures the underlying duration dependence common to all subjects, and a parametric exponential function. As shown in Equation 11, the latter component incorporates the vector of explanatory variables $\mathbf{X}$ and its associated coefficients $\tilde{\beta}$:

$$h(t | \mathbf{X}) = h_0(t) \exp(\mathbf{X}'\tilde{\beta}) \quad (11)$$

Integrating the hazard function over the interval $[0, t]$ yields the cumulative hazard function $H(t | \mathbf{X})$, which represents the total accumulated risk up to time t . Under the proportional-hazards assumption, Equation 12 shows that this function can be expressed as the product of the baseline cumulative hazard $H_0(t)$ and the individual risk score $\exp(\mathbf{X}'\tilde{\beta})$:

$$H(t | \mathbf{X}) = \int_0^t h(u | \mathbf{X}) du = \exp(\mathbf{X}'\tilde{\beta}) H_0(t), \quad \text{where } H_0(t) = \int_0^t h_0(u) du \quad (12)$$

Finally, the survival function $S(t | \mathbf{X})$ denotes the probability that the event has

not occurred by time t . As derived in Equation 13, this function is expressed as the exponential of the negative cumulative hazard, implying that the specific values of the explanatory variables proportionally scale the baseline survival probability:

$$S(t | \mathbf{X}) = \exp(-H(t | \mathbf{X})) = \exp\left(-\exp(\mathbf{X}'\tilde{\beta}) H_0(t)\right) \quad (13)$$

The coefficients $\tilde{\beta}$ are estimated using maximum partial likelihood, following ?. This approach allows consistent estimation of covariate effects without requiring specification of the functional form of the baseline hazard $h_0(t)$, thereby making the model semiparametric. Since duration is measured in days, tied event times may arise, meaning that multiple payments can be observed at exactly the same duration; in such cases, the approximation proposed by ? is used to adjust the partial likelihood without altering the structure of the model. In addition, standard errors are computed using the robust sandwich estimator, which accounts for potential heteroskedasticity and ensures valid statistical inference without imposing additional distributional assumptions.

In this sense, it can be observed that during the first 20 days, the probability of remaining in captivity exceeds 0.85 for both the FARC and the ELN, while for common criminal groups it declines more rapidly toward values close to 0.70, as illustrated in Figure 12b. Around day 60, the probability of remaining in captivity is approximately 0.56 for the FARC, 0.50 for the ELN, around 0.37 for paramilitary groups, and only 0.10 for common criminals. By day 180, these differences persist, with probabilities close to 0.29 for the FARC, 0.23 for the ELN, 0.12 for paramilitaries, and practically zero for common criminal groups. These patterns indicate that insurgent organizations tend to maintain captivity for longer periods before payment occurs, while decentralized criminal groups resolve events more quickly. From a strategic perspective, the evidence suggests that kidnapping duration and the timing of ransom payment are closely linked to organizational capacity, territorial control, and operational structure, reflecting systematic differences in captivity management and negotiation dynamics across armed actors.

Similarly, the cumulative probability of state rescue can be estimated over captivity duration. In this case, the curves represent the probability that a rescue has occurred before each point in time, as derived from the duration model specified for the rescue event. The figure shows that rescue is a progressive process whose intensity varies across armed actors. During the initial weeks, the trajectories show relatively similar patterns; however, as time advances, systematic differences become evident, as illustrated in Figure 13a. By day 100, the cumulative probability of rescue reaches approximately 0.33 for paramilitary groups, 0.28 for the ELN, and 0.27 for the FARC, while common criminal groups present slightly lower values. At the end of the 180-day horizon, paramilitaries approach 0.38, followed by the ELN at around 0.32 and the FARC at 0.31. These findings indicate that the cumulative effectiveness of state rescue efforts is not homogeneous,

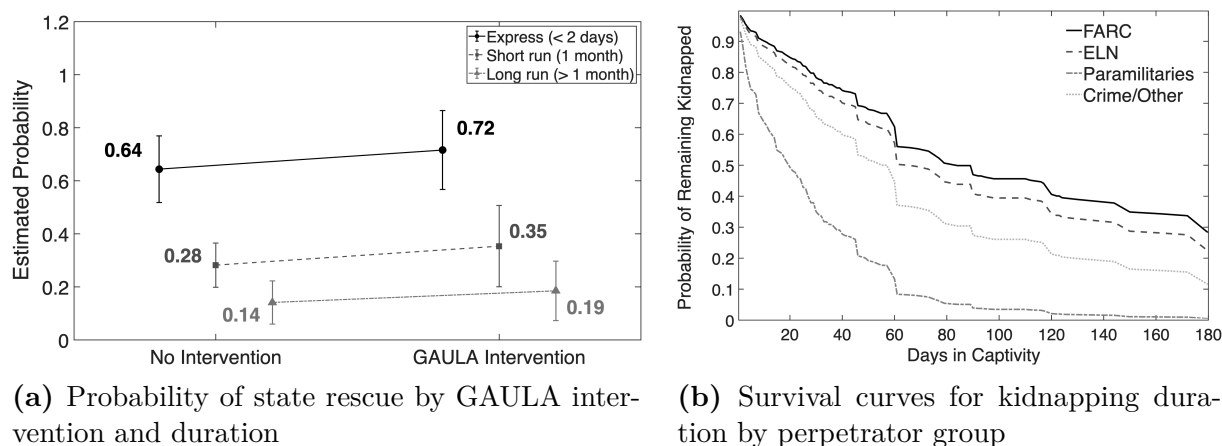


Figure 12. Temporal dynamics of state rescue and captivity.

Notes: Panel (a) presents the average predicted probabilities of state rescue, disaggregated by GAULA intervention and captivity duration. Panel (b) presents survival curves showing the estimated probability of remaining in captivity over time, broken down by perpetrator group.

Source: Authors' calculations using Stata 19, based on data from ?.

but rather depends on the organizational structure and operational environment of each group. The higher cumulative probability observed for paramilitary groups may reflect greater territorial exposure or more direct confrontation with state forces, while insurgent organizations with greater territorial control appear to experience a more gradual realization of rescue. Overall, the evidence confirms that the temporal dynamics of state rescue vary systematically according to the organizational profile of the armed actor.

Finally, unlike payment and rescue outcomes, whose probabilities evolve systematically with captivity duration due to negotiation processes, learning dynamics, and state intervention, death in captivity constitutes a rare event and does not appear to follow an optimal temporal strategy on the part of the captor. Given the low number of fatal events in the sample, representing less than five percent of cases, estimating proportional-hazard models would generate unstable survival curves with wide confidence intervals, thereby limiting the identification of statistically reliable patterns. For this reason, a descriptive approach based on simple lethality proportions is adopted, allowing a direct and transparent comparison across armed actors without imposing additional distributional assumptions.

The results reveal heterogeneity in the probability of death in captivity across organizational types. The FARC exhibit the lowest probability, 0.002, followed by the ELN at 0.005, while paramilitary groups reach 0.025 and common criminal or other actors show the highest value, 0.079, as shown in Figure 13b. Comparatively, the probability of death under common criminal groups is approximately forty times higher than that observed for the FARC and more than three times higher than that of paramilitary groups, suggesting that lethality reflects structural differences in organizational capacity and strategic incentives. Hierarchically structured insurgent organizations show lower execution rates,

consistent with stronger internal discipline and greater operational control, while common criminal groups display higher mortality rates, potentially associated with weaker organizational capacity, greater exposure to logistical errors, and more reactive responses under external pressure. Overall, the evidence indicates that the cost of failure in kidnapping varies systematically according to the organizational profile of the perpetrator.

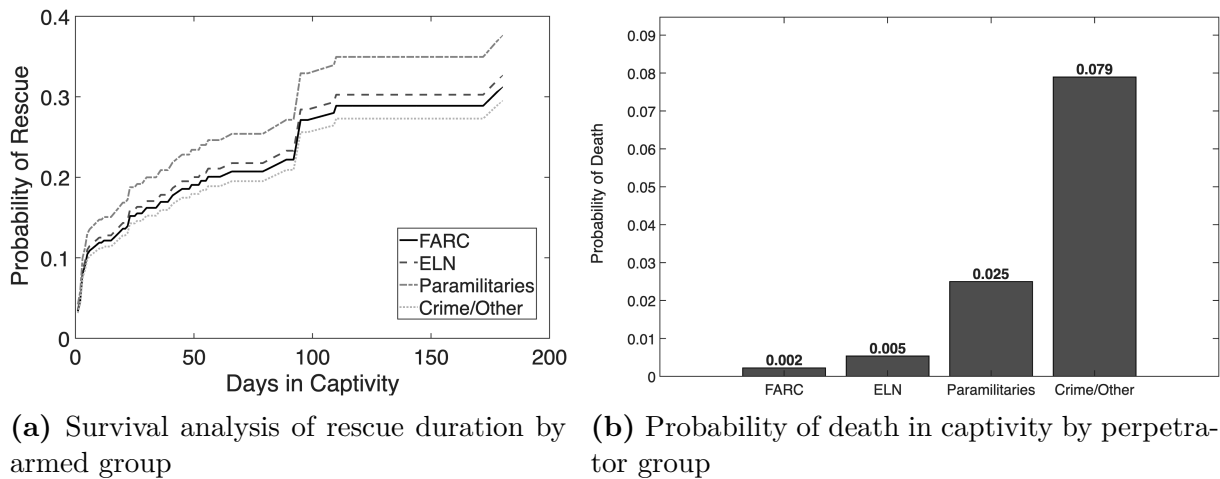


Figure 13. Temporal rescue patterns and captivity mortality.

Notes: Panel (a) shows the cumulative probability that state rescue has occurred at each point in time. Panel (b) presents the average probability of death in captivity by armed group, calculated as the mean of a binary death indicator within each category. The reported values are descriptive proportions and are not derived from a duration model.

Source: Authors' calculations using Stata 19, based on data from ?.

The temporal dimension of kidnapping events constitutes a fundamental component for understanding the strategic interaction between perpetrators and the State, since the analysis should not be limited to determining whether a ransom payment occurs, but must also account for the speed at which it materializes. While MNL models estimate the relative probabilities of alternative outcomes, the survival curves proposed by ? provide a dynamic representation of the probability that captivity persists beyond a given duration, thereby characterizing the temporal evolution of the event process. By contrast, the proportional hazards model developed by ? identifies differences in the instantaneous rate at which ransom payments occur over time, as formally expressed in Equation 11. Consequently, this latter approach is essential for capturing the strategic timing of negotiation outcomes, allowing a more precise assessment of how explanatory variables influence not only the probability that payment occurs but also the speed at which negotiations are resolved.

Within this framework, the hazard coefficients derived from the proportional hazards model developed by ? allow the identification of systematic differences in the instantaneous rate of ransom payment across armed actors. Taking the FARC as the reference category, the results reveal substantial heterogeneity in the retention technology employed by different groups. Kidnappings perpetrated by paramilitary groups show a hazard ratio

of 4.28 ($p < 0.001$), indicating that, holding all other variables constant, the instantaneous rate of ransom payment is more than four times higher than that observed for the FARC. In practical terms, if on a given day both groups still hold the victim captive, the immediate probability that ransom is paid at that moment is substantially higher in kidnappings carried out by paramilitary groups than in those perpetrated by the FARC. This pattern indicates that paramilitary organizations display greater urgency to conclude the event through ransom collection, reflecting shorter negotiation horizons and a faster monetization strategy.

Similarly, common criminal groups present a hazard ratio of 1.71 ($p = 0.021$), indicating a faster resolution of the event. By contrast, the ELN does not show statistically significant differences relative to the FARC, suggesting symmetry in insurgent operational practices. From a strategic perspective, these findings reveal a structural dichotomy in organizational behavior. Paramilitary and common criminal groups tend to operate under a high-turnover logic characterized by shorter time horizons, while consolidated insurgent organizations such as the FARC and the ELN maintain longer negotiations, consistent with greater territorial control and a more developed bureaucratic capacity to withstand the operational pressure exerted by the State.

A fundamental finding for public-policy design is the drastic reduction in the payment hazard rate observed when the victim belongs to the public sector. As shown in Table 4, this indicator records a value of 0.199 ($p < 0.001$) relative to the upper class, which serves as the reference category. In this context, the hazard rate represents the instantaneous probability that a payment event occurs at a given moment, conditional on the captivity situation having persisted up to that moment. This ratio indicates that the speed of completion of a payment for a public official decreases by 80.1% relative to the comparison baseline.

This economic disincentive is reinforced by an asymmetry in the State's tactical response capacity. When rescue events are examined, the results reported in the second column of Table 4 show that public-official cases present a hazard ratio of 0.237 ($p = 0.001$). Although this value indicates that rescues in this segment tend to unfold over a longer time horizon relative to civilians, its joint interpretation with the ransom-payment hazard reveals a form of operational paralysis for the captor. The criminal organization faces a strategic dilemma in which the probability of securing an economic transaction is very low, while the risk of eventual state intervention persists throughout a captivity that becomes increasingly costly and difficult to sustain.

Finally, the analysis of the contemporary period identifies an efficiency scissors that maximizes the deterrent effect by contrasting the current era with the pre-1998 period. In the Peace Process/Current period, beginning in 2011, resolution through payment has become 49% slower, registering a hazard ratio of 0.510 ($p = 0.035$). Simultaneously, the speed of successful rescues has increased drastically, reaching a hazard ratio of 4.863

($p < 0.001$). As shown in Table 4, this dynamic generates a structural collapse in the profitability of kidnapping; the perpetrator faces a scenario in which the time required to obtain a benefit is significantly longer, while the window of vulnerability to state intervention has closed exponentially, drastically reducing the strategic viability of this activity.

Table 4. *Proportional hazards regression results for kidnapping outcomes*

Predictors	Outcome: Payment (Hazard ratio)	Outcome: Rescue (Hazard ratio)
<i>Perpetrator group (Ref.: FARC)</i>		
ELN	1.186 (0.317)	1.057 (0.837)
Paramilitaries	4.284*** (0.000)	1.262 (0.530)
Common crime	1.713* (0.021)	0.935 (0.862)
<i>Victim profile (Ref.: upper class)</i>		
Middle class	0.780 (0.274)	1.071 (0.823)
Vulnerable population	1.542* (0.046)	0.512 (0.053)
Public sector	0.199*** (0.000)	0.237*** (0.001)
<i>Geographic zone (Ref.: Metropolis)</i>		
Andean region	1.290 (0.450)	1.709 (0.465)
Caribbean	1.170 (0.649)	1.024 (0.975)
Eastern Plains/Jungle	0.815 (0.644)	0.625 (0.530)
<i>Modus operandi (Ref.: direct attack)</i>		
Roadblock (illegal checkpoint)	1.780* (0.012)	2.339 (0.069)
Deception	1.330 (0.309)	3.275*** (0.000)
<i>Kidnapping characteristics</i>		
Group structure (vs. individual)	0.950 (0.801)	0.577 (0.114)

Continued on next page

Table 4 – continued from previous page

Predictors	Outcome: Payment (Hazard ratio)	Outcome: Rescue (Hazard ratio)
GAULA intervention (vs. none)	0.716 (0.267)	0.597 (0.244)
Ln(total victims)	1.205 (0.191)	1.593* (0.052)
Historical period (Ref.: pre-1998 rise)		
Crisis/Caguan (1998–2002)	1.112 (0.514)	0.677 (0.243)
Democratic Security (2003–2010)	0.735 (0.125)	2.022* (0.036)
Peace process/Current (2011+)	0.510* (0.035)	4.863*** (0.000)
Model statistics		
Observations (N)	461	461
Wald χ^2	117.57	182.68
Prob χ^2	0.0000	0.0000

Note. Coefficients are reported as Hazard Ratios (HR). Robust p-values clustered by municipality are shown in parentheses. * $p < .05$. ** $p < .01$. *** $p < .001$. *Technical note on sample size:* The reduction from $N = 1,125$ (Logit) to $N = 461$ (Cox) is due to the exclusion of 602 observations with missing duration data and 64 left-censored/zero-day observations, as required by the Breslow method for survival-time data. *Source:* Authors' elaboration using Stata 19.

2.6 Population-Level Kidnapping Risk: Evidence from a National Survey

This section draws on the Survey on Coexistence and Citizen Security (ECSC) ?, whose official methodology and operational guidelines are documented in the Fieldwork Manual and Basic Concepts ? and in the General Methodological Framework of the ECSC ?. The ECSC is a nationally representative probabilistic survey of the population aged 15 and over, with national, urban, and rural coverage. It is designed to measure direct victimization, perceptions of security, and reporting behavior, including extortion and, to a lesser extent, kidnapping for ransom. Compared with ?, its main strength lies in its sampling design and in the use of expansion factors that allow population-level inference, as well as in its capacity to capture unreported incidents and provide standardized, territorially comparable evidence on victimization dynamics.

On this basis, the section develops a statistical profile of ransom-kidnapping victims in Colombia and identifies individual characteristics that are not observable in ?. A logistic regression model is then estimated to examine whether these idiosyncratic attributes

are associated with patterns of victim selection by criminal networks, although their statistical relevance remains limited. The dependent variable is binary and indicates whether a surveyed person has been a kidnapping victim. The explanatory variables include educational attainment, gender, socioeconomic stratum, age-group fixed effects, and year fixed effects for the 2018–2021 period.

In addition, the ECSC is not designed to document ransom kidnapping in detailed case-by-case fashion, and the reported events show low statistical incidence within the sample. The information is self-reported and may therefore be subject to memory bias or underreporting. Moreover, because its unit of observation is the person rather than the specific criminal act, the survey does not allow reconstruction of the organizational structures, operational modalities, or long-run conflict patterns with the same depth as the case-by-case methodology employed by ? and presented in the previous section. Consequently, the ECSC should be interpreted as a complementary population-based source rather than as a substitute for specialized conflict records.

The descriptive and econometric results based on the ECSC survey indicate that, after applying population expansion factors, the estimated number of ransom-kidnapping victims between 2018 and 2021 amounts to 34,255 cases, with a clear increase in 2020 and 2021. Men represent 58.0% of victims, especially from 2020 onward, while cases are concentrated in socioeconomic strata 2 and 3, suggesting that victimization is not limited to the lowest or highest income levels. The largest shares are observed among persons with secondary and higher education. However, once the logistic model is estimated, most explanatory variables lose statistical significance. The only exception is the category “Up to primary education”, which is associated with a significantly lower probability of victimization relative to the reference group, defined as persons with complete secondary education. Average marginal effects are extremely small in absolute terms, reflecting the low prevalence of the event in the general population. Nevertheless, the model shows moderate discriminatory capacity, with an area under the ROC curve (AUC) of 0.72, indicating that when a victim and a non-victim are randomly compared, the model assigns a higher predicted probability to the actual victim in approximately 72% of cases. Overall, these findings suggest that ransom kidnapping is a low-prevalence phenomenon that is only weakly explained by idiosyncratic sociodemographic characteristics, although it is not entirely random, consistent with its strategic and selective nature in the Colombian context.

3 Relevance of the Empirical Framework for the Mechanism Model

The empirical evidence does not constitute a block separate from the theoretical model, but rather the source of its restrictions and parameters. The VEC system identifies macroeconomic persistence and long-run relationships that discipline retention and financing costs. The multinomial logit and competing-risks models provide probabilities of payment, death, rescue, and release, as well as heterogeneity by perpetrator, victim, region, and duration. These estimates determine the priors μ_0 , the intensities $\tilde{\lambda}_j$, the structural footprints $\beta_{K,j}(\theta_K)$, and the social loss margins used by the mechanism.

This connection fulfills two functions. First, it prevents type identification from depending on arbitrarily imposed criminal categories: types are distinguished by observable differences in captivity technologies, outcomes, and sensitivity to state instruments. Second, it makes it possible to interpret policy as dual control: α_t and γ_t simultaneously modify the material risk of the episode and the information available for later decisions. Thus, the empirical part anchors the prior and the theoretical part transforms that information into an implementable dynamic rule.

4 Theoretical Model

In Colombia, ransom kidnapping follows a non-linear trajectory that reveals **dynamic strategic interaction** under **incomplete information**. In this environment, heterogeneity among captors, variability in the state response, and household vulnerability configure a system of interdependent decisions. The mechanism formalized in this section models the chaining of payments, rescues, and operational pressure as a robust Bayesian learning process whose fundamental result is the **Theorem on the Identification of Rational Types in Unknown Environments**. Under this framework, state inference does not collapse in the face of surprising actions; instead, it distinguishes between strategic intention, imperfect physical execution, and statistical signal. To do so, the “Mano de Dios–Guadalupe” (MDG) process incorporates a perturbation akin to **trembling-hand** perfection in ?, refined through the notion of **sequential equilibrium** in ?. This integration allows initially low-probability events to be treated as infinitesimal deviations (trembles), ensuring that the planner maintains a consistent and rational belief system even off the equilibrium path. Consistent with the principles of **Perfect Bayesian Equilibrium** in ?, beliefs about the captor type θ_K are disciplined by Bayes updating and uninterrupted sequential consistency. If public policy generates asymptotically separable competing risks, the theorem shows that the accumulation of signals audits transitory volatility and forces convergence of the posterior distribution toward the structural identity of the true type. Thus, the normative objective goes beyond **reducing criminal**

profitability and **minimizing social loss**, in line with ? and ?; it seeks to **optimize the architecture of inference** that allows the State to identify the nature of the adversary, in continuity with the strategic approach of ?.

4.1 Environment, Types, and Deterrence Mechanism

Let $\mathcal{J} = \{S, K, F\}$ be the set of **agents**, where S denotes the State, K the kidnapper, and F the family. Each episode is characterized by the following vector of **types**:

$$\theta = (\theta_K, \theta_F, \theta_V, \theta_S) \in \Theta = \Theta_K \times \Theta_F \times \Theta_V \times \Theta_S.$$

Each component of θ captures an economic dimension of the episode. Thus, $\theta_K \in \Theta_K$ describes the **kidnapper type** and its organization or criminal technology, with $\Theta_K = \{\text{DC, PAR, ELN, FARC}\}$; $\theta_F \in \Theta_F$ represents the **family type** and approximates payment capacity, with $\Theta_F = \{\text{High wealth, Low wealth}\}$; $\theta_V \in \Theta_V$ identifies the **victim type** according to observable exposure and visibility attributes, with $\Theta_V = \{\text{Public, Private}\}$; and $\theta_S \in \Theta_S$ summarizes the **institutional environment**, with $\Theta_S = \{\text{Hard, Lax}\}$. In addition, each episode is located in an observable geographic zone $z \in \mathcal{R}$, with $\mathcal{R} = \{\text{Metropolitan, Andean, Caribbean, Pacific, East-Jungle}\}$. Location z is not a private strategic type, but a public initial condition of the episode. Its first role is to discipline the prior over the captor type, $\mu_0(\theta \mid z, \theta_V) = \mathbb{P}(\theta_K = \theta \mid z, \theta_V)$; in this mechanism only θ_K is taken to be **private** to K , the State and the family do not observe it, and they maintain a belief $\mu_t \in \Delta(\Theta_K)$. The other components and the geographic location z are **observable** at the opening of the episode.

At date t , the **public history** $h_t \in H_t$ recursively records public instruments, executed actions, and outcomes up to the beginning of t ((14)–(15)). By construction, h_t does not include the type vector θ nor the belief μ_t ; the types $(\theta_F, \theta_V, \theta_S)$ and z are observable at the opening of the episode but are not components of the tuple. The belief $\mu_t \in \Delta(\Theta_K)$ is updated by Bayes throughout the episode.

The **information sets** when choosing at t are

$$\mathcal{I}_t^S = (h_t, z, \theta_F, \theta_V, \theta_S, \mu_t), \quad \mathcal{I}_t^F = (h_t, z, \theta_F, \theta_V, \theta_S), \quad \mathcal{I}_t^K = (h_t, z, \theta_K, \theta_F, \theta_V, \theta_S).$$

Although μ_t is internal to the State, its exclusion from the private information sets creates no inconsistency: in the PBE, private agents share the same prior and the same public history h_t , so they can replicate the planner's Bayesian update ?. For the captor, μ_{t+1} is a strategic object that anticipates future state policy, not a source of uncertainty about its own type.

The period protocol is decomposed into a sequence of allocations, intentions, and stochastic materializations. At each instant t , agents formulate optimal strategic in-

tentions a_t^{i*} , derived from their respective optimization programs subject to Individual Rationality (IR) and Incentive Compatibility (IC) constraints. Define the **space of institutional allocations** as the product $\mathcal{X}_t := \mathcal{A}^{S,\text{disc}} \times \mathcal{A}^\alpha \times \mathcal{A}^\gamma$, where the State selects both the discrete action a_t^{S*} and the continuous instrument vector (α_t^*, γ_t^*) . The continuous instruments take values in the unit interval: $\alpha_t \in [0, 1]$ is the **financial blocking rate** (the fraction of the ransom blocked by the State) and $\gamma_t \in [0, 1]$ is the **intensity of operational pressure** (normalized between zero and maximum institutional capacity); both are observable public-policy instruments.

The transition between agents' strategic intentions and their effective materialization is mediated by a stochastic implementation operator called the **MDG process**. In each period, this operator transforms the discrete intentions $(a_t^{F*}, a_t^{K*}, a_t^{S*})$ into executed actions $\tilde{A}_t = (\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$, which enter the public record and determine h_{t+1} and the public signal ζ_t . The State's continuous instruments enter as the design optima (α_t^*, γ_t^*) , formally defined in Section 4.3.3; no additional continuous materialization is introduced. Conditional on those executed actions and on the optimal institutional policy, the system determines the physical outcome $m_t \in \mathcal{M}$ and the detection signal $d_t \in \{0, 1\}$ through a competing-risks block.

The physical outcome $m_t \in \mathcal{M}$ belongs to the continuation set $\mathcal{M}_{\text{cont}} = \{\text{cont}\}$ or to the absorbing set $\mathcal{M}_{\text{term}} = \{\text{pay}, \text{res}, \text{rel}, \text{kill}\}$. Separately, let $s_t \in \{0, 1\}$ be the **binary survival indicator of the victim** at the close of the period's operational margin, where $s_t = 1$ means that the victim survives and $s_t = 0$ means that the victim does not survive. This indicator is used in the rescue/cooperation branch and does not redefine the categorical support of m_t .

$$h_t = \left(t, \alpha_{[0,t-1]}^*, \gamma_{[0,t-1]}^*, \tilde{a}_{[0,t-1]}^F, \tilde{a}_{[0,t-1]}^K, \tilde{a}_{[0,t-1]}^S, m_{[0,t-1]}, d_{[0,t-1]} \right) \quad (14)$$

$$h_{t+1} = \left(h_t, \alpha_t^*, \gamma_t^*, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, m_t, d_t \right). \quad (15)$$

The horizon is the endogenous stopping time⁶

$$\tau = \inf\{t \geq 0 : m_t \in \mathcal{M}_{\text{term}}\}. \quad (16)$$

The dynamic kidnapping-deterrence mechanism is formally defined as the tuple

$$\mathcal{D} = \left\langle \mathcal{J}, \mathbb{T}, \Theta, \mathcal{R}, (\mathcal{X}_t)_{t \in \mathbb{T}}, (H_t)_{t \in \mathbb{T}}, (\mu_t)_{t \in \mathbb{T}}, \mathbb{P} \right\rangle.$$

The social choice rule is implementable if it constitutes the path of a **PBE ?**: the pair (s, μ) with $s = (s^S, s^K, s^F)$ and $\mu_t \in \Delta(\Theta_K)$. Given $z \in \mathcal{R}$ and $\theta_K \sim \mu_0$, the protocol

⁶The stopping-time condition guarantees that $\{\tau = t\}$ is measurable with respect to h_t .

deploys in each t the measurable public-policy rule

$$\chi_t: H_t \times \mathcal{R} \longrightarrow \mathcal{A}^{S, \text{disc}} \times \mathcal{A}^\alpha \times \mathcal{A}^\gamma, \quad s^S = \chi = (\chi_t)_{t \in \mathbb{T}},$$

interacting with private responses in $\mathcal{I}_t^K$ and $\mathcal{I}_t^F$. The PBE is the appropriate solution concept because it guarantees that the State cannot be systematically deceived by the captor: its beliefs μ_t are coherent with the actions actually observed, so operational pressure γ_t^* is calibrated on correct inferences about the type θ_K and not on assumptions inconsistent with equilibrium.

1. **Sequential rationality.** For each player $i \in \{K, F\}$, the optimal equilibrium **intention** a_t^{i*} satisfies

$$a_t^{i*} \in \arg \max_{a_t^i \in \mathcal{A}^i} \mathcal{U}_t^i(a_t^i \mid \mathcal{I}_t^i, \chi, s^{-i}),$$

while the planner selects the triple $(a_t^{S*}, \alpha_t^*, \gamma_t^*)$ to minimize expected social loss weighted by μ_t . The **social choice rule** is formalized through the measurable map $f_t: \mathcal{I}_t^S \rightarrow \mathcal{X}_t$, so that

$$x_t^* := f_t(\mathcal{I}_t^S) = (a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \mathcal{X}_t.$$

This vector summarizes the public policy in equilibrium; the MDG process then transforms intentions into executed actions $\tilde{A}_t$ entering the public record.

2. **Bayesian consistency on the path.** Let $\mathbb{P}^s$ denote the ex ante law over trajectories induced by strategy profile s , prior μ_0 , and captivity dynamics. At every node reached with positive probability under $\mathbb{P}^s$, beliefs coincide with Bayesian inference. For every $\theta \in \Theta_K$ and $t \geq 1$,

$$\mu_t(\theta) = \mathbb{P}^s(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}), \quad \mathcal{F}_t^{\text{pub}} = \sigma(z, \theta_V, \zeta_0, \dots, \zeta_{t-1}).$$

This condition is the core of the identification mechanism: by requiring μ_t to be coherent with the actions observed under s , it guarantees that γ_t^* is informative about θ_K .

Although the Revelation Principle guarantees equivalence with a direct mechanism, the indirect approach is deliberately adopted: it allows observable public-policy instruments—financial blocks and operational pressure—to be modeled as they are deployed in practice, rather than reducing the State to a designer of counterfactual messages.

4.2 Probabilistic Technology, Implementation, and Likelihoods

This subsection groups the stochastic technology of the mechanism before presenting the agents' optimization problems. The logical sequence is: optimal intentions a_t^{i*} materialize as executed actions $\tilde{a}_t^i$ through the MDG process; executed actions, together with instruments (α_t, γ_t) , induce competing risks and public signals; finally, those signals generate likelihoods and update belief μ_t by Bayes. Thus, when the objective functions of K , F , and S are introduced, all probabilistic objects entering their payoffs have already been defined.

4.2.1 Stochastic Implementation: MDG Process

This section formalizes the stochastic architecture that transforms strategic plans into observable realizations. The MDG implementation filter converts each agent's optimal intention a_t^{i*} into an executed action $\tilde{a}_t^i$ with full support over $\mathcal{A}^i$; the logit distribution implementing it is axiomatically justified as the unique law compatible with full support, dominance, and maximum entropy (Remark 4.2). The physical-outcome law $\mathbb{P}(\cdot \mid \theta_K, \mathcal{C}_t)$ determines the observable outcome m_t conditional on the material state of the period, including the executed triple. Finally, the inverse-transform mechanism operationalizes both layers in a reproducible sequence of draws.

The MDG process transforms intentions a_t^{i*} into executed actions $\tilde{a}_t^i$ through an implementation law concentrated on a_t^{i*} but with full support over $\mathcal{A}^i$, in the spirit of the *trembling hand* of ? and ?. The executed action of each agent $i \in \{K, S, F\}$ is

$$\tilde{a}_t^i \sim \mathbb{P}_i^{\text{imp}}(\cdot \mid a_t^{i*}, X_t), \quad (17)$$

where X_t summarizes the state of the system and $\mathbb{P}_i^{\text{imp}}$ concentrates dominant mass on a_t^{i*} . The executed profile $\tilde{A}_t = (\tilde{a}_t^K, \tilde{a}_t^S, \tilde{a}_t^F)$ is observable; the strategic intention remains latent. The physical outcome is drawn with the competing law

$$m_t \sim \mathbb{P}(\cdot \mid \theta_K, \mathcal{C}_t), \quad (18)$$

where $\mathcal{C}_t$ incorporates competing risks, structural parameters, optimal instruments, and materialized actions. Table 5 presents three representative configurations of the support of m_t .

Definition 4.1. MDG stochastic implementation filter. Let $\mathcal{A}^i$ be the finite set of admissible actions of agent i . Given the plan $a_t^{i*} \in \mathcal{A}^i$ and state vector X_t , execution is determined by the implementation law

$$\tilde{a}_t^i \sim \mathbb{P}_i^{\text{imp}}(\cdot \mid a_t^{i*}, X_t).$$

Table 5. Examples of the support of the physical outcome m_t subject to execution of the triple $\tilde{A}_t$

$\tilde{a}_t^K$	$\tilde{a}_t^S$	$\tilde{a}_t^F$	Support of m_t
kill	rescue	collude	death; rescue; payment; release; continuation
release	negotiate	cooperate	release; rescue; payment; death; continuation
continue	rescue	cooperate	continuation; rescue; payment; death; release

Source: Author's elaboration.

The stochastic structure of the model is formalized on a measurable space $(\Omega, \mathcal{F})$, where Ω is the sample space of states of nature and $\mathcal{F}$ the σ -algebra defining measurable events. Under this framework, $\mathbb{P}_i^{\text{imp}}$ assigns implementation probabilities, while the measurable function $\mathcal{H}_t^i : \mathcal{A}^i \times \Omega \rightarrow \mathcal{A}^i$ materializes the sequential draw, so that for each elementary event $\omega \in \Omega$ the executed action is $\tilde{a}_t^i(\omega) = \mathcal{H}_t^i(a_t^{i*}, \omega)$. This formulation makes the aggregate profile of executed actions

$$\tilde{A}_t = (\tilde{a}_t^K, \tilde{a}_t^S, \tilde{a}_t^F) \quad (19)$$

a random object that incorporates the operational perturbations of the period in a way compatible with the underlying probability measure.

- (i) **Plan dominance and error support.** For all $t < \tau$, the distribution $\mathbb{P}_i^{\text{imp}}$ concentrates most of its mass on a_t^{i*} , assigning strictly positive probability to alternative actions treated as *implementation frictions*. Under **full support**, $\mathbb{P}_i^{\text{imp}}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$.
- (ii) **Bounded institutional maturation.** The informational component $[H(\mu_t)/H(\mu_0)]e^{-\eta_{\text{cal}}t}$ decreases as posterior uncertainty falls, while the structural floor $\underline{c}$ persists in (21). When $\underline{c} > 0$, full support over $\mathcal{A}^i$ is preserved in the long run.

The implementation law $\mathbb{P}_i^{\text{imp}}$ adopts a logit structure with hybrid temperature, whose axiomatic justification is established in Remark 4.2:

$$\mathbb{P}_i^{\text{imp}}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) = \frac{\exp(\mathbf{1}\{a = a_t^{i*}\}/T_t)}{\sum_{a' \in \mathcal{A}^i} \exp(\mathbf{1}\{a' = a_t^{i*}\}/T_t)}. \quad (20)$$

Remark 4.2 (Axiomatic justification of the logit as the unique implementation law). Equation (20) is not an ad hoc choice: it is the *unique* distribution over $\mathcal{A}^i$ that simultaneously satisfies: (A1) **full support**, $\mathbb{P}_i^{\text{imp}}(a \mid a_t^{i*}, X_t) > 0$ for all $a \in \mathcal{A}^i$; (A2) **dominance**, $\mathbb{P}_i^{\text{imp}}(a_t^{i*} \mid a_t^{i*}, X_t) > \mathbb{P}_i^{\text{imp}}(a \mid a_t^{i*}, X_t)$ for all $a \neq a_t^{i*}$; and (A3) **maximum entropy**, among all distributions satisfying full support and dominance. By the characterization

theorem of ??, the entropy maximizer under these restrictions is precisely the logit form in (20).

Here, $T_t = T_t(\mu_t, t) > 0$ represents the **system temperature**; as long as it is positive, law (20) maintains full support over $\mathcal{A}^i$ and prevents likelihood degeneracy. To jointly capture strategic learning and institutional maturation, define

$$T_t = T_0 \max \left\{ \frac{H(\mu_t)}{H(\mu_0)} e^{-\eta_{\text{cal}} t}, \underline{c} \right\}, \quad (21)$$

where $T_0 > 0$ is initial temperature, $\eta_{\text{cal}} > 0$ the calendar decay rate, $H(\mu_t) = -\sum_{\theta} \mu_t(\theta) \ln \mu_t(\theta)$ Shannon entropy, and $\underline{c} \in [0, 1]$ the structural noise floor.

Definition 4.3 (Law of the physical outcome). The **physical outcome** m_t is an observable categorical variable on the support

$$\mathcal{M} = \{\text{Cont}, 1, 2, 3, 4\},$$

where Cont represents continuation of captivity at the close of the period and $j \in \{1, 2, 3, 4\}$ represents closure by competing cause j . Conditional on the captor type and the period's material state,

$$\mathcal{C}_t(\theta_K) = (\theta_K, \theta_F, \theta_V, \theta_S, z, \tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S, \alpha_t^*, \gamma_t^*, p_{\text{det},t}),$$

its distribution is defined from effective intensities $\tilde{\lambda}_j(t | \theta_K)$. The probability of continuation is

$$\mathbb{P}(m_t = \text{Cont} | \theta_K, \mathcal{C}_t) = p_{\text{Cont},t|\theta_K} = \exp \left(- \sum_{j=1}^4 \tilde{\lambda}_j(t | \theta_K) \Delta t \right). \quad (22)$$

The total daily closure probability and each cause's relative weight are

$$q_t(\theta_K) = 1 - p_{\text{Cont},t|\theta_K}, \quad \xi_j(t | \theta_K) = \frac{\tilde{\lambda}_j(t | \theta_K)}{\sum_{\ell=1}^4 \tilde{\lambda}_\ell(t | \theta_K)}, \quad h_j(t | \theta_K) = q_t(\theta_K) \xi_j(t | \theta_K). \quad (23)$$

Therefore,

$$\mathbb{P}(m_t = j | \theta_K, \mathcal{C}_t) = h_j(t | \theta_K), \quad j = 1, \dots, 4. \quad (24)$$

Materialization of m_t is obtained by inverse transform. Let $v_t \sim \text{Unif}(0, 1)$ and fix a total order on $\mathcal{M}$. Then

$$m_t = r \iff \sum_{\ell < r} \mathbb{P}(m_t = \ell | \theta_K, \mathcal{C}_t) \leq v_t < \sum_{\ell \leq r} \mathbb{P}(m_t = \ell | \theta_K, \mathcal{C}_t). \quad (25)$$

Thus, m_t enters the Bayesian likelihood only once:

$$\mathbb{P}(m_t \mid \theta_K, \mathcal{C}_t) = p_{\text{Cont}, t \mid \theta_K}^{\mathbf{1}\{m_t = \text{Cont}\}} \prod_{j=1}^4 h_j(t \mid \theta_K)^{\mathbf{1}\{m_t = j\}}. \quad (26)$$

4.2.2 Materialization Through Inverse Transform

The model distinguishes between **strategic intention**, derived from the dynamic program, and the **observed realization** in the data. This transition is operationalized through a hierarchical inverse-transform draw. First, $\mathcal{H}_t^i$ is the measurable materialization function of agent i , converting the rational intention a_t^{i*} and a uniform draw $u_{t,i} \sim \text{Unif}(0, 1)$ into an executed action $\tilde{a}_t^i$ according to $\mathbb{P}_i^{\text{imp}}$. Second, once the executed triple $\tilde{A}_t$ is consolidated, the measurable rule $\mathcal{G}_t$ materializes the physical outcome m_t from material state $\mathcal{C}_t$ and an additional uniform draw $v_t \sim \text{Unif}(0, 1)$ according to $\mathbb{P}(\cdot \mid \theta_K, \mathcal{C}_t)$. The executed action is determined by

$$\tilde{a}_t^i = a_k \iff \sum_{j=1}^{k-1} \mathbb{P}_i^{\text{imp}}(a_j \mid a_t^{i*}, X_t) \leq u_{t,i} < \sum_{j=1}^k \mathbb{P}_i^{\text{imp}}(a_j \mid a_t^{i*}, X_t). \quad (27)$$

Given $\tilde{A}_t$ and $\mathcal{C}_t$, the final outcome is materialized within $\mathcal{M}$ by

$$m_t = j \iff \sum_{l < j} \mathbb{P}(m_t = l \mid \theta_K, \mathcal{C}_t) \leq v_t < \sum_{l \leq j} \mathbb{P}(m_t = l \mid \theta_K, \mathcal{C}_t). \quad (28)$$

Table 6. Materialization through inverse-transform intervals

Phase	Case 1: logistical failure	Case 2: family panic	Case 3: surgical rescue
Intention a_t^{i*}	S : rescue	F : cooperate	S : rescue; K : kill
Executed triple $\tilde{A}_t$	(cont, neg, coop)	(lib, neg, colud)	(kill, res, coop)
Draw v_t	0.92	0.40	0.60
Outcome m_t	accidental rescue	materialized ment	pay- successful rescue

Note: The table illustrates the logic of the inverse transform; probabilities are illustrative intervals.

4.2.3 Competing Risks and Survival

Modeling the outcome of captivity begins by distinguishing between the **instantaneous intensity** and its daily probabilistic projection, in line with proportional hazards ? and econometric treatments for interval duration data ??. The **cause-specific intensity** $\lambda_j(t)$ is the structural variable measuring the marginal propensity for closure by cause j ; as an instantaneous rate, it is measured in units of 1/time and is not bounded above by

one. By contrast, for simulation and daily observation, the **discrete hazard** $h_j(t)$ is a probability in $[0, 1]$, representing the likelihood that event j materializes during interval t conditional on survival at the start of that period.

From a **material** perspective, risk intensities depend on the vector of **executed** actions $\tilde{A}_t = (\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$ generated by the MDG stochastic process. The continuous instruments modifying hazards, detection, closure probabilities, and likelihoods are the State's optima (α_t^*, γ_t^*) , chosen by the control problem with informational motive (69).

For each cause $j \in \{1, 2, 3, 4\}$, let $\lambda_{j0}(t)$ be the baseline intensity and let z denote the vector of observables. Effective intensity is

$$\tilde{\lambda}_j(t | \theta_K) = \lambda_{j0}(t) \exp(\beta_{K,j}(\theta_K) \cdot z + \zeta_{\alpha,j} \alpha_t^* + \zeta_{\gamma,j} \gamma_t^*). \quad (29)$$

The maturation filter, payment/continuation probability, and relative cause weights are

$$M(t) := \min \left\{ 1, \left(\frac{t}{T_{\text{mad}}} \right)^2 \right\}, \quad (30)$$

$$p_{\text{Cont},t} = \exp \left(- \sum_{j=1}^4 \tilde{\lambda}_j(t) \Delta t \right), \quad (31)$$

$$q(t) = 1 - p_{\text{Cont},t}, \quad (32)$$

$$\xi_j(t) = \frac{\tilde{\lambda}_j(t)}{\sum_{\ell=1}^4 \tilde{\lambda}_\ell(t)}, \quad j = 1, \dots, 4, \quad \text{where } \sum_{j=1}^4 \xi_j(t) = 1. \quad (33)$$

If the sum of effective intensities is zero, $q(t) = 0$ and no terminal cause is drawn. The daily marginal probability of closure by cause j is then $h_j(t) = q(t)\xi_j(t)$, so that $\sum_j h_j(t) = q(t) \leq 1$ by construction.

4.2.4 Survival Under Focal Rescue, Bayesian Accuracy, and Human Loss

At the beginning of each period t , the State observes the public information flow $\mathcal{I}_t^S$ and updates its posterior belief $\mu_t(\theta_K) = \mathbb{P}(\theta_K | \mathcal{I}_t^S)$ by Bayesian recursion. To operationalize information quality in rescue planning, define the precision index ι_t as the **probability mass of the modal type**:

$$\iota_t := \max_{\theta \in \Theta_K} \mu_t(\theta), \quad \text{with estimated profile } \hat{\theta}_t = \arg \max_{\theta \in \Theta_K} \mu_t(\theta). \quad (34)$$

Under this scheme, ι_t acts as an **operational confidence index**. However, rescue effectiveness depends not only on posterior concentration but also on the truthfulness of identification. Conditional on a focal rescue attempt and on the true type θ_K , survival probability is modeled through a logistic response penalizing miscalibration between the

estimated and true profiles:

$$\mathbb{P}(s_t = 1 \mid \iota_t, \hat{\theta}_t, \theta_K) = \Lambda\left(\alpha_{\text{leth}}(\theta_K) + \beta_R \iota_t \mathbf{1}\{\hat{\theta}_t = \theta_K\}\right). \quad (35)$$

If the State correctly identifies the captor, the intelligence input ι_t enhances the probability of success according to productivity β_R ; if identification is wrong, accumulated intelligence becomes ineffective or miscalibrated, reducing survival probability to the base level determined by intrinsic lethality $\alpha_{\text{leth}}(\theta_K)$.

This subsection projects the daily decision mechanism into a **longitudinal framework**, establishing the mapping between effective intensities $\tilde{\lambda}_j(t)$, marginal risks $h_j(t)$, and continuation probability $p_{\text{Cont},t}$ with historical duration and closure-mode records. Let $S(0) = 1$. For every $t \geq 1$,

$$S(t) = \prod_{t'=1}^t p_{\text{Cont},t'}, \quad F_j(t) = \sum_{t'=1}^t h_j(t') S(t' - 1), \quad \text{with } S(t) + \sum_{j=1}^4 F_j(t) = 1. \quad (36)$$

Proposition 4.4 (Expected captivity time and operational pressure). *Define the **net sensitivity** of type θ_K to optimal operational pressure γ_t^* as*

$$\kappa_h(\theta_K, t) := \left(\zeta_{\gamma,2} \tilde{\lambda}_2(t \mid \theta_K) + \zeta_{\gamma,3} \tilde{\lambda}_3(t \mid \theta_K)\right) - \zeta_{\gamma,1} \tilde{\lambda}_1(t \mid \theta_K). \quad (37)$$

If $\kappa_h(\theta_K, t) > 0$ for all t , the expected captivity time $\mathbb{E}[\tau \mid \theta_K]$ is strictly decreasing in γ_t^ . If $\kappa_h(\theta_K, t) < 0$ for all t , it is strictly increasing. The model therefore generates the **testable differential prediction***

$$\text{sgn}\left(\frac{\partial \mathbb{E}[\tau \mid \theta_K]}{\partial \gamma_t^*}\right) = -\text{sgn}(\kappa_h(\theta_K, t)).$$

Proof. The proof follows three steps: differentiating daily survival, propagating the derivative through the product rule, and concluding by dominated convergence. By Fubini's theorem, the expectation of a non-negative integer-valued random variable can be written as the sum of its survival function:

$$\mathbb{E}[\tau \mid \theta_K] = \sum_{t'=0}^{\infty} P(\tau > t' \mid \theta_K) = \sum_{t'=0}^{\infty} S(t' \mid \theta_K). \quad (38)$$

Since only Λ_t depends on γ_t^* ,

$$\frac{\partial p_{\text{Cont},t|\theta_K}}{\partial \gamma_t^*} = -\left(\sum_{j=1}^3 \zeta_{\gamma,j} \tilde{\lambda}_j(t \mid \theta_K)\right) p_{\text{Cont},t|\theta_K} \Delta t. \quad (39)$$

Regrouping terms by whether the channel accelerates or delays closure yields the sign

summarized by $\kappa_h(\theta_K, t)$. The derivative of survival is

$$\frac{\partial S(t' | \theta_K)}{\partial \gamma_t^*} = S(t' | \theta_K) \frac{1}{p_{\text{Cont}, t | \theta_K}} \frac{\partial p_{\text{Cont}, t | \theta_K}}{\partial \gamma_t^*}, \quad t \leq t', \quad (40)$$

and zero for $t > t'$. Therefore,

$$\frac{\partial \mathbb{E}[\tau | \theta_K]}{\partial \gamma_t^*} = \sum_{t'=t}^{\infty} \frac{\partial S(t' | \theta_K)}{\partial \gamma_t^*}, \quad (41)$$

whose sign is the opposite of $\kappa_h(\theta_K, t)$ under the stated uniform sign condition. $\square$

4.2.5 Capture Probability

The probability of capture or detection is governed by operational pressure and by the institutional environment. In reduced form,

$$p_{\text{cap}, t} = \mathbb{P}(d_t = 1 | \theta_K, \alpha_t^*, \gamma_t^*, z, \theta_S), \quad (42)$$

with technical specification

$$p_{\text{cap}, t} = \Lambda(\rho_0 + \rho_\gamma \gamma_t^* + \rho_\alpha \alpha_t^* + \rho_z z + \rho_S \theta_S - \rho_K(\theta_K)). \quad (43)$$

4.2.6 Physical Likelihood

The physical likelihood separates continuation and terminal-outcome cases:

$$\mathcal{L}_{F, t}(\theta | m_t = \text{Cont}) = p_{\text{Cont}, t | \theta}^{\mathbf{1}\{m_t = \text{Cont}\}}, \quad (44)$$

$$\mathcal{L}_{F, t}(\theta | m_t = j) = h_j(t | \theta)^{\mathbf{1}\{m_t = j\}}, \quad (45)$$

and compactly,

$$\mathcal{L}_{F, t}(\theta) = p_{\text{Cont}, t | \theta}^{\mathbf{1}\{m_t = \text{Cont}\}} \prod_{j=1}^4 h_j(t | \theta)^{\mathbf{1}\{m_t = j\}}. \quad (46)$$

The detection likelihood is Bernoulli:

$$\mathcal{L}_{d, t}(\theta) = p_{\text{cap}, t}(\theta)^{d_t} (1 - p_{\text{cap}, t}(\theta))^{1-d_t}. \quad (47)$$

The executed-action likelihood induced by the MDG layer is

$$\mathcal{L}_{I, t}(\theta) = \prod_{i \in \{K, S, F\}} \mathbb{P}_i^{\text{imp}}(\tilde{a}_t^i | a_t^{i*}(\theta), X_t), \quad (48)$$

so the joint physical likelihood is

$$\mathcal{L}_{F,t}^{\text{joint}}(\theta) = \mathcal{L}_{I,t}(\theta) \mathcal{L}_{F,t}(\theta) \mathcal{L}_{d,t}(\theta). \quad (49)$$

4.2.7 Voice

The communication channel is decomposed as

$$\mathcal{L}_{C,t}(\theta) = \mathbb{P}(c_t \mid \theta, \alpha_t^*, \gamma_t^*, d_t), \quad (50)$$

with full likelihood

$$\mathcal{L}_{C,t}(\theta) = \pi_{\text{call}}(d_t, \gamma_t^*, \theta)^{c_t} \left(1 - \pi_{\text{call}}(d_t, \gamma_t^*, \theta)\right)^{1-c_t}, \quad (51)$$

and diagonal shorthand

$$\pi_{\text{call}}(d_t, \gamma_t^*, \theta) = \Lambda(\delta_\theta + \zeta_{\gamma,\theta} \gamma_t^* + \zeta_d d_t). \quad (52)$$

The combined communication likelihood is denoted

$$\mathcal{L}_{C,t}(\theta). \quad (53)$$

4.2.8 Bayesian Updating

The full likelihood used by the mechanism is

$$\mathcal{L}_t(\theta) = \mathcal{L}_{F,t}^{\text{joint}}(\theta) \mathcal{L}_{C,t}(\theta), \quad (54)$$

and the posterior update is

$$\mu_{t+1}(\theta) = \frac{\mu_t(\theta) \mathcal{L}_t(\theta)}{\sum_{\theta' \in \Theta_K} \mu_t(\theta') \mathcal{L}_t(\theta')}. \quad (55)$$

4.3 Agents: Optimization and Equilibrium

In this way, the mechanism is structured under a **Principal-Agent** architecture in a dynamic environment. The **State** (S) acts as the designer (*Principal*) that seeks to align the choices of the **Agents** (Kidnapper and Family) toward the termination of the event.⁷

⁷**Probability and utility convention.** Unless another measure is explicitly indicated (e.g., $\mathbb{Q}$ in martingale definitions), $\mathbb{P}(\cdot)$ and $\mathbb{E}[\cdot]$ denote probabilities and expectations under the ex ante law of the mechanism; $\mathbb{P}_i^{\text{imp}}$ denotes player i 's stochastic implementation law (MDG process); $\mathbb{P}_\theta$ denotes the **counterfactual** episode law that fixes $\theta_K = \theta$. In the utility functions $\mathcal{U}_t^i$, the conditional probabilities that weight outcomes are written with $\mathbb{P}(\cdot \mid \cdot)$ unless the context explicitly indicates $\mathbb{P}_i^{\text{imp}}$ or $\mathbb{P}_\theta$. The continuous uniform distribution on $[0, 1]$ is denoted $\text{Unif}(0, 1)$, separately from utilities $\mathcal{U}$. Primitive scalar probabilities by type (e.g., contact intensity) are written $\pi_{\text{call}}(\theta_K)$ so as not to confuse them with

4.3.1 Analysis of the Kidnapper (K), Origin and Maximization of Actions

In each period t , the captor solves an optimal stopping problem by selecting $a_t^K \in \mathcal{A}^K = \{a_{rel}, a_{kill}, a_{cont}\}$ that maximizes its value function:

$$a_t^{K*} = \arg \max_{a_t^K \in \mathcal{A}^K} \left\{ \mathcal{U}_t^K(a_t^K \mid h_t, \theta_K, \theta_S) \right\} \quad (56)$$

Death (a_{kill}). The return from executing the hostage weighs the reputational benefit $\eta(\theta_i)$ against the expected penalty $\tilde{p}_{cap,t}(\theta_i)F_{cap}(\theta_i, \theta_S)$ if the captor is intercepted at the margin of the period⁸:

$$\begin{aligned} U_{kill}^K(\theta_i, \theta_S) = & \left(1 - \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}} [p_{cap}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*)] \right) \eta(\theta_i) \\ & - \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}} [p_{cap}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*)] F_{cap}(\theta_i, \theta_S). \end{aligned} \quad (57)$$

Continuation (a_{cont}). The captor's Bellman equation combines the current flow—expected ransom net of blocking, operational cost, and capture penalty—with the type-discounted future value. Since θ_K is the captor's private information, μ_t is not uncertainty for him but rather a strategic object: the evolution of μ_{t+1} conditions the State's future policy.

$$\begin{aligned} V_{cont,t}^K = & \mathbb{E}_{\tilde{A}_t \mid \mathcal{Q}_t^{\text{Cont}}} \left[\mathbb{P}(m_t = \text{pay} \mid \tilde{A}_t, X'_t, \theta_K = \theta_i) \right] R(1 - \alpha_t^*) \\ & - C_t(\gamma_t^*, \theta_i) \\ & - \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}} \left[p_{cap}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*) \right] F_{cap}(\theta_i, \theta_S) \\ & + \beta(\theta_i) \left(1 - \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}} [p_{cap}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*)] \right) \mathbb{E}_{\theta_i} [V_{t+1}^K(\mu_{t+1}(a_{cont}))]. \end{aligned} \quad (58)$$

where the captor's effective probabilities are written as

$$\tilde{p}_{cap,t}(\theta_i) := \mathbb{E}_{\tilde{a}_t^S \mid \mathcal{Q}_t^{\text{Cap}}} \left[p_{cap}(\tilde{a}_t^S, \theta_i, \theta_S, \alpha_t^*, \gamma_t^*) \right], \quad (59)$$

$$\tilde{p}_{pay,t}(\theta_i) := \mathbb{E}_{\tilde{A}_t \mid \mathcal{Q}_t^{\text{Cont}}} \left[\mathbb{P}(m_t = \text{pay} \mid \tilde{A}_t, X'_t, \theta_K = \theta_i) \right], \quad (60)$$

with

$$\begin{aligned} \mathcal{Q}_t^{\text{Cap}} &:= (a_t^{S*}, \alpha_t, \gamma_t, \theta_K, \theta_S), \\ \mathcal{Q}_t^{\text{Cont}} &:= (a_t^{S*}, a_t^{F*}, a_{cont}, \alpha_t, \gamma_t, \mu_t, \theta_K). \end{aligned}$$

$\mathbb{P}$, $\mathbb{P}_i^{\text{imp}}$, $\mathbb{P}_\theta$, or with the action label P in operational tables.

⁸**Conditioning.** In all branches, the effective probabilities $\tilde{p}_{\cdot,t}$ are defined as conditional expectations over the MDG realizations induced by the relevant tuple $\mathcal{Q}_t$. The explicit expressions for $\tilde{p}_{cap,t}$, $\tilde{p}_{pay,t}$, $\tilde{p}_{surv,t}$, $\tilde{p}_{rel,t}$, and $\tilde{p}_{kill,t}$ are reported below in the equations for the kidnapper, the family, and the State.

The convex-exponential specification (61) is designed to induce a separating equilibrium. The parameter $\kappa_c(\theta_i) > 0$ generates the *single-crossing property*: variations in γ_t affect different types θ_i asymmetrically, turning the persistence or termination of captivity into an informative signal for the State's Bayesian updating.

$$C_t(\gamma_t, \theta_i) = \phi(\theta_i) \exp(\kappa_c(\theta_i) \gamma_t) + \nu(\theta_i) \quad (61)$$

Release (a_{rel}). This branch collects the utility associated with the strategic closure of the kidnapping without obtaining rent. Let $\kappa_{rel}(\theta_i) \geq 0$ be the parameter, or function, that represents the disutility tied to release, reflected in reputational loss or foregone rent. By contrast, $\eta(\theta_i)$ summarizes the reputational benefit, or other private arrangements of the captor, when he materializes a_{kill} without being captured at the margin of t . Under this notation, release utility is expressed as

$$U_{rel}^K(\theta_i) = -\kappa_{rel}(\theta_i).$$

Compactly, the kidnapper's utility can be written as

$$\mathcal{U}_t^K(a_t^K \mid h_t, \theta_K, \theta_S) = \begin{cases} U_{rel}^K(\theta_i), & \text{if } a_t^K = a_{rel}, \\ U_{kill}^K(\theta_i, \theta_S), & \text{if } a_t^K = a_{kill}, \\ V_{cont,t}^K, & \text{if } a_t^K = a_{cont}. \end{cases} \quad (62)$$

In the numerical laboratory—where m_t is not observable— K 's program is closed with an **intrinsic stopping rule**: in each t , $V_{cont,t}^K$ is contrasted with the terminal utilities of release and killing until continuation no longer dominates the best *one-shot* payoff. It is useful to distinguish three objects: **(i)** the **physical** stopping time τ in (16), the anchor of the observable episode and basis of $\mathbb{E}[\tau \mid \theta_K]$ in Proposition 4.4; **(ii)** the integer that bounds $\mathbb{T} = \{0, 1, \dots, \tau\}$ when formalizing the mechanism, merely a calendar *ceiling* with no mechanical link to (16); and **(iii)** the simulated truncation rule stated below. We denote

$$\tau_K^{\text{sim}} := \min \left\{ t \geq 0 : V_{cont,t}^K \leq \max \left\{ -\kappa_{rel}(\theta_K), U_{kill}^K(\theta_K, \theta_S) \right\} \right\}, \quad (63)$$

where $-\kappa_{rel}(\theta_K) = U_{rel}^K(\theta_K)$ in the a_{rel} branch and U_{kill}^K appears in (57). The symbol τ_K^{sim} summarizes the first margin at which continuation no longer exceeds the maximum between surrender and lethality; it serves as a **truncation criterion** in simulation. The τ in (16), by contrast, marks when the physical outcome absorbs the episode in $\mathcal{M}$.

4.3.2 The Family (F), Rational Maximization of Survival

In each period t , the family chooses $a_t^F \in \mathcal{A}^F = \{a_{\text{coop}}, a_{\text{col}}\}$ by solving

$$a_t^{F*} \in \arg \max_{a^F \in \{a_{\text{coop}}, a_{\text{col}}\}} \mathbb{E}_{\theta_K | \mathcal{I}_t^F} [\mathcal{U}_t^F(a^F | \theta_K, \theta_F)]. \quad (64)$$

Cooperation (a_{coop}). The family cooperates when the expected value of the victim's survival, weighted by $V_L > 0$, exceeds the cost of following the institutional route $e_t(\gamma_t, \theta_F)$:

$$\mathcal{U}_t^F(a_{\text{coop}} | \theta_K, \theta_F) = \mathbb{E}_{\theta_K | \mathcal{I}_t^F} \left[\mathbb{E}_{\tilde{A}_t | \mathcal{Q}_t^{\text{coop}}} \left[\mathbb{P}(s_t = 1 | \gamma_t, \tilde{A}_t, \theta_K) \right] \right] V_L - e_t(\gamma_t, \theta_F). \quad (65)$$

The institutional cost adopts the **exponential-convex** form, analogous to the captor's $C_t(\gamma_t, \theta_i)$ in (61):

$$e_t(\gamma_t, \theta_F) = \phi_F(\theta_F) \exp(\kappa_F(\theta_F) \gamma_t) + \nu_F(\theta_F), \quad (66)$$

with $\phi_F(\theta_F) > 0$ and $\kappa_F(\theta_F) > 0$ for every $\theta_F \in \Theta_F = \{\text{High wealth}, \text{Low wealth}\}$, and $\nu_F(\theta_F) \in \mathbb{R}$ a type-specific fixed shift.⁹

where

$$\begin{aligned} \tilde{p}_{\text{surv},t}(\theta_i) &:= \mathbb{E}_{\tilde{A}_t | \mathcal{Q}_t^{\text{coop}}} \left[\mathbb{P}(s_t = 1 | \gamma_t, \tilde{A}_t, \theta_K = \theta_i) \right], \\ \mathcal{Q}_t^{\text{coop}} &:= (a_t^{S*}, a_{\text{coop}}, a_t^{K*}, \alpha_t, \gamma_t, \mu_t, \theta_K). \end{aligned}$$

Collusion (a_{col}). The family colludes by clandestinely paying the ransom R ; utility discounts the probability of detection and the fine F_{col} :

$$\begin{aligned} \mathcal{U}_t^F(a_{\text{col}}) &= \mathbb{E}_{\theta_K | \mathcal{I}_t^F} \left[\mathbb{E}_{\tilde{A}_t | \mathcal{Q}_t^{\text{col}}} \left[\mathbb{P}(m_t = \text{rel} | \tilde{A}_t, R, \theta_K) \right] \right] V_L \\ &\quad - R - \mathbb{E}_{\tilde{a}_t^F | \mathcal{Q}_t^{\text{col}}} \left[\mathbb{P}(d_t = 1 | \alpha_t, \tilde{a}_t^F, \mathcal{I}_t^F) \right] F_{\text{col}}. \end{aligned} \quad (67)$$

where

$$\begin{aligned} \tilde{p}_{\text{rel},t}(\theta_i) &:= \mathbb{E}_{\tilde{A}_t | \mathcal{Q}_t^{\text{col}}} \left[\mathbb{P}(m_t = \text{rel} | \tilde{A}_t, R, \theta_K = \theta_i) \right], \\ \mathcal{Q}_t^{\text{col}} &:= (a_t^{S*}, a_{\text{col}}, a_t^{K*}, \alpha_t, \gamma_t, \mu_t, \theta_K). \end{aligned}$$

⁹As with $\kappa_c(\theta_K)$ in (61), the exponent $\kappa_F(\theta_F)$ governs marginal sensitivity to operational hardening: families with higher κ_F suffer a sharper increase in the cost of cooperation when γ_t grows, generating *single-crossing* in γ_t if κ_F varies with θ_F . In that case, the cooperation path ceases to be neutral with respect to Θ_F and may become informative about the family's type, opening a **second dimension** of Bayesian learning complementary to the identification of θ_K . When heterogeneity in F is not an object of inference, κ_F and ϕ_F may be taken as constant in θ_F without loss of parsimony.

Compactly,

$$\mathcal{U}_t^F(a_t^F \mid \theta_K, \theta_F) = \begin{cases} \mathcal{U}_t^F(a_{\text{coop}}), & \text{if } a_t^F = a_{\text{coop}}, \\ \mathcal{U}_t^F(a_{\text{col}}), & \text{if } a_t^F = a_{\text{col}}. \end{cases} \quad (68)$$

4.3.3 The State (S), Mechanism Design and Optimal Decision

In each period t , given the accumulated public history $\mathcal{F}_t^{\text{pub}}$ and the belief vector $\mu_t \in \Delta(\Theta_K)$, the State S solves a Bayesian dual-control problem. Its choice combines a discrete institutional decision, $a_t^S \in \{a_{\text{res}}, a_{\text{neg}}\}$, and two continuous instruments, $(\alpha_t, \gamma_t) \in [0, 1]^2$, where α_t measures financial blocking and γ_t operational pressure. The decision is taken with the modal type $\hat{\theta}_t$ and informational precision ι_t defined in (34). The objective is not to maximize a utility of the State, but to minimize an expected social loss that combines human risk, economic costs, operational costs, and the value of learning.

General problem of the State. The admissible set $\Gamma_t(\mu_t)$ incorporates institutional feasibility and the belief-weighted incentive compatibility and participation constraints (IC_t^K , IR_t^K , IR_t^F). The executable problem of the State is:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)} \left\{ \sum_{\theta \in \Theta_K} \mu_t(\theta) L_t(a_t^S, \alpha_t, \gamma_t, \theta, \iota_t) + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}. \quad (69)$$

The first term is the expected material loss; Π_{t,a^S}^S disciplines how far instruments depart from a Bayesian anchor by branch; and $-\psi_H \Delta H_t$ introduces the informational motive. Given the mixed nature of the choice, the solution is obtained by partition: first, the continuous instruments are minimized within each branch $b \in \{R, N\}$; then the optimal rescue and negotiation values are compared.

Objective functions under rescue and negotiation. The instantaneous loss conditional on the latent type is defined by branch:

$$L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) = \begin{cases} V_t^R(\iota_t, \hat{\theta}_t, \theta_K, \alpha_t, \gamma_t), & \text{if } a_t^S = a_{\text{res}}, \\ V_t^N(\theta_K, \alpha_t, \gamma_t), & \text{if } a_t^S = a_{\text{neg}}. \end{cases} \quad (70)$$

Under tactical rescue, the State minimizes the expected cost of lethal failure of the operation plus the deployment cost:

$$V_t^R(\iota_t, \hat{\theta}_t, \theta_K, \alpha_t, \gamma_t) := \omega_k \left[1 - \mathbb{P}(s_t = 1 \mid \iota_t, \hat{\theta}_t, \theta_K) \right] + C_{\text{ops}}(\gamma_t, \alpha_t; \theta_K). \quad (71)$$

Here $\mathbb{P}(s_t = 1 \mid \cdot)$ is the probability of survival under rescue, specified in (35); therefore, lower precision ι_t makes branch R more costly because of greater human risk.

Under negotiation, the State minimizes the expected cost of criminal transfer net of blocking, captivity lethality, and maintenance of the cordon:

$$V_t^N(\theta_K, \alpha_t, \gamma_t) := \omega_p R(1 - \alpha_t) + \omega_k h_2(t \mid \theta_K, \mathcal{C}_t) + C_{\text{maint}}(\gamma_t, \alpha_t; \theta_K). \quad (72)$$

The rate $h_2(t \mid \theta_K, \mathcal{C}_t)$ corresponds to the daily death hazard derived from the competing-risks system (33). Under uncertainty, the expected value of the negotiation branch is:

$$V_t^N(\mu_t, \alpha_t, \gamma_t) := \sum_{\theta \in \Theta_K} \mu_t(\theta) [\omega_p R(1 - \alpha_t) + \omega_k h_2(t \mid \theta, \mathcal{C}_t) + C_{\text{maint}}(\gamma_t, \alpha_t; \theta)]. \quad (73)$$

The optimized floors of each branch are:

$$V_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) = \min_{(\alpha_t, \gamma_t) \in \Gamma_t^R(\mu_t)} \left\{ \sum_{\theta \in \Theta_K} \mu_t(\theta) V_t^R(\iota_t, \hat{\theta}_t, \theta, \alpha_t, \gamma_t) + \Pi_{t,R}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}, \quad (74)$$

$$V_t^{N,*}(\mu_t) = \min_{(\alpha_t, \gamma_t) \in \Gamma_t^N(\mu_t)} \left\{ V_t^N(\mu_t, \alpha_t, \gamma_t) + \Pi_{t,N}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}. \quad (75)$$

Operational cost components. The functions C_{ops} and C_{maint} are quadratic, continuous, and convex in (α_t, γ_t) . Their parameters depend on type θ_K to capture logistical, territorial, and financial heterogeneity:

$$C_{\text{ops}}(\gamma_t, \alpha_t; \theta_K) = c_0(\theta_K) + c_1(\theta_K)\gamma_t + \frac{c_2(\theta_K)}{2}\gamma_t^2 + c_3(\theta_K)\alpha_t + \frac{c_4(\theta_K)}{2}\alpha_t^2 + c_5(\theta_K)\gamma_t\alpha_t, \quad (76)$$

$$C_{\text{maint}}(\gamma_t, \alpha_t; \theta_K) = m_0(\theta_K) + m_1(\theta_K)\gamma_t + \frac{m_2(\theta_K)}{2}\gamma_t^2 + m_3(\theta_K)\alpha_t + \frac{m_4(\theta_K)}{2}\alpha_t^2 + m_5(\theta_K)\gamma_t\alpha_t. \quad (77)$$

The cross terms $c_5(\theta_K)$ and $m_5(\theta_K)$ capture complementarity or substitutability between financial blocking and operational pressure.

Perfect-information anchors and deviation function. To prevent the State from choosing arbitrarily extreme instruments under uncertainty, counterfactual references are constructed by type and branch. These references are not executable policies when the type is unknown; they are anchors indicating what the State would do if it knew θ with certainty:

$$(\alpha_{t,R}^{\text{bench}}(\theta), \gamma_{t,R}^{\text{bench}}(\theta)) \in \arg \min_{(\alpha, \gamma) \in [0,1]^2} V_t^R(\iota_t, \hat{\theta}_t, \theta, \alpha, \gamma), \quad \forall \theta \in \Theta_K, \quad (78)$$

$$(\alpha_{t,N}^{\text{bench}}(\theta), \gamma_{t,N}^{\text{bench}}(\theta)) \in \arg \min_{(\alpha, \gamma) \in [0,1]^2} V_t^N(\theta, \alpha, \gamma), \quad \forall \theta \in \Theta_K. \quad (79)$$

The loss frontiers under perfect information are:

$$\begin{aligned} B_{t,R}(\mu_t) &:= \sum_{\theta \in \Theta_K} \mu_t(\theta) V_t^R(\iota_t, \hat{\theta}_t, \theta, \alpha_{t,R}^{\text{bench}}(\theta), \gamma_{t,R}^{\text{bench}}(\theta)), \\ B_{t,N}(\mu_t) &:= \sum_{\theta \in \Theta_K} \mu_t(\theta) V_t^N(\theta, \alpha_{t,N}^{\text{bench}}(\theta), \gamma_{t,N}^{\text{bench}}(\theta)). \end{aligned} \quad (80)$$

The Bayesian center of branch $b \in \{R, N\}$ is obtained by weighting those references by the current belief:

$$\alpha_{t,b}^\mu := \sum_{\theta \in \Theta_K} \mu_t(\theta) \alpha_{t,b}^{\text{bench}}(\theta), \quad \gamma_{t,b}^\mu := \sum_{\theta \in \Theta_K} \mu_t(\theta) \gamma_{t,b}^{\text{bench}}(\theta). \quad (81)$$

The deviation penalty measures how far the candidate policy moves away from that Bayesian center:

$$\Pi_{t,b}^S(\alpha_t, \gamma_t; \mu_t) = \chi_\alpha (\alpha_t - \alpha_{t,b}^\mu)^2 + \chi_\gamma (\gamma_t - \gamma_{t,b}^\mu)^2, \quad b \in \{R, N\}, \quad \chi_\alpha, \chi_\gamma \geq 0. \quad (82)$$

Thus, $\Pi_{t,b}^S$ fulfills a regulatory function: it allows exploration, but charges a premium when the instrument departs from the Bayesian center that summarizes the perfect-information references.

Role of Shannon entropy. Entropy introduces the active exploration motive. The State recognizes that (α_t, γ_t) modifies the Cox intensities and the outcome probabilities of Table 10; therefore, a policy can be valuable not only because it reduces material loss today, but because it separates likelihoods and reduces uncertainty tomorrow. Belief entropy is:

$$H(\mu_t) := - \sum_{\theta \in \Theta_K} \mu_t(\theta) \log \mu_t(\theta), \quad (83)$$

with the convention $0 \log 0 = 0$. For each pair of public signals (m, d) with $m \in \{\text{Cont}, \text{rel}, \text{res}, \text{pay}, \text{kill}\}$ and $d \in \{0, 1\}$, the State computes the counterfactual posterior that would result from observing them under the candidate policy:

$$\mu_{t+1}^{m,d}(\theta) = \frac{\mu_t(\theta) \mathbb{P}(m_t = m \mid \theta, \alpha_t, \gamma_t, \mathcal{C}_t) \mathbb{P}(d_t = d \mid \theta, \alpha_t, \gamma_t)}{\sum_{\theta' \in \Theta_K} \mu_t(\theta') \mathbb{P}(m_t = m \mid \theta', \alpha_t, \gamma_t, \mathcal{C}_t) \mathbb{P}(d_t = d \mid \theta', \alpha_t, \gamma_t)}. \quad (84)$$

The expected information gain is the expected reduction of entropy over the joint public signal (m_t, d_t) :

$$\Delta H_t(\alpha_t, \gamma_t) = H(\mu_t) - \sum_m \sum_{d \in \{0,1\}} \mathbb{P}(m_t = m, d_t = d \mid \mu_t, \alpha_t, \gamma_t, \mathcal{C}_t) H(\mu_{t+1}^{m,d}), \quad (85)$$

where the joint predictive probability factorizes by conditional independence given the type: $\mathbb{P}(m, d \mid \mu_t, \alpha_t, \gamma_t, \mathcal{C}_t) = \sum_{\theta} \mu_t(\theta) \mathbb{P}(m_t = m \mid \theta, \mathcal{C}_t) \mathbb{P}(d_t = d \mid \theta, \alpha_t, \gamma_t)$. The term

$-\psi_H \Delta H_t$ operates as an informational subsidy: if $\psi_H = 0$, the State is purely myopic; if $\psi_H > 0$, policies that buy learning receive a discount in the loss. When μ_t concentrates on the true type, $H(\mu_t)$ and ΔH_t fall, endogenously extinguishing the exploratory motive.

Discrete decision rule and equilibrium. Once the two floors have been calculated, the State compares:

$$\mathcal{L}_t^{S*} = \min \left\{ V_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t), V_t^{N,*}(\mu_t) \right\}. \quad (86)$$

If $(\alpha_{\text{res}}^*, \gamma_{\text{res}}^*)$ and $(\alpha_{\text{neg}}^*, \gamma_{\text{neg}}^*)$ denote the continuous argminima of branches R and N , the executed policy is:

$$(a_t^{S*}, \alpha_t^*, \gamma_t^*) = \begin{cases} (a_{\text{res}}, \alpha_{\text{res}}^*, \gamma_{\text{res}}^*), & \text{if } V_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) \leq V_t^{N,*}(\mu_t), \\ (a_{\text{neg}}, \alpha_{\text{neg}}^*, \gamma_{\text{neg}}^*), & \text{if } V_t^{R,*}(\iota_t, \hat{\theta}_t, \mu_t) > V_t^{N,*}(\mu_t). \end{cases} \quad (87)$$

In the case of strict numerical equivalence between the loss floors, rescue prevalence is assumed by the weak inequality in the first condition.

4.4 Incentive Compatibility and Individual Rationality Constraints

The State's loss minimization is bounded by three design constraints.

Bayesian incentive compatibility for the kidnapper (IC^K).

$$\mathbb{E}_{\theta \sim \mu_t} \left[V_t^K(a^*(\theta) \mid \theta, \alpha_t, \gamma_t) - V_t^K(a^*(\theta_j) \mid \theta, \alpha_t, \gamma_t) \right] \geq 0, \quad \forall \theta_j \in \Theta_K. \quad (88)$$

The expectation is taken over the true type; θ_j indexes a fixed mimicry deviation. When $\mu_t(\theta^*) \rightarrow 1$, this collapses to standard incentive compatibility of the true type against each rival imitation.

Bayesian participation constraint for the kidnapper (IR^K)—deterrence. This constraint operates under a punitive logic of strategic deterrence. It requires the utility from bloodless release to dominate the best criminal alternative:

$$\mathbb{E}_{\theta \sim \mu_t} \left[U_{\text{rel}}^K(\theta) - \max \left\{ V_{\text{cont},t}^K(\theta, \alpha_t, \gamma_t), U_{\text{kill}}^K(\theta, \theta_S) \right\} \right] \geq 0. \quad (89)$$

This is consistent with the logic of ? adapted to armed conflict: public policy erodes the expected profitability of persisting or escalating violence relative to surrender.

Family participation constraint (IR^F).

$$\mathcal{U}_t^F(a_{\text{coop}}) \geq \mathcal{U}_t^F(a_{\text{col}}). \quad (90)$$

The feasible policy set simultaneously satisfying all three constraints is:

$$\Gamma_t(\mu_t) := \left\{ (a_t^S, \alpha_t, \gamma_t) \in \mathcal{A}^{S, \text{disc}} \times [0, 1]^2 : IC^K, IR^K, IR^F \text{ are satisfied} \right\}. \quad (91)$$

Its non-emptiness—**robust feasibility**—is the primitive condition of the implementability theorem.

5 Bayesian Learning and Type Identification

Bayesian convergence does not imply identification: beliefs may stabilize at a random limit without concentrating on the true type if different profiles generate observationally equivalent signals ????. This section establishes the conditions under which the public history of the mechanism discriminates types in the limit, combining Bayesian coherence of beliefs, statistical separation between signal laws, and the singularity criterion of ??.

The proof strategy requires: **(a)** a well-defined public filtration on which μ_t is a martingale (Definition 5.1 and Definition 5.3); **(b)** a long-run identification condition on the prolonged captivity event $\mathcal{T}$, which prevents the signal laws of different types from explaining the same observable trajectory (Definition 5.6); and **(c)** uniform total-variation separation provided by heterogeneity in $\beta_{K,j}$ (Lemma 5.7), which guarantees mutual singularity of trajectory laws and the collapse of posterior masses on false types.

5.1 Public Filtration, Likelihood, and the Role of IR/IC

Fixing the perfect Bayesian equilibrium of the mechanism, $\mathbb{P}$ is the ex ante law on the sample space Ω and describes the complete experiment before knowing the captor type. First, after observing location z and victim profile θ_V , the type is drawn according to the common structural prior, $\theta_K \sim \mu_0$, with $\mu_0(\theta \mid z, \theta_V) = \mathbb{P}(\theta_K = \theta \mid z, \theta_V)$. Then, conditional on that type, the PBE protocol, public policy, executed actions, MDG filter, outcomes (m_t, d_t) , and evolution of the public history h_t unfold. Under $\mathbb{P}$, the type is ex ante random and the observable trajectory is interpreted as a **mixture** of realizations compatible with different types, with initial marginal conditional on (z, θ_V) given by $\mu_0(\theta \mid z, \theta_V)$.

Definition 5.1 (Public signal and filtration). **(i) Public signal ζ_t .** The **public signal** ζ_t is the random vector summarizing what is observable at the close of period t . The order of its components reproduces the way in which the public history incorporates, at the close of each period, the chosen instruments, the executed actions, and the observed results: $\alpha_t \in [0, 1]$ and $\gamma_t \in [0, 1]$ are the State's continuous **instruments**; $(\tilde{a}_t^F, \tilde{a}_t^K, \tilde{a}_t^S)$ are the **executed discrete actions** that appear as public observables after the MDG

process and do not replace equilibrium intentions a_t^{i*} in the record; $m_t \in \mathcal{M}_{\text{out}}$ is the **physical outcome**; and $d_t \in \{0, 1\}$ is the **detection signal**.

(ii) **Value space.** ζ_t takes values in the product space

$$\mathcal{Z}_t \subseteq \mathbb{R}^2 \times \mathcal{A}^F \times \mathcal{A}^K \times \mathcal{A}^{S, \text{disc}} \times \mathcal{M}_{\text{out}} \times \{0, 1\},$$

where the discrete components are symbols in finite sets and the continuous components are real coordinates. Thus, ζ_t codes what is observable at the close of t and summarizes the public innovation that feeds the next history of the episode.

(iii) **Public filtration** $\{\mathcal{F}_t^{\text{pub}}\}_{t \geq 0}$. The initial instant is fixed by

$$\mathcal{F}_0^{\text{pub}} := \sigma(z, \theta_V),$$

that is, the σ -algebra generated by the observable geographic location at the opening of the episode and the public victim profile. For each integer $t \geq 1$,

$$\mathcal{F}_t^{\text{pub}} := \sigma(z, \theta_V, \zeta_0, \dots, \zeta_{t-1}),$$

that is, the σ -algebra generated by the initial location, the public victim profile, and the signals observed **strictly before** period t begins (equivalently, “at the opening of day t ” in the document’s temporal reading).

(iv) **Measurability.** Each $\zeta_t : (\Omega, \mathcal{F}) \rightarrow (\mathcal{Z}_t, \mathcal{B}(\mathcal{Z}_t))$ is assumed measurable, and z and θ_V are observable at the opening of the episode, so that $\sigma(z, \theta_V, \zeta_0, \dots, \zeta_{t-1})$ is well-defined as a sub- σ -algebra of $\mathcal{F}$.

Definition 5.2 (Predictive likelihood under a counterfactual type). Given a perfect Bayesian equilibrium with profile (χ, s^F, s^K) , the stochastic dynamics of the mechanism, and a counterfactual type $\theta \in \Theta_K$, the **predictive likelihood** of period t is defined by

$$\ell_t(\theta; J) := \mathbb{P}_\theta(\zeta_t \in J \mid \mathcal{F}_t^{\text{pub}}), \quad J \in \mathcal{B}(\mathcal{Z}_t), \quad (92)$$

where $\mathcal{Z}_t$ is the value space of the public signal and J represents an observable subset of that space. Conditioning is interpreted as the conditional expectation of $\mathbf{1}\{\zeta_t \in J\}$ given the public information accumulated up to t , defined up to $\mathbb{P}_\theta$ -null sets through standard regular versions. The likelihoods $\mathcal{L}_{F,t}$ and $\mathcal{L}_{C,t}$ are the operational realization of this kernel for the physical block and the communication block of the period.

The architecture of the model postulates that the likelihood of observable trajectories, denoted by $\mathcal{L}_t(\theta) = \mathcal{L}_{F,t}(\theta)\mathcal{L}_{C,t}(\theta)$, is not a statistically invariant object, but is subject to the **Critique of ?**. With any modification in the mechanism χ , whether through the hardening of instruments or greater operational aggressiveness by the State (a_t^S), agents

reoptimize their strategies in response to the new institutional environment, thereby recomposing the equilibrium map (χ, s^F, s^K) and the predictive likelihood of behaviors encoded in the informational flow ζ_t . In this framework, the conditional kernel ℓ_t acts as the formal bridge between the latent state space $(\theta \in \Theta_K)$ and the observable space (Z_t) . It is essential to note that the **participation (IR)** and **incentive compatibility (IC)** restrictions ?? do not replace this process, but **anchor** the functional form of ℓ_t by restricting the set of admissible strategic profiles; only under a profile satisfying these conditions does the observed likelihood coincide with that induced by an implementable mechanism.

5.2 Belief Martingale and Almost Sure Convergence

The **martingale** structure of $\mu_t(\theta)$ under the ex ante law $\mathbb{P}$ guarantees that each posterior mass may stabilize at a random limit Z_θ as $t \rightarrow \infty$. Long-run identification does not create those limits; rather, it determines their content under the law conditioned on the true type and on the prolonged captivity event, making the masses on false types vanish and the remaining mass concentrate on the realized type. Thus, two problems are separated: convergence of beliefs as coherent sequential prediction, and collapse to the truth when the public experiment discriminates types.

5.3 Formalization of the Initial Belief: the Structural Prior

To guarantee the operational consistency of Bayesian learning, the State must start from a grounded conjecture that collects the historical trajectory of the conflict. Based on the empirical evidence, the initial belief μ_0 is formalized through a **Multinomial Logit (Softmax)** structure, which parametrizes the probability of each type $\theta_K \in \Theta_K$ as a function of the observable attributes at the opening of the episode:

$$\mu_0(\theta \mid z, \theta_V) = \frac{\exp(\varpi_\theta + \eta_{\theta,z} + \xi_{\theta,V})}{\sum_{j \in \Theta_K} \exp(\varpi_j + \eta_{j,z} + \xi_{j,V})}. \quad (93)$$

This specification allows the initial probability mass to be filtered through three structural dimensions linking the already defined types to the kidnapping environment. First, **historical prevalence** (ϖ_θ) acts as a base-frequency intercept for each group $\theta \in \Theta_K$. Second, the **geographic specialization filter** ($\eta_{\theta,z}$) modulates probability as a function of location z : a metropolitan location increases belief mass toward common crime, whereas presence in peripheral regions favors the identification of structures with consolidated territorial control. Finally, the **visibility filter** ($\xi_{\theta,V}$) captures the strategic correlation between the victim profile θ_V and the captor agenda, so the weight function tilts toward insurgent types with political-pressure objectives whenever the victim has a public profile.

Definition 5.3 (Belief process as martingale). For each type $\theta \in \Theta_K$, the posterior belief process is

$$\mu_t(\theta) = \mathbb{P}(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}}), \quad t \geq 0. \quad (94)$$

The process $\{\mu_t(\theta)\}_{t \geq 0}$ is adapted to $\{\mathcal{F}_t^{\text{pub}}\}_{t \geq 0}$, bounded in $[0, 1]$, integrable, and satisfies the tower property

$$\mathbb{E}^{\mathbb{P}}[\mu_{t+1}(\theta) \mid \mathcal{F}_t^{\text{pub}}] = \mu_t(\theta) \quad \mathbb{P}\text{-a.s.}$$

Thus, for every fixed θ , $\{\mu_t(\theta)\}_{t \geq 0}$ is a bounded martingale under the ex ante law $\mathbb{P}$.

Lemma 5.4. Asymptotic stability of beliefs. (Doob martingale convergence theorem) *Under the ex ante law $\mathbb{P}$, the belief process $\{\mu_t(\theta)\}_{t \geq 0}$ is a **bounded martingale** that converges almost surely to a limit random variable Z_θ with $Z_\theta \in [0, 1]$ $\mathbb{P}$ -a.s.*

Proof. By Definition 5.3, $\mu_t(\theta) = \mathbb{P}(\theta_K = \theta \mid \mathcal{F}_t^{\text{pub}})$. The tower property gives

$$\mathbb{E}^{\mathbb{P}}[\mu_{t+1}(\theta) \mid \mathcal{F}_t^{\text{pub}}] = \mu_t(\theta). \quad (95)$$

Since $\mu_t(\theta)$ is a conditional probability,

$$0 \leq \mu_t(\theta) \leq 1 \quad \mathbb{P}\text{-a.s.} \quad (96)$$

Thus the process is uniformly bounded in L^1 . Doob's martingale convergence theorem implies the existence of Z_θ such that

$$\mu_t(\theta) \longrightarrow Z_\theta \quad \mathbb{P}\text{-a.s.} \quad (97)$$

and the bound passes to the limit, so $Z_\theta \in [0, 1]$ almost surely. $\square$

5.4 Identification and Concentration on the True Type

Lemma 5.4 guarantees that, under $\mathbb{P}$, each belief process $\mu_t(\theta)$ is a bounded martingale converging almost surely to a limit Z_θ . However, statistical convergence *per se* does not imply that the limit is binary ($Z_\theta \in \{0, 1\}$). In Bayesian learning models, several types may remain indistinguishable if the public signal structure does not allow them to be discriminated in the limit ?.

Definition 5.5 (Prolonged captivity event). Let $\mathcal{T}$ be the event in which the captivity episode remains active indefinitely:

$$\mathcal{T} := \{\omega \in \Omega : \iota_t(\omega) = 1 \text{ for all } t \geq 0\}.$$

Equivalently, $\mathcal{T}$ is the tail event in which no absorbing outcome terminates the episode in finite time. It is the asymptotic regime under which the public experiment supplies an

unbounded sequence of signals.

Definition 5.6 (Asymptotic identification on $\mathcal{T}$). The mechanism is **asymptotically identifiable** on $\mathcal{T}$ if, for the true type θ_K^* and every false type $\theta \neq \theta_K^*$, the laws induced by the public signal process $\{\zeta_t\}_{t \geq 0}$ under θ_K^* and under θ are mutually singular when restricted to $\mathcal{T}$. Equivalently, there exists a tail event A_θ such that

$$\mathbb{P}_{\theta_K^*}(A_\theta \mid \mathcal{T}) = 1, \quad \mathbb{P}_\theta(A_\theta \mid \mathcal{T}) = 0.$$

Lemma 5.7 (Hazard separation and total variation). *Suppose there exist two distinct types $\theta \neq \theta'$ and a set of periods $\mathcal{S}_{\theta, \theta'} \subseteq \mathbb{N}$ infinite on $\mathcal{T}$ such that the competing-risk coefficients differ on at least one closing channel, $\beta_{K,j}(\theta) \neq \beta_{K,j}(\theta')$ for some $j \in \{1, 2, 3\}$, and the policy path keeps the corresponding channel active. Then there exists $\varepsilon_{\theta, \theta'} > 0$ such that, for all $t \in \mathcal{S}_{\theta, \theta'}$,*

$$\left\| p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}}) - p_{\theta'}(\cdot \mid \mathcal{F}_t^{\text{pub}}) \right\|_{\text{TV}} \geq \varepsilon_{\theta, \theta'}. \quad (98)$$

Consequently, the induced trajectory laws are mutually singular under the repeated experiment on $\mathcal{T}$.

Proof. Differences in $\beta_{K,j}$ change the proportional hazards $\lambda_j(t \mid \mathcal{C}_t, \theta)$ and therefore the one-period predictive distribution of $Y_t = (m_t, d_t)$. When the channel remains active along infinitely many periods, the one-step distributions remain separated in total variation by a positive amount on an infinite subsequence. Coupling this recurrent separation with the product experiment and Kakutani's criterion yields mutual singularity of the infinite trajectory laws. $\square$

5.5 MDG Process, Materialization of Actions, and the True-Types Theorem

The theorem is stated over laws induced by the full PBE, including the intention-to-execution layer. Conditioning on $\mathcal{F}_t^{\text{pub}}$ means conditioning on histories where the realized actions $\tilde{a}_s^i$ have already been observed, not only the intended equilibrium actions a_s^{i*} . The convergence $\mu_t(\theta_K^*) \rightarrow 1$ on $\mathcal{T}$ is convergence of beliefs about the type **after** each day's public actions have materialized.

Theorem 5.8 (Identification of rational types in unknown environments). *Suppose the mechanism is asymptotically identifiable on $\mathcal{T}$ under Definition 5.6, the prior has full support, $\mu_0(\theta) > 0$ for all $\theta \in \Theta_K$, and beliefs are updated consistently with the PBE as in Definition 5.3. Then, on the event $\mathcal{T}$,*

$$\mu_t(\theta_K^*) \longrightarrow 1 \quad \text{and} \quad \mu_t(\theta) \longrightarrow 0 \quad \forall \theta \neq \theta_K^*,$$

almost surely under the law of the true type restricted to $\mathcal{T}$. Equivalently, μ_t converges weakly to the Dirac measure $\delta_{\theta_K^*}$.

Proof. By Lemma 5.4, $\{\mu_t(\theta)\}_{t \geq 0}$ is a bounded martingale and converges almost surely to a limit $Z_\theta \in [0, 1]$. Asymptotic identification implies that for every false type $\theta \neq \theta_K^*$ there is a tail event A_θ with probability one under the true type on $\mathcal{T}$ and probability zero under θ . The likelihood ratio of θ against θ_K^* therefore converges to zero on A_θ by the standard Bayesian consistency argument under mutual singularity. Since the prior has full support,

$$\mu_t(\theta) = \frac{\mu_0(\theta)L_t(\theta)}{\sum_{\theta' \in \Theta_K} \mu_0(\theta')L_t(\theta')} \rightarrow 0 \quad \forall \theta \neq \theta_K^*.$$

Finiteness of Θ_K allows the convergence to hold simultaneously for all false types. Since posterior masses sum to one, the remaining mass converges to the true type: $\mu_t(\theta_K^*) \rightarrow 1$. $\square$

Remark 5.9 (Finite horizon and interpretation of the theorem). The theorem is an asymptotic identification result on the event $\mathcal{T}$. In a finite episode, especially one that ends quickly, the mechanism may generate only a finite number of observations; in that case the relevant conclusion is martingale convergence and stabilization of beliefs, not necessarily point identification of the true type. Point identification in short episodes would require repeated independent episodes or an extension of the horizon.

5.6 Properties, Limitations, and Dynamics of Learning

The posterior ratio between two types evolves according to the accumulation of log likelihood ratios. For a true high-risk type and a low-risk alternative, the hazard ratio

$$\frac{\lambda_j(t \mid \theta_{high})}{\lambda_j(t \mid \theta_{low})} = \exp(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})) \quad (99)$$

summarizes the structural wedge that feeds learning. When realized outcomes repeatedly favor the high-risk law, the posterior tilts toward θ_{high} ; when the likelihoods remain close, learning is slow or incomplete.

Under the hypotheses of Theorem 5.8, this evidence flow is not a random walk on the belief simplex but a systematic tilt toward θ_K^* . When the materialized action coincides with rational equilibrium, the posterior concentrates through direct recognition of preferences and validation of IC. When the *MDG process* introduces a deviation, the Bayesian filter compares the hazard structures compatible with the observed trajectory and assigns mass to the type whose risk technology is most coherent with it.

Outside these limits, the mechanism transcends mere statistical convergence to reach asymptotic identification. Perfect Bayesian Equilibrium guarantees, by Lemma 5.4, that

μ_t converges to some random limit; but only when Definition 5.6 is satisfied does that limit collapse to the Dirac measure $\delta_{\theta_K^*}$.

5.7 Equilibrium of the Mechanism, Integration of PBE and Implementability

The learning result must be read together with equilibrium existence and implementability. Bayesian updating is meaningful only on histories generated by a strategy profile that is sequentially rational and compatible with the informational structure of the mechanism. For this reason, the next definitions state the PBE object and the implementability property that connects the social choice rule to incentive and participation constraints.

Definition 5.10 (Perfect Bayesian Equilibrium of the mechanism). A **Perfect Bayesian Equilibrium** of the mechanism is a pair (s, μ) , where $s = (\chi, s^K, s^F)$ is a strategy profile for the State, the captor, and the family, and $\mu = \{\mu_t\}_{t \geq 0}$ is a system of beliefs, such that:

- (a) **Sequential rationality.** At every information set reached with positive probability, each agent's strategy is optimal given beliefs μ_t and the continuation strategies of the other agents.
- (b) **Bayesian consistency.** At every reached node,

$$\mu_t(\theta) = \mathbb{P}^s(\theta_K = \theta \mid \sigma(z, \theta_V, \zeta_0, \dots, \zeta_{t-1})), \quad (100)$$

and off-path beliefs are admissible and compatible with sequential rationality.

Definition 5.11 (Implementable mechanism). Given a PBE (s^*, μ^*) , the mechanism **implements** the social choice rule f_t if the equilibrium outcome induced by s^* coincides with the State's policy rule that minimizes expected social loss subject to incentive compatibility and participation constraints. In particular, the equilibrium policy must satisfy the Bayesian incentive compatibility constraint for the captor IC^K , the participation constraint for the captor IR^K , and the participation constraint for the family IR^F .

5.8 Economic Reading of the Implementability Result

The implementability analysis involves two logically distinct layers. The first, formalized in Definition 5.10, is one of **strategic consistency**: the existence of a profile (s, μ) in which the State, the captor, and the family are sequentially rational given μ_t , and in which beliefs μ_t are consistent with Bayes' rule at reached nodes. The second, articulated in Definition 5.11, is one of **implementability**: that this same profile also satisfies IC^K , IR^K , and IR^F , so that the social choice rule f_t , the rescue or negotiation policy plus instruments (α_t, γ_t) that minimize expected loss, is sustainable as an equilibrium *outcome*.

without any agent having an incentive to deviate. The PBE fixes who acts, with what beliefs, and with what rationality; implementability determines whether the public *outcome* desired by the planner can be maintained when types are private and participation is voluntary.

Theorem 5.12 establishes that both layers are compatible under verifiable conditions. The central hypothesis is **robust feasibility** (v): the non-emptiness of the set $\Gamma_t(\mu_t)$ of policies simultaneously admissible under IC^K , IR^K , and IR^F at each date and for each belief. Since Γ_t is compact, with continuous and convex sections in the continuous instruments, and since the correspondence $\mu_t \mapsto \Gamma_t(\mu_t)$ is continuous, the State's restricted program admits a solution by the Weierstrass theorem and the optimal correspondence is upper hemicontinuous. Quasiconvexity of the restricted objective (hypothesis (vii)) makes that correspondence convex-valued, which enables Kakutani's fixed point theorem (?) and guarantees the existence of the PBE. Full support of the MDG process (hypothesis (iii)) ensures that all relevant nodes are reached with positive probability, making Bayesian updating operative on the path and reducing *off-path* indeterminacy. The result does not add a PBE to a previously designed mechanism: the social rule that minimizes expected loss subject to Γ_t emerges directly as an equilibrium trajectory.

Theorem 5.12 and Theorem 5.8 guarantee substantively distinct properties that operate on complementary planes. The **existence of the PBE** (enabled by hypotheses (ii), (v), and (vii)) is a necessary condition for coherence of the model as a game: without it there is no equilibrium path on which to formulate optimal policy. **Nash–Bayes implementability** with IC^K , IR^K , and IR^F is the design condition: it delimits the set of sustainable allocations when the captor's type is private information and the family's participation cannot be forced. **Asymptotic learning** (Lemma 5.7, Theorem 5.8) is a property of the informational flow: under uniform total-variation separation between types, beliefs converge almost surely to the true type in the prolonged captivity regime $\mathcal{T}$. These three properties are logically independent: a mechanism may implement a social rule and nevertheless generate observationally equivalent types in the short run; Bayesian convergence, in turn, neither implies nor requires that the implemented policy be the one that minimizes expected loss. Theorem 5.12 should therefore be read as a *conditional* guarantee on primitive feasibility (v): it states that the policy defined by the State's restricted program can be sustained as a PBE of the given mechanism, not that every admissible policy is sustainable independently of the incentive structure.

Theorem 5.12 (Conditional implementability of the mechanism in PBE). *Consider the mechanism of Definition 5.10 under the following conditions:*

- (i) $|\Theta_K| < \infty$, with strictly positive prior $\mu_0(\theta) > 0$ for every $\theta \in \Theta_K$.
- (ii) The action spaces $\mathcal{A}^i$ are finite for every $i \in \{K, S, F\}$ and the game tree has a finite

number of nodes in each period t , so that each agent's behavioral strategy space is a finite product of simplexes.

- (iii) The implementation law $\mathbb{P}_i^{\text{imp}}$ of the MDG process assigns strictly positive probability to every action $a \in \mathcal{A}^i$ for every plan a_t^{i*} and state X_t (full support, (17)).
- (iv) The instrument sets are $\mathcal{A}^\alpha = [0, 1]$ and $\mathcal{A}^\gamma = [0, 1]$, compact and convex; the functions L_t , G , and $\mathcal{U}_t^i$ are measurable, uniformly bounded in $\mathbb{T}$, and continuous in the continuous decision variables.
- (v) **Robust feasibility.** For each t and each $\mu_t \in \Delta(\Theta_K)$, the feasible set $\Gamma_t(\mu_t)$ defined in (91) is non-empty and compact. For each discrete branch a_t^S , its continuous section

$$\Gamma_t^{a^S}(\mu_t) := \{(\alpha_t, \gamma_t) : (a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)\}$$

is convex, and the correspondence $\mu_t \mapsto \Gamma_t(\mu_t)$ is **continuous** (upper and lower hemicontinuous) and compact-valued on $\Delta(\Theta_K)$.¹⁰ Non-emptiness is a primitive condition of simultaneous compatibility between IC^K , IR^K , and IR^F .

- (vi) **Parametric separability.** For every $t \in \mathbb{T}$, the captor's value function admits the representation

$$V_t^K(a^*(\theta') \mid \theta_K, \alpha_t, \gamma_t) = \sum_j w_j(a^*(\theta'), \alpha_t, \gamma_t) \cdot \exp(\beta_{K,j}(\theta_K) \cdot z),$$

where the weights $w_j(\cdot)$ depend on the action path $a^*(\theta')$ and on the instruments, but not on θ_K . This condition makes it possible to audit the economic direction of incentives; it does not replace the verification of IC^K in (v) nor the statistical identification of Lemma 5.7.

- (vii) **Quasiconvexity of the restricted objective.** For each t , each $\mu_t \in \Delta(\Theta_K)$, and each discrete branch $a_t^S \in \mathcal{A}^{S,\text{disc}}$, the functional

$$(\alpha_t, \gamma_t) \mapsto \sum_{\theta_K \in \Theta_K} \mu_t(\theta_K) L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) + \Pi_{t,a^S}^S(\alpha_t, \gamma_t; \mu_t) - \psi_H \Delta H_t(\alpha_t, \gamma_t)$$

is quasiconvex in (α_t, γ_t) over $\Gamma_t^{a^S}(\mu_t)$. Combined with the convexity of $\Gamma_t^{a^S}(\mu_t)$ (hypothesis (v)), this condition guarantees that the restricted optimum correspondence $\Psi_t(\mu_t) := \arg \min_{\Gamma_t^{a^S}(\mu_t)} \{\cdot\}$ has convex values, making Kakutani's fixed point theorem applicable to the State's program.

Then there exists a pair (s^*, μ^*) , with $s^* = (\chi^*, s^{K*}, s^{F*})$, that:

¹⁰This is the standard framework for applying Berge's maximum theorem to the restricted policy; see ?.

- (a) Constitutes a **Perfect Bayesian Equilibrium** in the sense of Definition 5.10: strategies in s^* are mutual best responses given μ^* , and beliefs μ_t^* are Bayes-consistent at nodes reached with positive probability.
- (b) **Implements the social choice rule** f_t : on the equilibrium path, $\chi_t^*(h_t) = (a_t^{S^*}, \alpha_t^*, \gamma_t^*)$ coincides with the allocation that minimizes expected social loss (86) subject to $\Gamma_t(\mu_t^*)$, according to rule (87).
- (c) **Satisfies** IC^K , IR^K , and IR^F in the profile (s^*, μ^*) .

Proof. The proof proceeds in five steps: existence of the fixed point in the restricted game, Bayesian consistency on the path, implementation of the social rule, incentive compatibility, and participation constraints.

1. Existence of equilibrium. By hypothesis (ii) and the finite horizon of Definition 5.10, each agent's behavioral strategy space Σ^i is a finite product of simplexes, compact and convex. The State's restriction is formulated through the feasible correspondence $\Gamma_t(\mu_t)$: by hypothesis (v), Γ_t is compact and non-empty, with continuous convex sections in each discrete branch and **continuous** in μ_t .

Given a profile s , the on-path beliefs $\mu_t(s)$ are measurable at reached nodes. The State's program over $(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t)$ has a continuous objective in (α_t, γ_t) by hypothesis (iv) and a compact feasible set; by the maximum theorem (Berge; see ?), the optimal value is continuous in μ_t and the correspondence of optimal sets is upper hemicontinuous and compact. By hypothesis (vii), the objective functional is quasiconvex in (α_t, γ_t) over each convex section $\Gamma_t^{a^S}(\mu_t)$; hence the argmin over a non-empty compact convex set is convex, so the planner's best-response correspondence in mixed strategies has convex values.

The expected utilities of K and F are multilinear in behavioral strategies. Together, the restricted best-response correspondence

$$\text{BR}(s) := \text{BR}_\Gamma^S(s^{-S}) \times \text{BR}^K(s^{-K}) \times \text{BR}^F(s^{-F})$$

is upper hemicontinuous, with non-empty and convex values over Σ ?. By Kakutani's fixed point theorem ?, there exists s^* such that $s^* \in \text{BR}(s^*)$. Hypothesis (v) explicitly carries the simultaneous compatibility between IC^K , IR^K , and IR^F ; for this reason the result is conditional and not an unconditional proof of non-emptiness of $\Gamma_t(\mu_t)$.

2. Bayesian consistency on the path. Hypothesis (iii) implies that $\mathbb{P}_i^{\text{imp}}(\tilde{a}_t^i = a \mid a_t^{i*}, X_t) > 0$ for every $a \in \mathcal{A}^i$, so the ex ante law $\mathbb{P}^{s^*}$ assigns strictly positive probability to realized actions compatible with the physical dynamics and with nodes reachable under

the public history. At those nodes, Bayes' formula

$$\mu_t^*(\theta) = \mathbb{P}^{s^*}(\theta_K = \theta \mid \sigma(z, \theta_V, \zeta_0, \dots, \zeta_{t-1}))$$

is well defined and satisfies condition (b) of Definition 5.10. For off-path nodes, beliefs are taken to be admissible and compatible with sequential rationality in the remaining subgames. The pair (s^*, μ^*) thus constructed is a PBE in the sense of ??; uniqueness of off-path beliefs is not asserted.

3. Implementation of the social choice rule. At the fixed point, S 's strategy solves at every on-path node the restricted program

$$(a_t^{S^*}, \alpha_t^*, \gamma_t^*) \in \arg \min_{(a_t^S, \alpha_t, \gamma_t) \in \Gamma_t(\mu_t^*)} \left\{ \sum_{\theta_K \in \Theta_K} \mu_t^*(\theta_K) L_t(a_t^S, \alpha_t, \gamma_t, \theta_K, \iota_t) + \Pi_{t, a^S}^S(\alpha_t, \gamma_t; \mu_t^*) - \psi_H \Delta H_t(\alpha_t, \gamma_t) \right\}$$

This solution coincides with program (86) subject to IR/IC and rule (87); defining $f_t(\mathcal{I}_t^S) := \chi_t^*(h_t)$, the equilibrium policy implements the social choice rule on the path.

4. Incentive compatibility (IC^K): reformulation by cycles (?). By Step 3, the equilibrium policy belongs to $\Gamma_t(\mu_t^*)$ at every on-path node. By hypothesis (v) and Definition 5.11, $\Gamma_t(\mu_t^*)$ incorporates IC^K as a **constitutive** condition: for every rival imitation $\theta_j \in \Theta_K$,

$$\mathbb{E}_{\theta \sim \mu_t^*} \left[V_t^K(a^*(\theta) \mid \theta, \alpha_t^*, \gamma_t^*) - V_t^K(a^*(\theta_j) \mid \theta, \alpha_t^*, \gamma_t^*) \right] \geq 0.$$

Thus IC^K is established directly by membership in Γ_t , without appealing to the State's loss-minimization program (86).

For an equivalent algebraic check in the finite menu, let $\Theta_K = \{\theta_1, \dots, \theta_n\}$ and define

$$\Delta_{ij}^t := V_t^K(a^*(\theta_j) \mid \theta_i, \alpha_t^*, \gamma_t^*) - V_t^K(a^*(\theta_i) \mid \theta_i, \alpha_t^*, \gamma_t^*).$$

The Bayesian version of IC^K is equivalent to requiring

$$\mathbb{E}_{\theta_i \sim \mu_t^*} [\Delta_{ij}^t] \leq 0 \quad \forall j = 1, \dots, n.$$

In particular, for every cycle $(\theta_{k_0}, \theta_{k_1}, \dots, \theta_{k_m}, \theta_{k_0})$ in Θ_K , expected verification takes the finite form

$$\sum_{\ell=0}^m \mu_t^*(\theta_{k_\ell}) \Delta_{k_\ell, k_{\ell+1}}^t \leq 0 \quad (\text{indices mod } m+1). \quad (101)$$

5. Participation constraints (IR^K, IR^F). By construction in Step 3, the equilibrium instruments belong to $\Gamma_t(\mu_t^*)$, a set that by hypothesis (v) simultaneously guarantees

IR^K (89) and IR^F (90) in the equilibrium profile. Having completed the five steps, (s^*, μ^*) is a PBE that implements f_t and satisfies IC^K , IR^K , and IR^F simultaneously. $\square$

Remark 5.13 (Separation of results: implementability and asymptotic identification). The present theorem and Theorem 5.8 establish distinct properties that complement one another. Theorem 5.12 guarantees, under robust feasibility (v), that the mechanism has a PBE in which the equilibrium policy implements the social choice rule; IC^K enters through the definition of $\Gamma_t(\mu_t)$ and Definition 5.11, and the cyclic form (101) is a verifiable reformulation when grouping pairs is convenient. The dynamic revelation principle of ? typically assumes an ordered continuum $\Theta_K \subseteq \mathbb{R}$. Parametric separability (vi) helps interpret and audit incentives from $\beta_{K,j}$, but does not replace (v). Theorem 5.8, in turn, guarantees that *under that same equilibrium*, if heterogeneity in $\beta_{K,j}$ generates uniform total-variation separation between types (Lemma 5.7), then beliefs converge almost surely to the true type θ_K^* in regime $\mathcal{T}$.

6 Mechanism Calibration and Quantitative Results

This section calibrates the dynamic mechanism of Section 4 for the four captor types, $\Theta_K = \{\text{DC}, \text{PAR}, \text{ELN}, \text{FARC}\}$. Following Bernal’s calibration structure, the exercise studies the evolution of beliefs $\mu_t(\theta)$, the response of instruments (α_t, γ_t) , the information gain ΔH_t , and compliance with individual rationality and incentive compatibility constraints. All scenarios are evaluated with the same seed and a fixed horizon of **150 periods**. This horizon is a simulation parameter distinct from the endogenous stopping time τ defined in (16); after the first terminal outcome, trajectories represent the counterfactual extension conditional on non-absorption that makes it possible to compare learning across types.

The scenarios are: *I*, common crime (DC), hard State, high-wealth family, private victim, and metropolitan region; *II*, paramilitary groups (PAR), lax State, high-wealth family, private victim, and East–Jungle region; *III*, ELN, lax State, low-wealth family, public victim, and Caribbean region; and *IV*, FARC, hard State, high-wealth family, private victim, and East–Jungle region. The four cases preserve the available calibration parameters and trajectories; only the common cutoff $t \leq 150$ is applied, without extrapolating observations.

6.1 Bayesian Learning and Type Separation

Figure 14 presents the evolution of beliefs. In DC, the posterior of the true type goes from 0.350 to practically one. In PAR and ELN it starts from 0.050 and reaches, respectively, 0.999922 and 0.999911 at $t = 150$. These three scenarios exhibit effective separation in finite horizon. The FARC case functions as a stress test: the initial mass on FARC is

0.150, but the trajectory ends up concentrating on DC, with $\mu_{150}(\text{DC}) = 0.99999999$. Therefore, this calibration does not correctly identify FARC; far from hiding the result, the exercise shows that posterior consistency requires the observed signals to maintain informative separation from rival types.

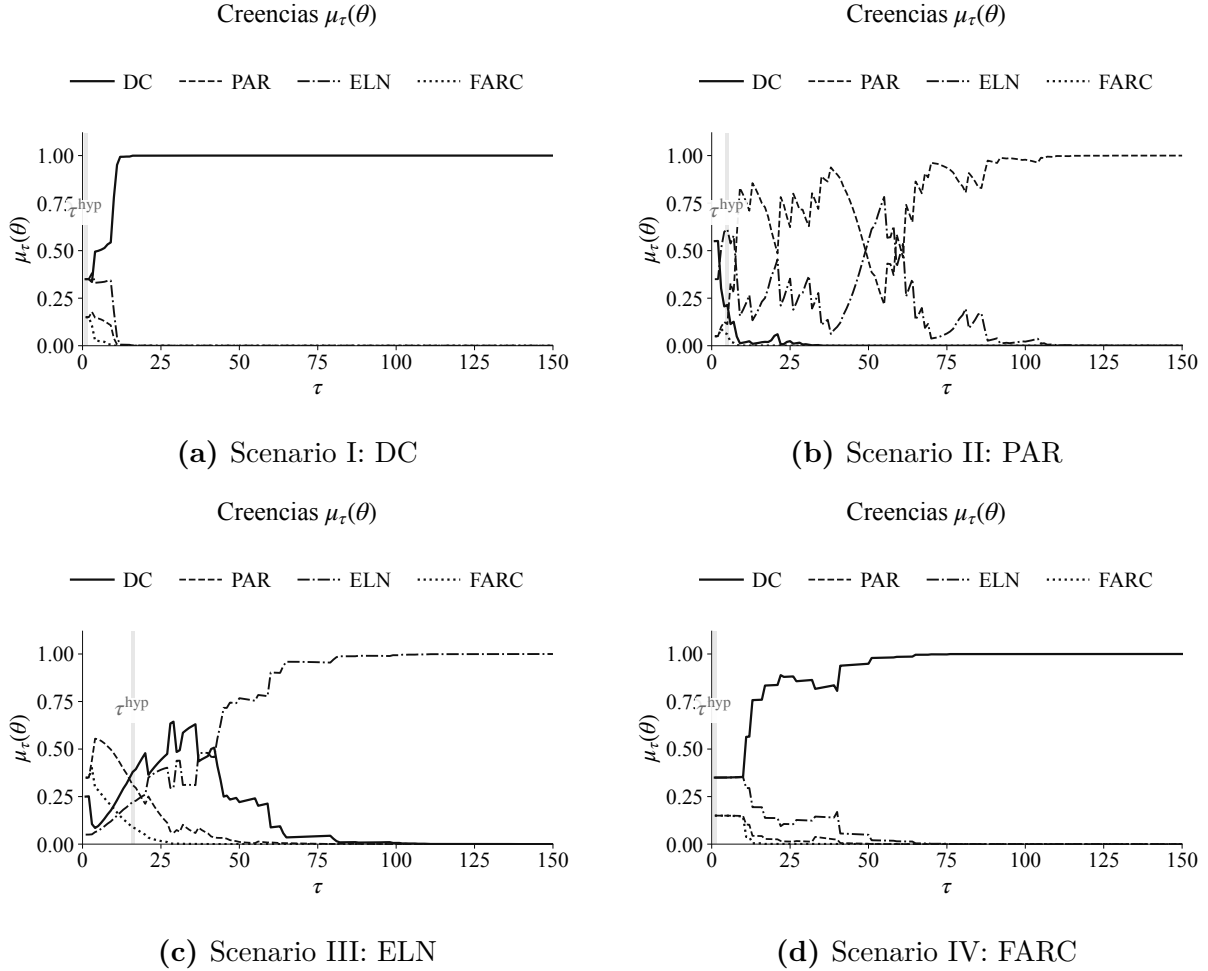


Figure 14. Bayesian beliefs for the four captor types, $t \leq 150$.

Note: the gray band indicates the first outcome different from continuation. The posterior trajectory corresponds to the counterfactual extension conditional on non-absorption.

6.2 Operational Pressure and Financial Interdiction

Average operational pressure is 0.804 in DC, 0.871 in PAR, 0.903 in ELN, and 0.812 in FARC. Relative to rescue references, $\gamma^R = (0.80, 0.86, 0.92, 0.80)$, the trajectories show that the State does not mechanically replicate a pure branch. The policy combines the material effect of the intervention with the expected value of information, so intensity changes with the posterior and with the institutional conditions of the episode.

Average financial interdiction is 0.790 in DC, 0.921 in PAR, 0.910 in ELN, and 0.803 in FARC. Rescue reference values are $\alpha^R = (0.78, 0.84, 0.90, 0.78)$. The highest average

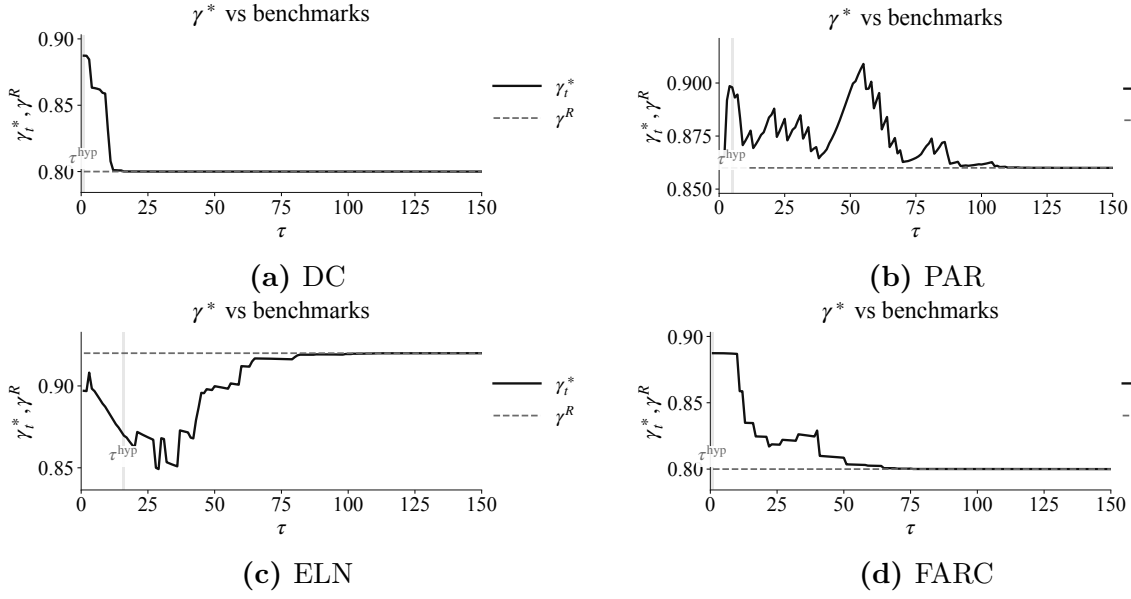


Figure 15. Optimal operational pressure γ_t^* and mechanism references.

intensity appears in PAR, where the combination of high family wealth, private victim, and East-Jungle environment preserves the strategic value of restricting payments. In ELN, the instrument also remains high, but operates in an episode with a public victim and low payment capacity.

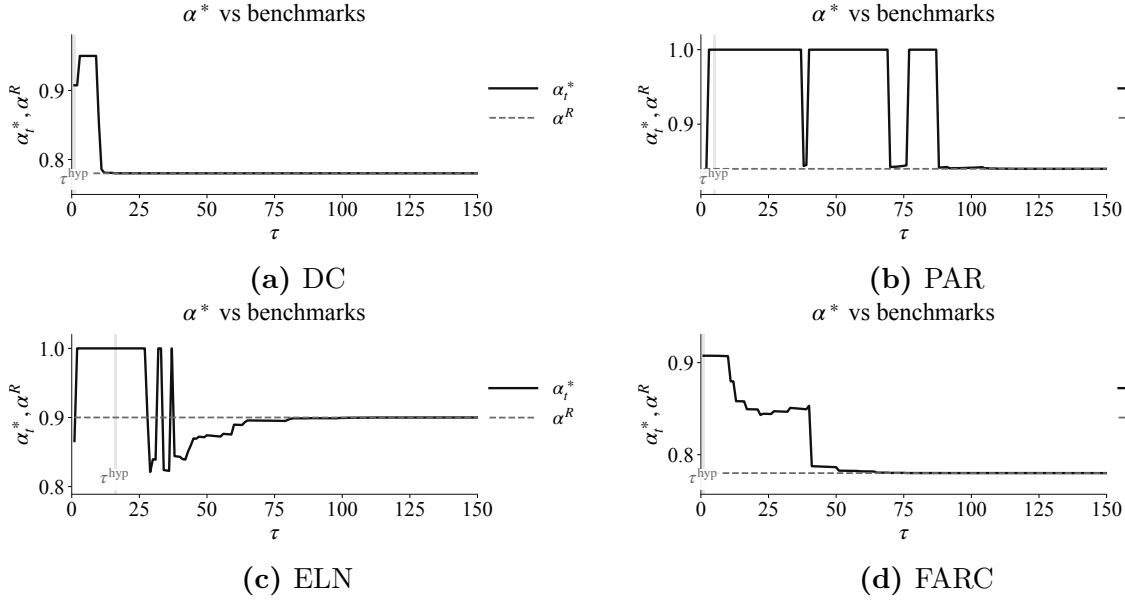


Figure 16. Optimal financial interdiction α_t^* and mechanism references.

6.3 Information Gain and Viability of the Mechanism

Information gain is concentrated in the first periods. Its maximum is 0.0381 at $t = 7$ for DC, 0.0353 at $t = 6$ for PAR, 0.0430 at $t = 10$ for ELN, and 0.0319 at $t = 10$ for

FARC. As ΔH_t approaches zero, the marginal contribution of new observations declines and policy becomes governed mainly by material costs and accumulated belief.

The checks IR^K , IC^K , and IR^F are satisfied in all 150 cycles of the four scenarios. This result establishes internal viability of the simulated trajectories, but by itself does not guarantee correct identification: FARC satisfies the constraints and nevertheless its signals induce posterior classification toward DC. The distinction confirms that mechanism implementability and statistical separation of types are related but conceptually different conditions.

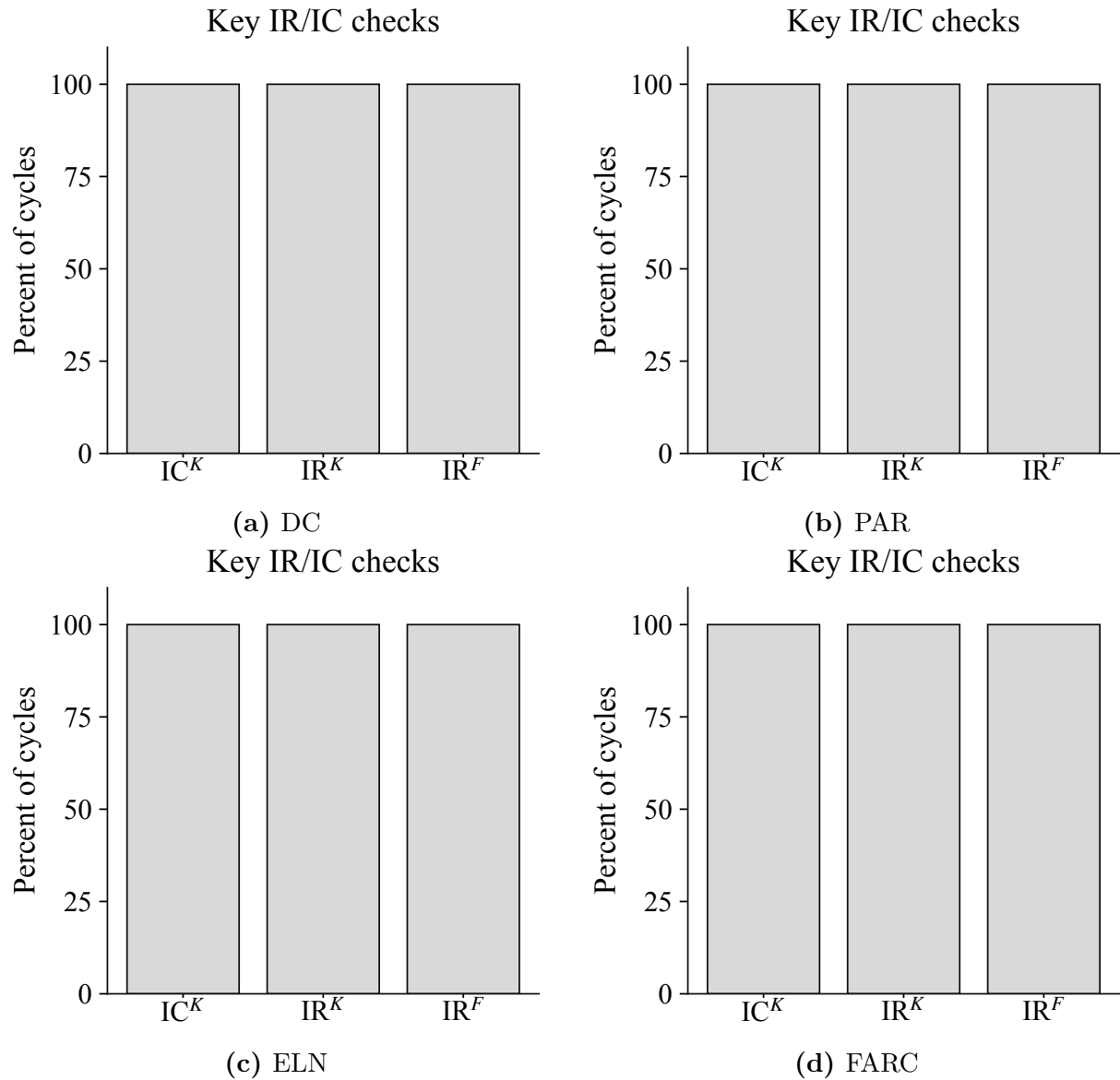


Figure 17. Compliance with participation and incentive constraints.

6.4 Quantitative Synthesis

The first terminal outcomes occur at $t = 16$ for DC (rescue), $t = 5$ for PAR (death), $t = 16$ for ELN (release), and $t = 32$ for FARC (rescue). Table 7 summarizes the results computed over the same 150-period horizon.

Table 7. Calibration summary for the four types, $t \leq 150$

Scenario	$\mu_1(\theta^*)$	$\mu_{150}(\theta^*)$	$\bar{\alpha}^*$	$\bar{\gamma}^*$	τ^{hyp}	Outcome
DC	0.350	1.000000	0.790	0.804	16	Rescue
PAR	0.050	0.999922	0.921	0.871	5	Death
ELN	0.050	0.999911	0.910	0.903	16	Release
FARC	0.150	0.000000	0.803	0.812	32	Rescue

Note: in FARC the posterior mass concentrates on DC at the end of the horizon; therefore, the value close to zero is not interpreted as identification of the true type, but as evidence of lack of separation under that trajectory.

Taken together, the comparison of the four cases shows that the same mechanism can be implementable and generate differentiated policies, but posterior concentration depends on the informational capacity of the signals. DC, PAR, and ELN produce clear separation at 150 periods; FARC reveals a zone of empirical fragility that requires revising the calibration of likelihoods or incorporating additional signals before sustaining robust identification for that type.

7 Policy Implications

The results of the preceding sections have direct consequences for the design of antiskidnapping policy. Proposition 4.4 establishes that the effect of γ_t on expected captivity duration is governed by the sign of $\kappa_h(\theta_K, t)$: only when pressure accelerates closure outcomes more than it inhibits the payment channel is the episode shortened; otherwise, the State may paradoxically prolong it. Proposition 7.1 translates this heterogeneity into a Bayesian calibration rule: the State uses Bayes updating to discipline its policy, so as evidence ($\mathcal{I}_t$) increases weight on the high-risk type, the planner intensifies γ_t^* to exploit that sensitivity and force early closure. The magnitude of the response is proportional to the structural separation $\beta_{K,j}(\theta_{\text{high}}) - \beta_{K,j}(\theta_{\text{low}})$, which acts as a policy amplifier when new information arrives.

Proposition 7.1 (Optimal operational pressure and Bayesian type identification). *Suppose $\Theta_K = \{\theta_{\text{low}}, \theta_{\text{high}}\}$ with $\beta_{K,j}(\theta_{\text{high}}) > \beta_{K,j}(\theta_{\text{low}})$ for $j \in \{2, 3\}$ (the high-risk type exhibits greater propensity to rescue and death), and that the State's objective (86) is strictly convex in γ_t in the interior of $\Gamma_t(\mu_t)$. Then optimal operational pressure γ_t^* is*

strictly increasing in the posterior belief on the high-risk type:

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} > 0.$$

The elasticity of γ_t^* with respect to $\mu_t(\theta_{high})$ depends on the differential ratio of marginal losses between types, which in turn is proportional to

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})), \quad j \in \{2, 3\}.$$

Consequently, the range of variation of optimal pressure in response to Bayesian updates is larger the more separated the structural parameters $\beta_{K,j}$ are across types.

Proof. The proof proceeds in four steps: construction of the State's program, first-order condition, application of the implicit function theorem, and derivation of structural elasticity.

Step 1. State program and interior condition. With $\Theta_K = \{\theta_{low}, \theta_{high}\}$ and $\mu_t(\theta_{low}) = 1 - \mu_t(\theta_{high})$, the State objective with respect to γ_t , fixing μ_t and the remaining instruments, reduces to

$$\mathcal{W}(\gamma_t, \mu_t) := \mu_t(\theta_{high}) L_t(\gamma_t, \theta_{high}) + (1 - \mu_t(\theta_{high})) L_t(\gamma_t, \theta_{low}) + \chi_\gamma (\gamma_t - \gamma_t^\mu)^2. \quad (102)$$

By the strict convexity hypothesis in γ_t over the interior of $\Gamma_t(\mu_t)$, $\mathcal{W}$ has a unique interior minimizer $\gamma_t^*(\mu_t)$ satisfying

$$\Phi(\gamma_t, \mu_t) := \mu_t(\theta_{high}) \frac{\partial L_t}{\partial \gamma_t}(\theta_{high}) + (1 - \mu_t(\theta_{high})) \frac{\partial L_t}{\partial \gamma_t}(\theta_{low}) + 2\chi_\gamma (\gamma_t - \gamma_t^\mu) = 0, \quad (103)$$

with $\partial\Phi/\partial\gamma_t = \partial^2\mathcal{W}/\partial\gamma_t^2 > 0$.

Step 2. Implicit function theorem. Since $\Phi(\gamma_t^*, \mu_t) = 0$ and $\partial\Phi/\partial\gamma_t > 0$, the implicit function theorem guarantees that γ_t^* is continuously differentiable in $\mu_t(\theta_{high})$ around the optimum, with derivative

$$\frac{d\gamma_t^*}{d\mu_t(\theta_{high})} = -\frac{\partial\Phi/\partial\mu_t(\theta_{high})}{\partial\Phi/\partial\gamma_t} = -\frac{\frac{\partial L_t}{\partial \gamma_t}(\theta_{high}) - \frac{\partial L_t}{\partial \gamma_t}(\theta_{low})}{\partial\Phi/\partial\gamma_t}. \quad (104)$$

Because the denominator is strictly positive, the sign of (104) is determined by the sign of the marginal-loss difference in the numerator.

Step 3. Sign of the marginal-loss difference. Under proportional risks, $L_t(\gamma_t, \theta_K)$ depends on γ_t through the effective intensities $\tilde{\lambda}_j(t \mid \theta_K) = \lambda_{j0}(t) \exp(\beta_{K,j}(\theta_K) \cdot z + \zeta_{\gamma,j} \gamma_t)$.

Differentiating and forming the difference between types,

$$\begin{aligned} \frac{\partial L_t}{\partial \gamma_t}(\theta_{high}) - \frac{\partial L_t}{\partial \gamma_t}(\theta_{low}) &= \sum_{j \in \{2,3\}} \frac{\partial L_t}{\partial \tilde{\lambda}_j} \zeta_{\gamma,j} \left[\tilde{\lambda}_j(t \mid \theta_{high}) - \tilde{\lambda}_j(t \mid \theta_{low}) \right] \\ &\quad + \frac{\partial L_t}{\partial \tilde{\lambda}_1} \zeta_{\gamma,1} \left[\tilde{\lambda}_1(t \mid \theta_{high}) - \tilde{\lambda}_1(t \mid \theta_{low}) \right]. \end{aligned} \quad (105)$$

The hazard ratio between types for $j \in \{2, 3\}$ satisfies

$$\frac{\tilde{\lambda}_j(t \mid \theta_{high})}{\tilde{\lambda}_j(t \mid \theta_{low})} = \exp\left(\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})\right) > 1, \quad (106)$$

so $\tilde{\lambda}_j(t \mid \theta_{high}) > \tilde{\lambda}_j(t \mid \theta_{low})$ for $j \in \{2, 3\}$. Since $\partial L_t / \partial \tilde{\lambda}_j < 0$ for $j \in \{2, 3\}$ and $\zeta_{\gamma,j} > 0$, the first term of (105) is strictly negative. By the hypothesis $\beta_{K,j}(\theta_{high}) > \beta_{K,j}(\theta_{low})$, this differential dominates the effect of the payment channel ($j = 1$), so the difference in (105) is negative.

Step 4. Conclusion and structural elasticity. Substituting in (104), the numerator is negative and the denominator positive; therefore $d\gamma_t^* / d\mu_t(\theta_{high}) > 0$. The magnitude of this adjustment is proportional to the separation between high-risk and low-risk intensities, quantified by ratio (106): the larger $\beta_{K,j}(\theta_{high}) - \beta_{K,j}(\theta_{low})$, the larger the difference in $\partial L_t / \partial \gamma_t$ between types and, therefore, the larger the optimal response of γ_t^* to Bayesian updates. $\square$

7.1 Policy Recommendations

Integrate macrofinancial alerts into operational planning. Cointegration between ransom kidnapping, coercive capacity, and financial conditions indicates that changes in liquidity and the cost of capital may modify the criminal portfolio. Early-warning systems of GAULA units should incorporate regional indicators of credit, liquidity, and financial pressure to anticipate changes in the expected duration and profitability of kidnapping.

Apply financial interdiction adaptively. The block α_t^* disciplines captor participation and reduces the space for clandestine agreements, but indiscriminate application may impose unnecessary costs on the family. The severity of interdiction should respond to the posterior type, victim profile, and family wealth. When the posterior concentrates and the marginal value of experimentation falls, the block can be moderated without abandoning financial traceability or deterrent capacity.

Adopt intelligence protocols based on Bayesian dual control. Operational pressure γ_t^* should simultaneously pursue safe closure of the episode and likelihood separation

between organizations. The systematic combination of detection signals, proofs of survival, captivity duration, and competing-risk patterns makes it possible to classify the captor early, allocate tactical resources better, and reduce the human risk of operations.

8 Structural Identification: Singularity of Measures

This section presents, in an abstract framework consistent with Section 5 and Lemma 5.7, why **recurrent local separation** in total variation between daily laws couples with the criterion of ? to obtain **mutual singularity** of trajectory laws. The prolonged captivity event $\mathcal{T}$ (Definition 5.5) is a subset of Ω ; time indices are $t \in \{0, 1, 2, \dots\}$.

Notation (aligned with the body of the text). Given a policy path χ compatible with the PBE, let $Y_t = (m_t, d_t)$ be the daily public record and let $p_\theta^{(t)}$ be the mass function of Y_t conditional on the public information at the opening of t under type θ (same logic as $p_\theta(\cdot \mid \mathcal{F}_t^{\text{pub}})$ in Lemma 5.7). For two masses p, q on the same finite support, the **affinity** (Bhattacharyya coefficient) is $\rho(p, q) := \sum_y \sqrt{p(y)q(y)}$; squared Hellinger distance adopts here the normalization

$$\mathbf{H}^2(p, q) := \frac{1}{2} \sum_y \left(\sqrt{p(y)} - \sqrt{q(y)} \right)^2 = 1 - \rho(p, q).$$

Total variation distance satisfies $\|p - q\|_{\text{TV}} = \frac{1}{2} \sum_y |p(y) - q(y)|$ (as in (98)).

Lemma 8.1 (Product of experiments and singularity (Kakutani criterion)). *Fix two types $\theta \neq \theta'$, and suppose that, conditional on the policy fixed in equilibrium and on the relevant comparison regime (in particular, on event $\mathcal{T}$ when working with prolonged captivity), the sequence of daily experiments $\{Y_t\}_{t \geq 0}$ is independent and each Y_t takes values in a common finite support $\mathcal{Y}_t$. That is, for each finite horizon T , the law of the trajectory $(Y_0, \dots, Y_T)$ factorizes as the product measure $\bigotimes_{t=0}^T p_\theta^{(t)}$ under type θ , and analogously as $\bigotimes_{t=0}^T p_{\theta'}^{(t)}$ under θ' . Let $\rho_t := \rho(p_\theta^{(t)}, p_{\theta'}^{(t)})$. Then the affinity between the product laws at horizon T is*

$$\rho\left(\bigotimes_{t=0}^T p_\theta^{(t)}, \bigotimes_{t=0}^T p_{\theta'}^{(t)}\right) = \prod_{t=0}^T \rho_t. \quad (107)$$

*If there exists an infinite set of periods $\mathcal{S}_{\theta, \theta'} \subseteq \mathbb{N}$ such that $\|p_\theta^{(t)} - p_{\theta'}^{(t)}\|_{\text{TV}} \geq \varepsilon > 0$ for all $t \in \mathcal{S}_{\theta, \theta'}$, then $\rho_t \leq 1 - \varepsilon^2/2$ for all such t , and therefore $\prod_{t \in \mathcal{S}_{\theta, \theta'}} \rho_t = 0$. In the limit $T \rightarrow \infty$, the criterion of ? implies that the product measures over the infinite sequence are **mutually singular** ?.*

Proof. Identity (107) is obtained by expanding the masses of finite trajectories. Under

independence,

$$\rho\left(\bigotimes_{t=0}^T p_{\theta}^{(t)}, \bigotimes_{t=0}^T p_{\theta'}^{(t)}\right) = \sum_{(y_0, \dots, y_T)} \prod_{t=0}^T \sqrt{p_{\theta}^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)} = \prod_{t=0}^T \sum_{y_t \in \mathcal{Y}_t} \sqrt{p_{\theta}^{(t)}(y_t) p_{\theta'}^{(t)}(y_t)}.$$

For the second part, use the standard inequality $\|p - q\|_{\text{TV}} \leq \sqrt{2} \mathbf{H}(p, q)$ in finite spaces. Since $\mathbf{H}^2(p, q) = 1 - \rho(p, q)$, if $\|p - q\|_{\text{TV}} \geq \underline{\varepsilon}$, then

$$\rho(p, q) \leq 1 - \frac{\underline{\varepsilon}^2}{2}.$$

Thus, for each $t \in \mathcal{S}_{\theta, \theta'}$, $\rho_t \leq 1 - \underline{\varepsilon}^2/2 < 1$. Since there are infinitely many periods in $\mathcal{S}_{\theta, \theta'}$, the product of affinities contains infinitely many factors uniformly bounded below one; hence $\prod_{t \in \mathcal{S}_{\theta, \theta'}} \rho_t = 0$. By the criterion of ? for products of probabilities, the associated infinite product measures are mutually singular ?. $\square$

General version of Theorem 5.8. Lemma 8.1 isolates the *pure* measure-theoretic argument: under product structure and recurrent total-variation separation, the criterion of ? forces **mutual singularity** of trajectory laws. Theorem 5.8 in the body of the text invokes asymptotic identification (Definition 5.6) and, as a sufficient verifiable route, Lemma 5.7. The following result states the same Bayesian conclusion replacing that route with the hypotheses of Lemma 8.1 on the daily experiment Y_t .

Theorem 8.2 (Posterior consistency under singularity of product measures (Kakutani criterion)). *Suppose Θ_K is finite, the prior μ_0 has full support, and Bayesian updating is consistent with the PBE (Definition 5.3). Let θ_K^* be the realized type and $\mathcal{T}$ the prolonged captivity event (Definition 5.5). Suppose that, for every $\theta \in \Theta_K$ with $\theta \neq \theta_K^*$, the following conditions hold:*

- (i) **Trajectory compatibility.** $\mathbb{P}_{\theta_K^*}(\mathcal{T}) > 0$ and, for each false type, either $\mathbb{P}_{\theta}(\mathcal{T}) = 0$ (exclusion by survival), or the laws of signals are compared under the law conditioned/restricted to $\mathcal{T}$ as in the following clauses.
- (ii) **Product and recurrent TV separation.** For every false type with $\mathbb{P}_{\theta}(\mathcal{T}) > 0$, conditional on the policy fixed by equilibrium and on event $\mathcal{T}$, the sequence $\{Y_t\}_{t \geq 0}$ satisfies the product factorization and the existence of $\mathcal{S}_{\theta, \theta_K^*}$ and $\underline{\varepsilon}_{\theta} > 0$ described in Lemma 8.1.
- (iii) **Signal-experiment alignment.** In $\mathcal{T}$, the public signal component ζ_t contains $Y_t = (m_t, d_t)$; since Y_t is $\sigma(\zeta_t)$ -measurable, mutual singularity of the laws of $\{Y_t\}_{t \geq 0}$ transfers to that of $\{\zeta_t\}_{t \geq 0}$ in the sense of part (b) of Definition 5.6.

Then, in regime $\mathcal{T}$,

$$\mu_t(\theta) \xrightarrow[t \rightarrow \infty]{} 0 \quad \forall \theta \neq \theta_K^*, \quad \mathbb{P}_{\theta_K^*}\text{-a.s. in } \mathcal{T},$$

and

$$\mu_t(\theta_K^*) \xrightarrow[t \rightarrow \infty]{} 1 \quad \mathbb{P}_{\theta_K^*}\text{-a.s. in } \mathcal{T}.$$

Equivalently, in $\mathcal{T}$, $\{\mu_t\}$ converges weakly, $\mathbb{P}_{\theta_K^*}$ -a.s., to $\delta_{\theta_K^*}$.

Proof. Fix $\theta \neq \theta_K^*$. First we prove that the posterior of the false type converges to zero. If $\mathbb{P}_\theta(\mathcal{T}) = 0$, the comparison regime $\mathcal{T}$ does not occur under the false type, while $\mathbb{P}_{\theta_K^*}(\mathcal{T}) > 0$ by (i). Therefore, under the true law restricted to $\mathcal{T}$, the false type is ruled out by differential survival and its posterior likelihood vanishes.

Now suppose $\mathbb{P}_\theta(\mathcal{T}) > 0$. By (ii), the laws of $\{Y_t\}_{t \geq 0}$ factorize as products and exhibit recurrent total-variation separation. Lemma 8.1 implies that the infinite product laws induced by θ and by θ_K^* are mutually singular. By (iii), that singularity transfers to the public flow $\{\zeta_t\}_{t \geq 0}$ feeding Bayesian updating. Thus, there exists a tail event A_θ such that

$$\mathbb{P}_{\theta_K^*}(A_\theta \mid \mathcal{T}) = 1, \quad \mathbb{P}_\theta(A_\theta \mid \mathcal{T}) = 0.$$

Let

$$R_t^\theta := \frac{d\mathbb{P}_\theta|_{\mathcal{F}_t^{\text{pub}}, \mathcal{T}}}{d\mathbb{P}_{\theta_K^*}|_{\mathcal{F}_t^{\text{pub}}, \mathcal{T}}}$$

be the finite likelihood ratio of the false type against the true type, defined with respect to the Bayesian mixture with full-support prior. Tail singularity forces $R_t^\theta \rightarrow 0$ $\mathbb{P}_{\theta_K^*}$ -a.s. in $\mathcal{T}$?? By Bayes,

$$\mu_t(\theta) = \frac{\mu_0(\theta)R_t^\theta}{\mu_0(\theta_K^*) + \sum_{\theta' \neq \theta_K^*} \mu_0(\theta')R_t^{\theta'}} \leq \frac{\mu_0(\theta)}{\mu_0(\theta_K^*)} R_t^\theta,$$

and since $\mu_0(\theta_K^*) > 0$, $\mu_t(\theta) \rightarrow 0$ almost surely in $\mathcal{T}$. Since Θ_K is finite, all false coordinates converge to zero simultaneously outside a finite null set. Finally, $\sum_{\theta \in \Theta_K} \mu_t(\theta) = 1$ for every t , so passing to the limit in the finite sum gives $\mu_t(\theta_K^*) \rightarrow 1$. In a finite simplex, coordinate convergence to $\delta_{\theta_K^*}$ is equivalent to weak convergence. $\square$

9 Conclusion

This paper presents a dynamic deterrence mechanism under incomplete information for the analysis of ransom kidnapping, whose central contribution is the **Theorem on the Identification of Rational Types in Unknown Environments** (Theorem 5.8). The result establishes that, when public policy generates asymptotically separable risk intensities across captor types, the accumulation of daily signals forces almost sure convergence

of the State's posterior beliefs toward the true type θ_K^* in the prolonged captivity regime $\mathcal{T}$.

Three structural elements support this result. First, the MDG process (Mano de Dios–Guadalupe), axiomatically justified as an implementation law compatible with full support, dominance, and maximum entropy (Remark 4.2), ensures that the public history remains informative even off the equilibrium path. Second, the Implementability Theorem (Theorem 5.12) establishes that there exists a Perfect Bayesian Equilibrium implementing the social choice rule under incentive compatibility IC^K , individual rationality IR^K , and IR^F . Third, the structural identification section (Section 8) provides the measure-theoretic foundation: mutual singularity of trajectory laws, through the Kakutani criterion, converts local statistical separation into global asymptotic identification.

On the empirical plane, cointegration between kidnapping modalities and financial conditions reveals a persistent environment that conditions retention costs. At the microeconomic level, multinomial and competing-risk models show systematic heterogeneity by perpetrator, victim, region, duration, and institutional presence. These results turn the Colombian evidence into priors, intensities, and constraints directly usable by the mechanism.

The dynamic calibration illustrates its operational viability. The optimal policy combines financial interdiction, operational pressure, and informational value to accelerate posterior concentration without abandoning incentive and participation constraints. The policy implications (Section 7) are therefore operational: γ_t^* must be calibrated according to posterior beliefs and structural separation between types, while its effect on duration depends on the net sensitivity $\kappa_h(\theta_K, t)$ of competing risks.

The future agenda includes extension to continuous types $\Theta_K \subseteq \mathbb{R}$, endogenous strategic communication between family and State, structures with multiple private agents, and structural estimation with panel data on kidnapping episodes. These extensions would make it possible to transfer the framework to other asymmetric-information problems, such as financial supervision, labor selection, and surveillance of illicit organizations.

Statement on the Use of Generative Artificial Intelligence

In preparing this manuscript, generative artificial intelligence tools were used as support for bibliographic and writing review, programming and formal verification in Lean 4, quantitative code development in MATLAB, Stata, and Python, and computational implementation of the calibrated model. The author verified and validated the information, proofs, estimates, and codes, and assumes full responsibility for the content and academic integrity of the document.